

Next Best View Selections for Semantic and Dynamic 3D Gaussian Splatting

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Abstract

Understanding semantics and dynamics has been crucial for embodied agents in various tasks. Both tasks have much more data redundancy than the static scene understanding task. We formulate the view selection problem as an active learning problem, where the goal is to prioritize frames that provide the greatest information gain for model training. To this end, we propose an active learning algorithm with Fisher Information that quantifies the informativeness of candidate views with respect to both semantic Gaussian parameters and deformation networks. This formulation allows our method to jointly handle semantic reasoning and dynamic scene modeling, providing a principled alternative to heuristic or random strategies. We evaluate our method on large-scale static images and dynamic video datasets by selecting informative frames from multi-camera setups. Experimental results demonstrate that our approach consistently improves rendering quality and semantic segmentation performance, outperforming baseline methods based on random selection and uncertainty-based heuristics.

1. Introduction

Recent advancements in 3D Gaussian Splatting (3DGS) have enabled real-time rendering and semantic scene understanding for both static and dynamic environments. Extensions of 3DGS now incorporate features from large-scale 2D vision-language models for semantic reasoning. To enable dynamic scene representations, recent methods [16, 20, 21, 23, 30, 42] augment Gaussians with temporal modeling to capture changes in geometry and appearance over time. These developments make 3DGS as a promising backbone for applications in robotics, AR/VR, and digital content creation, where efficiency and semantic accuracy are critical. However, the potential of 3DGS in such real-world settings is limited by the substantial resources required for training, especially when scaling to large-scale environments, densely instrumented multi-camera systems, or long-duration dynamic sequences. The key challenge lies in the vast number of candidate viewpoints: although many are redundant or

uninformative, choosing which views are most informative is non-trivial. Unlike static reconstruction, dynamic environments introduce additional uncertainty due to evolving geometry, appearance changes, and semantic inconsistencies across time.

A natural solution is **next-best-view** (NBV) selection, which seeks to prioritize views that most improve reconstruction and understanding. Effective NBV selection can significantly improve both geometric reconstruction and semantic consistency, especially in dynamic environments where scene content changes over time. In practice, however, existing strategies often rely on random sampling, uncertainty heuristics [14, 40], or black-box selection models [29, 41], which do not explicitly maximize information gain and therefore can lead to suboptimal reconstruction and semantic coverage. Moreover, prior work predominantly focuses on static scenes, leaving dynamic NBV selection largely unexplored. Under dynamic scenes, these strategies struggle to generalize across scenes with complex motion and appearance variation, often lead to wasted computation, incomplete reconstructions, and semantic drift, leaving us without a principled way to allocate training resources effectively.

In this work, we present a unified framework that combines dynamic and semantic 3DGS and formulates NBV selection as an active learning problem. Building on this formulation, we develop the first NBV algorithm tailored to dynamic semantic 3DGS, jointly capturing geometric, semantic, and temporal deformation information. Unlike FisherRF [11], which considers only geometric information for static scenes, our method systematically leverages Fisher Information to identify views that maximally improve both reconstruction quality and dynamic scene modeling. To handle the computational challenges of large-scale dynamic semantic 3DGS, we propose an efficient diagonal approximation of the Fisher Information matrix and introduce a novel estimator for the Fisher Information of deformation networks using the trace of the gradient outer product. Beyond proposing a new algorithm, our framework integrates NBV selection directly into the 3DGS backbone, greatly extending the scenarios and applications of NBV to semantic and dynamic scene understanding.

We evaluate our framework on large-scale static datasets (Replica [36]) and dynamic multi-camera sequences (Neu3D [20]), showing consistent improvements over baseline methods. Our contributions include:

- The first Fisher Information-driven NBV selection framework for *dynamic semantic 3DGS*, capturing geometric, photometric, semantic, and deformation informativeness.
- An efficient diagonal approximation of the Fisher Information matrix, enabling tractable computation in large-scale semantic 3DGS models.
- A novel method to estimate Fisher Information for deformation MLPs, leveraging the trace of the gradient outer product as a proxy.

2. Related works

2.1. Semantic 3D Gaussians Splatting

Recent advancements have significantly improved semantic understanding in 3D scene representations. Early works build upon implicit radiance field representations such as NeRF. Methods like Semantic NeRF [49], Panoptic Lifting [35], and Contrastive Lift [1] pioneered the integration of semantic labels into NeRF-based frameworks, achieving sharp and accurate 3D semantic segmentation. Subsequent approaches—such as Distilled FFD [15], F3F [38], panoptic NeRF [6], NeRF-SOS [4], Interactive Segmentation of Radiance Fields [7], Weakly Supervised 3D Open-Vocabulary Segmentation [22], FFNF [24], VL-Fields [37], Featurenerf [45], Distilled Feature Fields [34], and LERF [13] leverage semantic features from pretrained vision-language models like CLIP [31], LSeg[17], or DINO[25] to enable open-vocabulary, pixel-level semantic understanding.

More recently, 3D Gaussian Splatting (3DGS) [12] has emerged as a compelling explicit alternative to NeRF due to its faster training, real-time rendering capabilities, and high-quality scene reconstruction. Semantic extensions of 3DGS have shown great promise. Semantic Gaussians [8] and ConceptFusion [9] project features from 2D pretrained encoders onto 3D Gaussians, enabling open-vocabulary understanding. OpenGaussian [43] and OpenScene [28] achieve point-level semantic segmentation and produce sharp and accurate semantic segmentation results. Feature 3DGS [50], CLIP-FO3D [47], and PLA [2] further distill semantic features from 2D foundation models into the 3D domain using differentiable rasterization. Thanks to the real-time rendering and optimization capabilities of 3DGS, notable examples like SGS-SLAM [19], SemGauss-SLAM [51], and GS³LAM [18] also integrate semantic features into 3DGS-based SLAM pipelines for robust and scalable scene reconstruction.

Motivated by the strong generalization and ease of training of Feature 3DGS [50], we adopt its framework as the foundation for our semantic Gaussian Splatting module.

2.2. Dynamic Gaussians Splatting

Several recent works have extended 3D scene representation to dynamic scenes by introducing temporal modeling, enabling novel view synthesis in dynamic environments. Based on NeRF, early approaches such as D-NeRF [30] introduced time as an explicit input to model non-rigid deformations, while Nerfies [26] leveraged deformation fields with elastic regularization to animate casually captured scenes. HyperNeRF [27] advanced this by modeling topological changes via a hyperspace embedding, handling complex non-rigid transformations. STaR [46] decomposes scenes into static and dynamic components using a self-supervised rigid-body model.

As for 3DGS-based models, 4D Gaussian Splatting (4D-GS) [42] introduces a unified representation that combines 3D Gaussians with 4D neural voxels with a lightweight MLP to efficiently predict Gaussian deformations over time. Gaussian-Flow [21] alternatively models temporal attribute deformations using time and frequency domain parameterizations. Dynamic 3D Gaussians [23] uses dense 6-DOF tracking and dynamic reconstruction through enforcing local rigidity constraints to enable Gaussians to maintain persistent attributes when they move and rotate over time. Explicit sampling strategies like Fully Explicit Dynamic Gaussian Splatting [16] have also been proposed to separate static and dynamic components during training to represent continuous motion and reduce computational cost.

In our work, we adopt the 4D-GS [42] framework as the foundation for our dynamic Gaussians module, leveraging its efficient modeling of dynamic scenes and real-time rendering capabilities.

2.3. Uncertainty and view selection

Driven by the need to reduce training costs and enable active training, a paradigm where models selectively acquire data based on informativeness, many recent works have focused on extracting useful cues from the trained model, existing data, and candidate viewpoints. A central theme across these methods is the estimation and exploitation of uncertainty. Magic Moments [3] introduces higher moments of rendering equations to quantify the uncertainty of radiance fields and show their relationship with rendered error and their usage as powerful NBV selection criteria. CSS [40] utilizes the expectation and variance of categorical distributions to probabilistically update semantic maps within the 3D Gaussian Splatting (3DGS) framework. Similarly, [14] proposes a colorized surface voxel to interpret color uncertainty and surface information. While these methods primarily adopt Bayesian formulations to explicitly quantify uncertainty, others infer uncertainty by learning data distributions via alternative models. DNRSelect [41] employs a reinforcement learning-based view selector trained on rasterized images to find optimal views for deferred neural rendering. ActiveInit-

Splat [29] defines a black-box objective function based on density and occupancy metrics and employs a Gaussian process as the surrogate model to predict informative views.

Beyond uncertainty estimation, several methods directly target maximizing information gain. GauSS-MI [44] utilizes Shannon Mutual Information to select informative viewpoints. FisherRF [11] introduces a Fisher Information-based metric to guide view selection in radiance fields without reliance on ground-truth data. AG-SLAM [10] further extends this idea by integrating it into an active SLAM framework, enabling online scene reconstruction and autonomous exploration via information-driven trajectory planning.

In this work, we adopt Fisher Information as the unified selection criterion for both semantic Gaussian Splatting and deformation-based dynamic modeling, facilitating view selection that balances scene semantics and motion-aware updates.

3. Method

In this section, we describe our active learning algorithm with Fisher Information for NBV selection in dynamic semantic 3DGS. We begin by reviewing the necessary preliminaries on semantic 3DGS and dynamic 3DGS, which form the foundation of our unified backbone. We then introduce our formulation for computing Fisher Information, first for Gaussian parameters, which capture both geometry and semantics features in Section 3.2 and subsequently for the deformation network, which captures temporal dynamics in Section 3.3. This decomposition allows us to quantify the informativeness of candidate views across all key aspects of the scene representation, enabling principled and efficient NBV selection.

3.1. Preliminaries

In this Section, we first introduce the two core components of our representation framework. Section 3.1.1 details the formulation of Semantic 3D Gaussian Splatting, which distills semantic features from 2D vision-language models into a 3D explicit representation. Section 3.1.2 extends this framework to 4D Gaussian Splatting, incorporating a deformation network to capture dynamic scene changes over time.

3.1.1 Semantic 3D Gaussians Splatting

Following the work of feature 3DGS [50], the semantic 3D Gaussian Splatting (3DGS) represents a scene using a collection of Gaussians $\{\mathcal{G}_i\}_{i=1}^N$, where each Gaussian $\mathcal{G}_i$ is defined by its location $x_i \in \mathbb{R}^3$, rotation $q_i \in \mathbb{R}^4$, scale $s_i \in \mathbb{R}^3$, opacity $\alpha_i \in \mathbb{R}$, color $c_i \in \mathbb{R}^3$, and a semantic feature vector $f_i \in \mathbb{R}^d$. The full parameter set is:

$$\Theta_i = \{x_i, q_i, s_i, \alpha_i, c_i, f_i\}.$$

To project the 3D Gaussians into image space, each covariance matrix Σ_i is transformed to camera space:

$$\Sigma'_i = JW\Sigma_iW^\top J^\top, \quad (1)$$

where W is the world-to-camera transformation and J is the Jacobian of the projection function. The original 3D covariance is decomposed as:

$$\Sigma_i = R_i S_i^2 R_i^\top, \quad (2)$$

where R_i is the rotation matrix, which can be converted from quaternions q_i and S_i is a diagonal matrix of scales s_i . In this way, the Σ_i can be guaranteed to be positive semi-definite during optimization.

Rendering is performed using volumetric α -compositing in front-to-back order. The color C and semantic feature F_s at a pixel are computed as:

$$C = \sum_{i \in \mathcal{N}} c_i \alpha_i T_i, \quad F_s = \sum_{i \in \mathcal{N}} f_i \alpha_i T_i, \quad (3)$$

where $\mathcal{N}$ is the set of Gaussians overlapping the pixel (sorted by depth), and the transmittance T_i is:

$$T_i = \prod_{j=1}^{i-1} (1 - \alpha_j). \quad (4)$$

In order to supervise the semantic features, we distill from the pretrained 2D foundation model Lseg [17] with a feature dimension of 512. To reduce training and rendering cost, we keep low-dimensional features $f_i \in \mathbb{R}^{128}$ at each 3D Gaussian and employ a lightweight convolutional decoder to upsample the rendered feature map $F_s \in \mathbb{R}^{H \times W \times 128}$ to $F'_s \in \mathbb{R}^{H \times W \times 512}$. The overall loss function is:

$$\mathcal{L} = \mathcal{L}_{\text{rgb}} + \gamma \mathcal{L}_{\text{feat}}, \quad (5)$$

with:

$$\mathcal{L}_{\text{rgb}} = (1 - \lambda) \|I - \hat{I}\|_1 + \lambda \mathcal{L}_{\text{D-SSIM}}, \quad \mathcal{L}_{\text{feat}} = \|F_t - F'_s\|_1, \quad (6)$$

where $\hat{I}$ is the rendered image and I is the ground truth.

3.1.2 Dynamic 3DGS

4D-GS [42] extends the static 3D Gaussian Splatting framework to dynamic scenes by introducing a deformation network that models the motion and shape changes of 3D Gaussians over time. At any timestamp t , the Gaussians are transformed via a deformation network:

$$G'_t = G + \Delta G_t = \{X_i + \Delta X_i(t), s_i + \Delta s_i(t), r_i + \Delta r_i(t)\}, \quad (6)$$

where $\Delta X_i(t)$, $\Delta s_i(t)$, and $\Delta r_i(t)$ are the time-dependent predicted deformations.

The deformation network F_ϕ consists of two components: a spatial-temporal structure encoder H and a multi-head deformation decoder D . The encoder H uses a multi-resolution voxel grid decomposition, inspired by K-Planes [5], over six 2D plane projections $(x, y), (x, z), (y, z), (x, t), (y, t), (z, t)$. Each Gaussian’s position and time (X_i, t) are queried from these planes via bilinear interpolation to produce a feature embedding:

$$F_i = \phi_d \left(\bigcup_{(i,j)} \text{interp}(R_{(i,j)}^{(l)}) \right), \quad (7)$$

where $R_{(i,j)}^{(l)}$ denotes the voxel planes at resolution level l , and ϕ_d is a small MLP used to fuse the plane features.

The decoder $D = \{\phi_x, \phi_r, \phi_s\}$ consists of three heads that output the deformation for position, rotation, and scale:

$$\Delta X_i(t) = \phi_x(F_i), \quad \Delta r_i(t) = \phi_r(F_i), \quad \Delta s_i(t) = \phi_s(F_i). \quad (8)$$

During rendering, the pretrained 3DGS are first deformed through $\{G'_i\}_{i=1}^N = F(\{G_i\}_{i=1}^N)$. Then, the deformed Gaussians G'_t are rendered via differentiable splatting to produce the final image:

$$\hat{y}_t = S(M_t, G'_t), \quad (9)$$

where M_t is the camera pose at time t and S denotes the splatting renderer.

3.2. Fisher information for semantic 3DGS

In this section, we will describe how to qualify Fisher information for Gaussian parameters for NBV selection. Following the instruction of FisherRF [11], let the radiance field be parameterized by G , representing the set of attributes for all 3D Gaussians in the scene, including location x , scale s , opacities α , colors c , and semantic feature f . At a given camera pose $M \in \text{SE}(3)$, the rendering function $S(M, G)$ produces a synthetic image. We consider the negative log-likelihood of the observed image y under the rendered prediction:

$$\mathcal{L}(y; G) = -\log p(y|M, G) = \|y - S(M, G)\|_2^2. \quad (10)$$

Under standard regularity conditions, the *observed* Fisher Information with respect to the parameters G at pose M is given by the Hessian of this loss:

$$\mathcal{I}(G; M) = \nabla_G S(M, G)^\top \nabla_G S(M, G) + \lambda I, \quad (11)$$

where λI is a log-prior regularization term.

However, due to the large number of parameters, computing the full Fisher Information matrix is intractable in practice. Following the approximation proposed in FisherRF [11], we apply a Laplace approximation by retaining

only the diagonal elements of the Hessian and adding a prior regularization term:

$$\mathcal{I}(G; M) \approx \text{diag}(\nabla_G S(M, G)^\top \nabla_G S(M, G)) + \lambda I, \quad (12)$$

This diagonal approximation significantly reduces computational cost and memory overhead, while preserving the essential structure of the Fisher Information needed for evaluating local sensitivity.

To select the most informative viewpoint from a candidate pool $\{M_i\}_{i=1}^N$, we pick the view that maximize the *Expected Information Gain* (EIG), defined as:

$$\text{EIG}(M_i) = \text{tr}(\mathcal{I}(G; M_i) \cdot \mathcal{I}_{\text{train}}^{-1}), \quad (13)$$

where $\mathcal{I}_{\text{train}}$ is the accumulated Fisher Information of current training views. The trace measures how much the candidate view would sharpen the current uncertainty landscape over parameters.

3.3. Fisher Information for Deformation Network

In this section, we introduce a novel estimator for the Fisher Information of multilayer perceptrons (MLPs), which we employ to quantify the informativeness of our deformation module.

For the deformation network F_ϕ , the informativeness of a candidate view $(\mathbf{x}^{acq}, \mathbf{y}^{acq})$ is quantified by the Fisher Information with respect to deformation network parameters ϕ and the current training data D^{train} :

$$\begin{aligned} \mathcal{I}[\phi; \mathbf{y}^{acq} | \mathbf{x}^{acq}, D^{\text{train}}] &= H[\phi | D^{\text{train}}] \\ &\quad - H[\phi | \mathbf{y}^{acq}, \mathbf{x}^{acq}, D^{\text{train}}]. \end{aligned} \quad (14)$$

Following FisherRF [11], this gain can be upper bounded by the curvature of the log-likelihood:

$$\begin{aligned} \mathcal{I}[\phi] &\leq \frac{1}{2} \text{tr} \left(\mathbf{H}_\phi'' [\log p(\mathbf{y}^{acq} | \mathbf{x}^{acq}, \phi)] \right. \\ &\quad \left. \cdot \mathbf{H}_\phi'' [\log p(\phi | D^{\text{train}})]^{-1} \right). \end{aligned} \quad (15)$$

where $\mathbf{H}_\phi''$ denotes the Hessian with respect to ϕ .

For view selection, the prior Hessian term $\mathbf{H}_\phi'' [\log p(\phi | D^{\text{train}})]^{-1}$ is constant across candidate views and can be dropped. Thus our acquisition criterion simplifies to maximizing

$$\mathcal{S}(\mathbf{x}^{acq}) = \text{tr}(\mathbf{H}_\phi'' [\log p(\mathbf{y}^{acq} | \mathbf{x}^{acq}, \phi)]). \quad (16)$$

Finally, we estimate the trace efficiently with Hutchinson’s estimator:

$$\mathcal{S}(\mathbf{x}^{acq}) \approx \mathbb{E}_{\mathbf{v} \sim \mathcal{N}(0, I)} [\|\mathbf{H}_\phi'' \mathbf{v}\|^2], \quad (17)$$

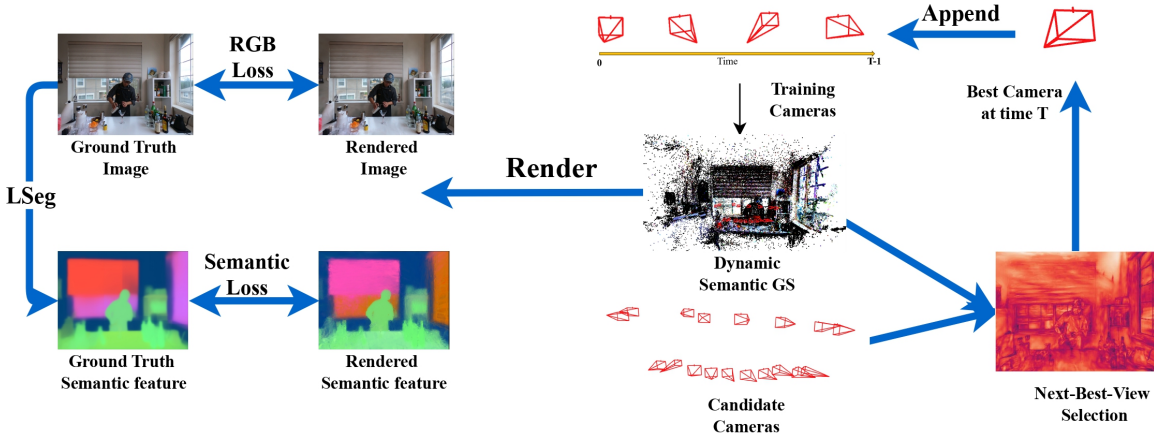


Figure 1. **Overview of our work.** At each timestep, we select the NBV from a set of candidate camera views using a Fisher Information-based criterion. The selection process evaluates each candidate view based on its expected contribution to both semantic feature learning and dynamic deformation modeling. Once selected, the chosen view is incorporated into the training process of our Dynamic and Semantic 3DGS. For rendering and supervision, the Gaussians are first deformed by an MLP deformation network based on the current timestep, then projected into RGB images and semantic feature maps. These outputs are compared against ground-truth images and semantic labels to jointly optimize both the dynamic deformation and semantic representation.

where $\mathbf{v}$ is a random probe vector and Hessian-vector products are computed via automatic differentiation.

This formulation explicitly ties Fisher Information to the deformation network parameters ϕ , ensuring that the selected views maximize temporal consistency and improve the modeling of dynamic scene evolution.

4. Experiments

In this section, we present a comprehensive evaluation of our proposed method. We first describe the experimental setup, including the datasets, baseline methods, and evaluation metrics. We then report results on two key scenarios: selecting a limited number of candidate camera poses from large-scale semantic datasets for static semantic 3DGS, and selecting informative frames from multi-camera sequences for dynamic and semantic 3DGS. Finally, we conduct ablation studies to analyze the impact of the log-prior regularization parameter and different feature choices for NBV selection, providing deeper insights into the design and effectiveness of our approach.

4.1. Experimental Set-up

Dataset For semantic 3DGS active training, we choose Replica [36], which composes various indoor environments with semantic labels. We follow the dataset setting of Feature 3DGS [50] to pick 75-80 images for each scene and leave 15 of them for testing.

For dynamic and semantic 3DGS active training, we choose Neu3D [20], which includes 15-21 synchronous camera

videos. From each view, we sample 40 temporally synchronized frames. The central view, `cam00`, is reserved as the test view, and the remaining views are used for training. To gain the ground truth semantic features, we apply LSeg [17] to semantic segmentation and attain pixel-level features for each image. We utilize Colmap [33] [32] to get sparse point clouds for training initialization.

Baselines We compare our method against random view selection and semantic uncertainty-based selection. Following the approach of Magic Moments [3], we use feature covariance as a measure of uncertainty and implement the computation efficiently in CUDA. To estimate the second moment of semantic features, we first square each feature vector associated with a Gaussian and accumulate the results to compute the per-pixel second moment. The covariance is then obtained by subtracting the squared mean (i.e., the square of the rendered semantic output) from the second moment.

Metrics We evaluate our rendering quality with Structural Similarity Index Measure (SSIM) [39], Peak-signal-to-noise ratio (PSNR), and Learned Perceptual Image Patch Similarity (LPIPS) [48], and semantic segmentation quality with mean intersection-over-union (mIoU) and mean accuracy (mAcc). To evaluate semantic segmentation, we first project Gaussian semantic features into rendered images to get per-pixel features. Then, we project feature maps into class probabilities by computing dot products between features

Method	SSIM ($\uparrow$)	PSNR ($\uparrow$)	LPIPS ($\downarrow$)	mAcc ($\uparrow$)	mIoU ($\uparrow$)
Random	0.9077	29.6600	0.1477	0.9189	0.7120
Magic Moments [3]	0.9615	33.5200	0.0591	0.9256	0.7593
FisherRF [11]	0.9593	32.1700	0.0614	0.9230	0.7441
Ours	0.9626	33.6000	0.0572	0.9289	0.7648

Table 1. Average performance on Replica dataset under different view selection strategies. The **best**, **second best**, and **third best** results are highlighted with light red, orange, and yellow backgrounds, respectively.

and text features from predefined labels and assigning every pixel to the class with the highest logit per pixel.

4.2. Static semantic 3DGS active training

For active training in static semantic Gaussian Splatting, we first start our training with two images. Then after 200 iterations, we compute Fisher Information for all the remained camera poses and add the one that yields the highest expected information gain to the training set, until adding 10 images every 200 iterations. The model is then trained for a total of 7,000 iterations. All experiments are conducted on an NVIDIA L40 GPU, requiring approximately 4GB of memory and 45 minutes of training per scene. For each NBV selection, it takes approximately 40 seconds.

Table 1 summarizes the average performance of different view selection strategies on the Replica dataset. Our method, which incorporates Fisher Information computed with semantic information, achieves the best results across all metrics, indicating improved visual quality and semantic accuracy.

Removing the semantic information from the Fisher Information calculation leads to a consistent decline in performance, highlighting the importance of incorporating semantic sensitivity when estimating view informativeness. The semantic covariance baseline Magic Moments [3], which relies on second-order statistics of categorical features, underperforms compared to our Fisher Information-based method with semantic information. The random view selection baseline results in the weakest performance across all metrics, further emphasizing the advantage of principled, information-driven view selection.

4.3. Dynamic and semantic 3DGS active training

For active training in dynamic and semantic 3DGS, we begin by training a static semantic model using the first frame from all cameras for 3000 iterations. Subsequently, to train the deformation network for dynamic, for each future timestep, we compute the Fisher Information across all candidate camera views and iteratively select the view with the highest information gain every 200 iterations. This process continues until all 39 remaining timesteps are each assigned a single informative camera frame. The dynamic training phase runs

Method	SSIM ($\uparrow$)	PSNR ($\uparrow$)	LPIPS ($\downarrow$)	mAcc ($\uparrow$)	mIoU ($\uparrow$)
Random	0.8831	26.7361	0.1933	0.8848	0.7284
Magic Moments [3]	0.9184	28.3795	0.1581	0.8890	0.7350
FisherRF [11]	0.9186	27.6743	0.1605	0.8863	0.7547
Ours	0.9239	28.6191	0.1553	0.8963	0.7604

Table 2. Average performance across Neu3D using different view selection strategies.

Method	SSIM ($\uparrow$)	PSNR ($\uparrow$)	LPIPS ($\downarrow$)	mAcc ($\uparrow$)	mIoU ($\uparrow$)
Ours (w/o reg.)	0.9173	28.5853	0.1589	0.8929	0.7413
Ours ($\lambda = 10^{-7}$)	0.9184	28.0638	0.1594	0.8881	0.7387
Ours ($\lambda = 10^{-6}$)	0.9239	28.6191	0.1553	0.8963	0.7604

Table 3. Average performance on Neu3D dataset under our view selection strategy with and without regularization.

Method	SSIM ($\uparrow$)	PSNR ($\uparrow$)	LPIPS ($\downarrow$)	mAcc ($\uparrow$)	mIoU ($\uparrow$)
Geom [11]	0.9186	27.6743	0.1605	0.8863	0.7547
Geom+Sem	0.9200	28.2788	0.1596	0.8885	0.7571
Geo+Def	0.9198	28.1501	0.1578	0.8905	0.7367
Ours	0.9239	28.6191	0.1553	0.8963	0.7604

Table 4. Average performance across Neu3D using different view selection strategies.

14000 iterations. For regularization, here we set $\lambda = 10^{-6}$. Training is conducted on an NVIDIA L40 GPU, requiring approximately 8GB of memory and 2 hours per scene. For each NBV selection, it takes approximately 20 seconds.

Table 2 presents the average performance across five dynamic scenes from the Neu3D dataset using different view selection strategies. The proposed method based on Fisher Information achieves the best performance across all evaluation metrics, presenting superior visual quality and semantic consistency. In contrast, the semantic covariance baseline consistently underperforms the Fisher-based method. The random selection baseline yields the weakest performance, especially in perceptual quality and segmentation accuracy, further validating the effectiveness of Fisher Information-guided view selection for dynamic and semantic 3DGS.

We also analyze the attention patterns of the three selection methods. As shown in Fig. 4, which visualizes the interest heatmaps, the subject is in the process of closing the kitchen tongs. Magic Moments [3] distributes attention evenly across the entire kitchen tongs. FisherRF [11] emphasizes the connection between the handles and the gripping ends. In contrast, our method focuses more on the handles, likely due to their dynamic motion during the closing action.

4.4. Ablation study

Ablation on choice of log-prior regularization parameter. We conduct an ablation study to evaluate the effect of the log-prior regularization parameter λ in the Fisher Information computation (Eq. 11). This term is critical for stabilizing

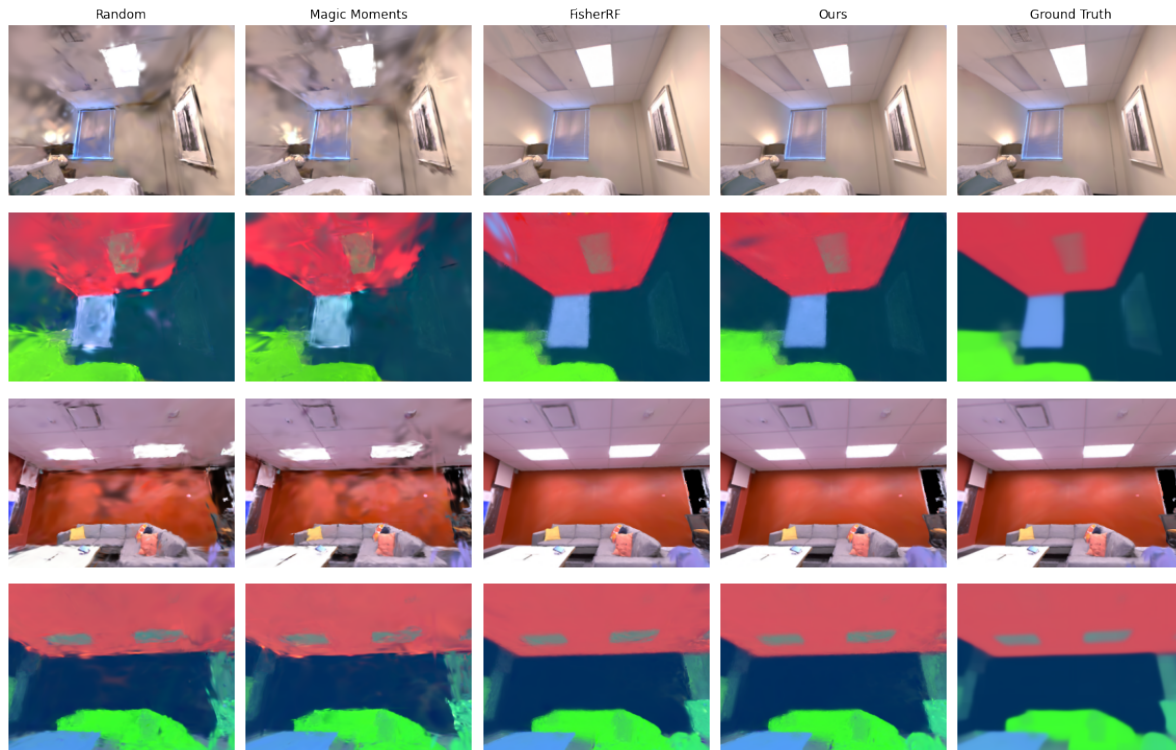


Figure 2. **Semantic segmentation results on Replica dataset.** The first and third rows of images are the rendered results. The second and forth rows of images are the results of visualizing the feature maps of test set after being trained with 12 training views. All the methods are based on Feature 3DGS [50] except for different view selection methods to augment the training data.

the diagonal approximation of the Hessian, especially in dynamic and semantic scenes. Table 3 compares three variants of our method on the Neu3D dataset: no regularization ($\lambda = 0$), weak regularization ($\lambda = 10^{-7}$), and our default setting ($\lambda = 10^{-6}$). The results show that the full model with $\lambda = 10^{-6}$ consistently outperforms the other two across all metrics. Without regularization, the Fisher scores become noisier and less reliable, leading to degraded view selection and semantic accuracy.

Ablation on Features Used for NBV Selection. We conduct an ablation study to evaluate the contribution of different features in the Fisher Information computation for NBV selection in 4DGS. Specifically, we compare four variants on the Neu3D dataset: **Geom**, which uses Fisher information only on geometric parameters following *FisherRF* [11]; **Geom+Sem**, which additionally incorporates semantic features; **Geom+Def**, which combines geometric and deformation parameters; and **Ours**, which jointly considers geometric, semantic, and deformation parameters. As shown in Table 4, adding semantic information improves fine-detail reconstruction and high-frequency texture fidelity, while adding deformation information enhances temporal consistency in dynamic regions. Our full model, which integrates all three components, achieves the best performance across

all metrics, confirming that semantic and deformation cues provide complementary benefits to purely geometric NBV selection.

5. Conclusion

To address the critical challenge of reducing training data requirements while preserving both rendering fidelity and semantic accuracy across static and dynamic scenes, we introduce a unified 3DGS backbone that jointly models semantic and temporal representations, together with an active learning framework based on Fisher Information for Next-Best-View selection. By quantifying the informativeness of candidate viewpoints with respect to geometric and semantic Gaussian parameters and the deformation network, our method identifies the most informative viewpoints to guide the training process. Our experimental results on Replica and Neu3D validate the effectiveness of the approach. Across diverse datasets, the proposed method consistently improves reconstruction quality, semantic consistency, and training efficiency compared to existing baselines. These findings highlight that view selection guided by explicit information measures can lead not only to faster convergence but also to more reliable modeling of dynamic and semantic scenes. By reducing the reliance on large amounts of training data,

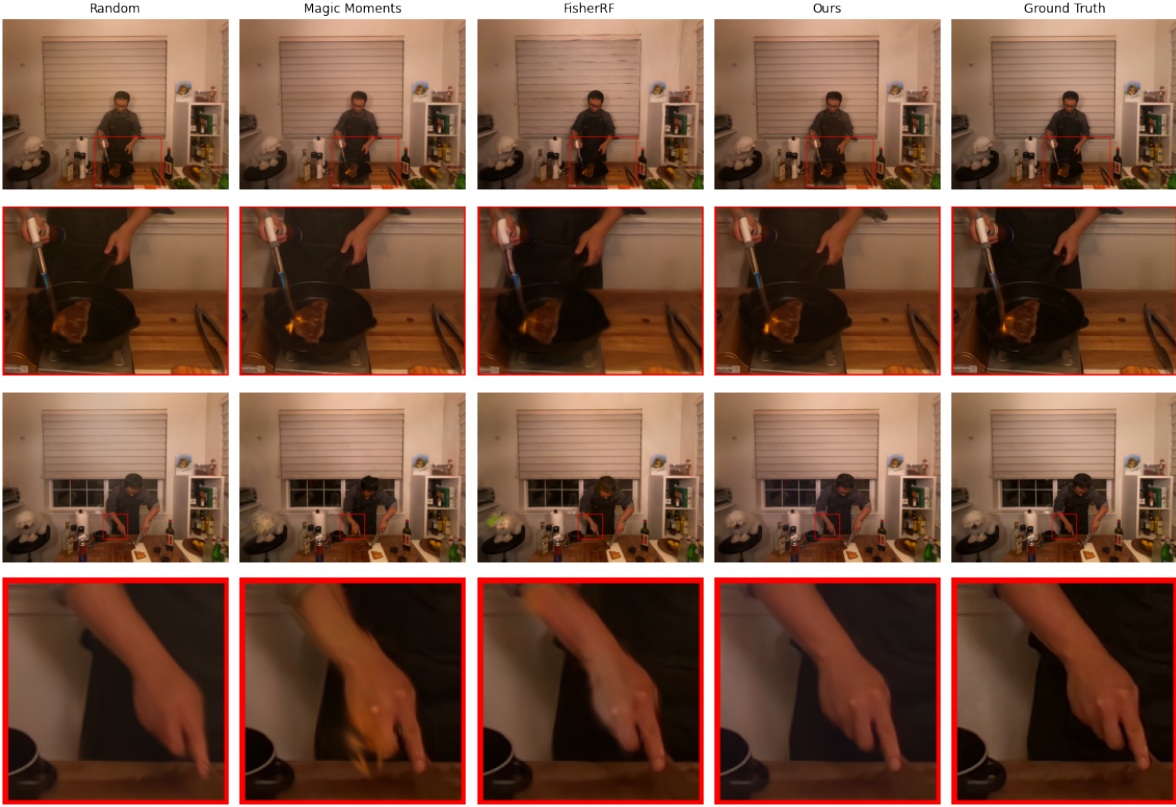


Figure 3. **Zoomed-in qualitative study of our method on Neu3D dataset.** The second and fourth rows are zoom-in figures. Visualizations are the results of the test set after being trained with 39 training views. All the methods are based on the same dynamic and semantic 3DGS, except for different view selection methods to augment the training data.

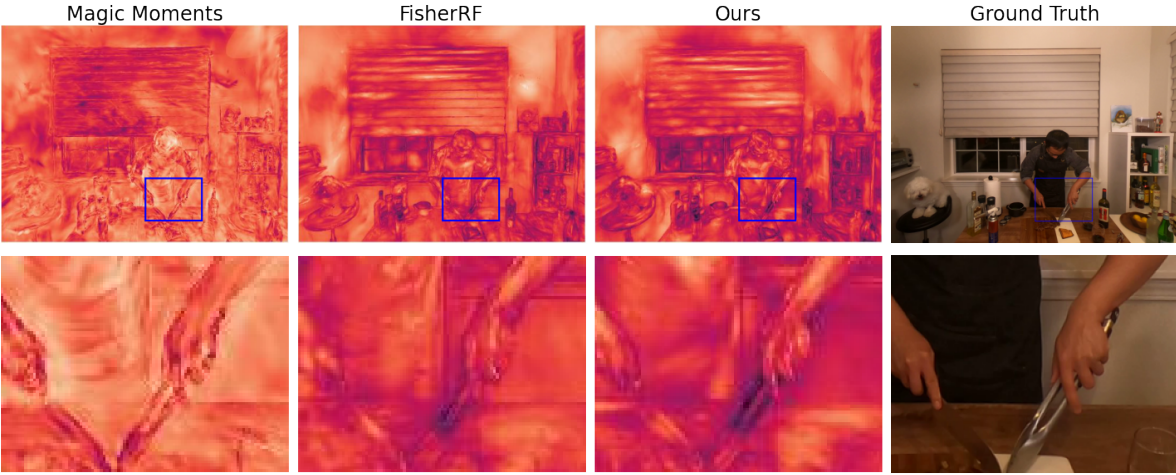


Figure 4. **Heatmap for selection method interest on Neu3D dataset.** The second row is zoom-in figures.

517 our method enables more efficient and scalable training of
518 high-quality 3DGS models.
519 Our framework offers a foundation for adaptive data acquisi-
520 tion. In robotics and SLAM, principled NBV selection can
521 help agents actively choose viewpoints to improve mapping

accuracy and semantic awareness under limited sensing bud-
gets. In AR/VR, efficient view selection accelerates scene
capture and dynamic updates. These impacts highlight the
potential of our approach to enable efficient, scalable, and
intelligent scene understanding across diverse domains.

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SCORE: Saturated Consensus Relocalization in Semantic Line Maps

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Abstract— This is the arXiv version of our paper published on IROS 2025. We present SCORE, a visual relocalization system that achieves unprecedented map compactness by adopting semantically labeled 3D line maps. SCORE requires only 0.01%–0.1% of the storage needed by structure-based or learning-based baselines, while maintaining practical accuracy and comparable runtime. The key innovation is a novel robust estimation mechanism, *Saturated Consensus Maximization* (Sat-CM), which generalizes classical *Consensus Maximization* (CM) by assigning diminishing weights to inlier associations according to maximum likelihood with probabilistic justification. Under extreme outlier ratios (up to 99.5%) arising from one-to-many ambiguity in semantic matching, Sat-CM enables accurate estimation when CM fails. To ensure computational efficiency, we propose an accelerating framework for globally solving Sat-CM formulations and specialize it for the Perspective-n-Lines problem at the core of SCORE.

I. INTRODUCTION

Visual relocalization refers to the task of estimating a camera’s pose in a known environment from an input image—a critical capability for mobile robotics, augmented reality, and related applications. This process relies on stored scene representations, which vary across methods: some employ 3D map points with visual descriptors [1]–[5], others leverage deep feature maps from posed reference images [6], [7], or encode scene geometry into weights of neural networks [8], [9]. While these paradigms differ in how they represent and utilize scene information, they face a shared trade-off between representation compactness and estimation accuracy. In this work, we advance towards unprecedented compactness while maintaining practical accuracy with two ingredients as illustrated in Fig. 1: a semantically labeled 3D line map as scene representation, and *Saturated Consensus Maximization* (Sat-CM) as an enabling robust mechanism for resolving one-to-many ambiguous association. We adopt 3D lines to represent scene geometry due to their ubiquity in man-made environments and ability to capture structural elements more compactly than points. And we adopt semantics as the visual descriptor of map lines for three reasons. Firstly, semantics is a naturally compressed descriptor which takes only an integer label to store given the dictionary, consuming far less memory than fine-grained descriptors like SIFT [10]

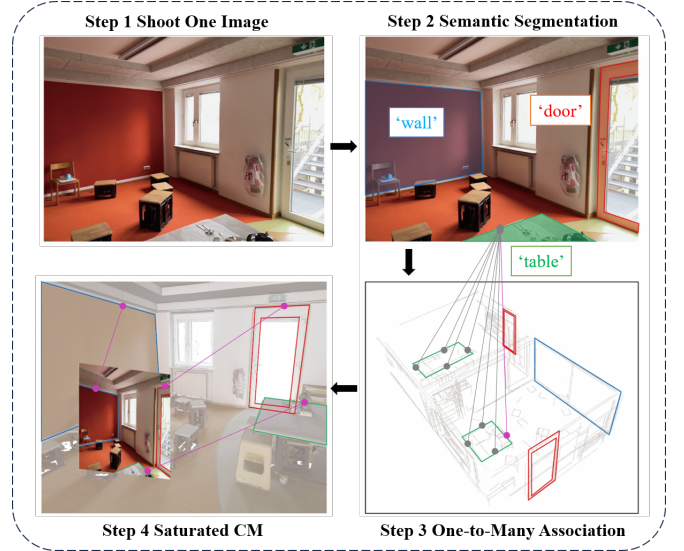


Fig. 1: A kidnapped robot relocalizes itself by associating 2D lines in an image and 3D map lines based on semantic labels, and solving a perspective-n-lines problem. We propose **Saturated Consensus Maximization** to address the extremely high outlier ratio caused by one-to-many associations.

and SuperPoint [11]. Secondly, semantics exhibit strong invariance to partial occlusions and viewpoint changes, which are common challenges to line descriptors [12]. Lastly, The increasing availability of accurate segmentation models (e.g., SAM2 [13]) and vision-language systems (e.g., GPT-4 [14]) makes semantic labeling increasingly practical.

However, this highly compact approach introduces a critical challenge: associating 2D-3D lines via semantic labels causes one-to-many ambiguity and extreme outlier ratios (up to 99.5% in our experiments) for pose estimation. Although prior works have acknowledged similar ambiguity when using simple or quantized visual descriptors [1]–[3], the field still lacks a principled methodology for handling one-to-many association. Current approaches typically avoid rather than address association ambiguity - for instance, the widely-used ratio test [10] explicitly rejects ambiguous matches by requiring a dominant best candidate. In contrast, we embrace ambiguous associations through a principled robust mechanism, Sat-CM, which generalizes the classical *consensus maximization* (CM) method by assigning a diminishing weight for inlier associations according to likelihood. To integrate Sat-CM into our relocalization pipeline, named SCORE, we develop a general accelerated global search

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framework for Sat-CM problems, and devise a specialized solution for the perspective-n-lines (PnL) problem. We summarize our **key contributions** as follows:

- 1) **Ultra-Compact Visual Relocalization:** we push the boundaries of map compactness for visual relocalization by adopting a semantically labeled 3D line map.
- 2) **Novel Robust Mechanism:** to address one-to-many ambiguity inherent in semantic association, we propose the Sat-CM method which generalizes classic CM method and evaluates inlier associations according to the maximum likelihood criteria.
- 3) **Accelerated Global Solver:** we develop an accelerated global search framework for general Sat-CM problems, and apply it to solve the PnL problem central to our pipeline based on rigorous interval analysis.
- 4) **Comprehensive Evaluation:** we demonstrate superiority of Sat-CM over CM, and evaluate practicality of SCORE with extensive experiments on the ScanNet++ [15] dataset. Enabled with the power of Sat-CM and an accelerated global search algorithm, SCORE achieves practical accuracy within comparable runtime, while consuming only 0.01% to 0.1% storage of representative baselines.

II. RELATED WORK

A. Visual Relocalization and Storage Burden

Structure-based visual relocalization methods establish explicit 2D-3D associations through detected local point features [10], [11] and their matches [16], often aided by image retrieval (IR) techniques [17]. Recent advances [18]–[20] incorporate line segments to exploit structural regularities in man-made environments. While accurate, these methods maintain memory-intensive 3D maps with heavy visual descriptors (128 bytes per SIFT or 1 KB per Super-Point descriptor vs. 12 bytes per 3D coordinate). **End-to-end learning methods** bypass explicit association by encoding scenes in neural network weights [8], [9]. Though elegant, these approaches are inherently scene-specific, requiring adaptation for new environments. Hybrid solutions [6], [7] mitigate this issue by regressing poses relative to retrieved reference images with pre-established 2D-3D associations. However, they still demand substantial storage for either the reference images or their deep feature maps, which consume hundreds of KB per image even after quantization [7].

B. Compact Representation and One-to-Many Ambiguity

To relieve the burden of storing memory-intensive visual descriptors, previous works propose to keep an informative subset of the 3D quantities [21], quantize the visual descriptors [1], [2], or adopt a hybrid approach [3], [4] to reduce the map size. The one-to-many ambiguity in association arises along with descriptor quantization, while existing solutions are fundamentally limited. Sattler et al. [1] prune non-unique associations using co-visibility graphs, but risk prematurely discarding real associations. Recent RANSAC variants [2], [3] generate hypotheses exclusively from sampling unambiguous correspondences and utilize ambiguous associations

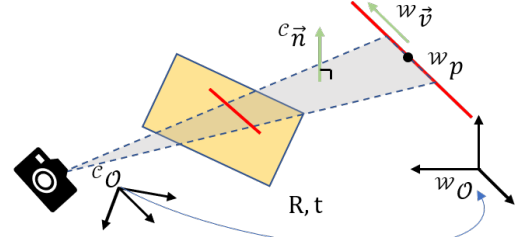


Fig. 2: Projection model underlying the PnL problem.

only for evaluating the pose hypotheses. These variants fail when unique associations are unavailable and remain constrained by a CM formulation, which is not suitable under one-to-many ambiguity.

III. PRELIMINARY

Notations: we use the notation $\mathbf{a} \bullet \mathbf{b} := \mathbf{a}^\top \mathbf{b}$, and use $(\mathbf{a}, \mathbf{b}) := \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix}$ to denote the concatenation of two vectors.

We use the notation ${}^{\mathcal{F}}\mathbf{a}$ to highlight that $\mathbf{a}$ is observed in the reference frame $\{\mathcal{F}\}$. Specifically, we denote the normalized camera frame as $\mathcal{C}$, and the world frame as $\mathcal{W}$. We use $\mathbf{1}\{\cdot\}$ to denote the indicator function, which equals one if the event inside the bracket happens and equals zero otherwise.

Consider a 2D line in the image which writes as follows in the pixel coordinate: $[A \ B \ C] (u, v, 1) = 0$. Given the camera intrinsic matrix $\mathbf{K}$, we can write it in the normalized image coordinate as $[A_c \ B_c \ C_c] (x, y, 1) = 0$, with $[A_c \ B_c \ C_c] = [A \ B \ C] \mathbf{K}$. We use the normalized coefficient vector ${}^{\mathcal{C}}\vec{\mathbf{n}}$ to parameterize a 2D line l in $\{\mathcal{C}\}$:

$${}^{\mathcal{C}}\vec{\mathbf{n}} = \frac{(A_c, B_c, C_c)}{\|(A_c, B_c, C_c)\|}, \quad l := \{\mathbf{p} \in \mathbb{R}^2 | {}^{\mathcal{C}}\vec{\mathbf{n}} \bullet ({}^{\mathcal{C}}\mathbf{p}, 1) = 0\}.$$

We refer to ${}^{\mathcal{C}}\vec{\mathbf{n}}$ as the normal vector since it is perpendicular to the plane passing through the camera origin and l . As for a 3D line L observed in the world coordinate, we parameterize it with a point ${}^{\mathcal{W}}\mathbf{p}_0 \in \mathbb{R}^3$ and a direction vector ${}^{\mathcal{W}}\vec{\mathbf{v}} \in \mathbb{S}^2$:

$$L := \{\mathbf{p} \in \mathbb{R}^3 | ({}^{\mathcal{W}}\mathbf{p} - {}^{\mathcal{W}}\mathbf{p}_0) \times {}^{\mathcal{W}}\vec{\mathbf{v}} = \mathbf{0}\}.$$

Assume the relative transformation from the normalized camera frame to the world coordinate writes as follows

$${}^{\mathcal{W}}\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}.$$

Assume a 2D line l in the image is the projection of a 3D line L in the world map, the following two equations [22] uniquely determine the projection:

$$(\mathbf{R} {}^{\mathcal{C}}\vec{\mathbf{n}}) \bullet {}^{\mathcal{W}}\vec{\mathbf{v}} = 0, \quad (1)$$

$$(\mathbf{R} {}^{\mathcal{C}}\vec{\mathbf{n}}) \bullet ({}^{\mathcal{W}}\mathbf{p}_0 - \mathbf{t}) = 0. \quad (2)$$

IV. SAT-CM EXEMPLIFIED BY THE ROBUST PnL PROBLEM WITH SEMANTIC MATCHING

We address a challenging scenario where a kidnapped robot relocalizes itself with one image and an aggressively compressed scene map, which consists of 3D lines parameterized by their endpoints and semantic labels. The robot

extracts 2D line segments from the image with semantic labels, and associates with map lines based on semantics. Without any other information, the robot associates each 2D line with all map lines of the same label, leading to a significant outlier ratio. While the CM method [23] traditionally underpins robust estimation in robotics, its power is restricted under such one-to-many ambiguity. We therefore propose the Sat-CM method, which introduces a saturation function to handle more than one inlier associations. In this section, we formalize Sat-CM and motivate it with the PnL problem, where the robot pose is estimated using matched 2D image lines and 3D map lines.

A. Saturated Consensus Maximization

Suppose we are given a sample set $\{\mathbf{x}_k\}$, a data set $\{\mathbf{y}_m\}$, and a residual function $f(\Theta|\mathbf{x}, \mathbf{y})$ w.r.t an unknown parameter Θ . Suppose under the ground truth Θ^o , the residual of a real sample-data association satisfies $|f(\Theta^o|\mathbf{x}, \mathbf{y})| \leq \epsilon$, where ϵ is a tolerance term to model against the effect of random noise. For each sample $\mathbf{x}_k$, there exists multiple matched data denoted as $\{\mathbf{y}_{m_k^{(i)}}\}$. The Sat-CM method estimates Θ^o by solving the following problem:

$$\max_{\Theta} \sum_{k=1}^K \sigma_k \left(\sum_i \mathbf{1}\{|f(\Theta|\mathbf{x}_k, \mathbf{y}_{m_k^{(i)}})| \leq \epsilon\} \right), \quad (3)$$

where $\sigma_k(N) = \sum_{n=1}^N w_k(n)$ is referred to as the *saturation function*, and one arrives at the classic CM formulation with $\sigma_k(N) = N$, $w_k(n) = 1 \forall n \geq 1$. For conciseness, we introduce several terminologies. Under a parameter hypothesis Θ , we refer to a sample-data pair $(\mathbf{x}_k, \mathbf{y}_{m_k^{(i)}})$ as an *inlier association* given that $|f(\Theta|\mathbf{x}_k, \mathbf{y}_{m_k^{(i)}})| \leq \epsilon$. We use the terminology *settled* to describe the state of a sample $\mathbf{x}_k$ having at least one inlier association, and use *unsettled* to describe the state of $\mathbf{x}_k$ having no inlier association. As the naming indicates, the saturation function assigns a diminishing weight $w_k(n)$ to each additional inlier of $\mathbf{x}_k$, and the total assigned weight saturates to a finite value.

B. PnL Problem under One-to-many Association

In the PnL problem, we treat 2D lines l_k extracted from the image as samples $\mathbf{x}_k$, and the 3D map lines L_m as data $\mathbf{y}_m$. We denote a 2D line with semantic label s_k , and a 3D line with semantic label s_m respectively as:

$$l_k : (\mathcal{C}\vec{\mathbf{n}}_k, s_k), \quad L_m : (\mathcal{W}\mathbf{p}_m, \mathcal{W}\mathbf{v}_m, s_m),$$

A semantic label s is an integer id corresponding to a word in the dictionary, e.g., (1, 'chair') and (2, 'table'). For simplicity of notation and implementation, we treat a line with multiple labels as separate lines sharing the same geometric position but with different labels. We match each image line l_k with all map lines of the same label, and denote this set of associations as $\{L_{m_k^{(i)}}\}$, with $\#\{L_{m_k^{(i)}}\} := M_k$.

The common practice in the robust PnL problem adopts the CM method and estimates rotation based on constraint (1)

first [24]. Under our setting, this problem writes:

$$\max_{\mathbf{R} \in SO(3)} \sum_{k=1}^K \sum_{i=1}^{M_k} \mathbf{1}\{|(\mathbf{R}^C \vec{\mathbf{n}}_k) \bullet \mathcal{W}\vec{\mathbf{v}}_{m_k^{(i)}}| \leq \epsilon_r\}, \quad (4)$$

where the outer summation is over all 2D lines l_k , and the inner summation is over the associated 3D lines $L_{m_k^{(i)}}$. As illustrated by the following toy example, CM may flavor an unreasonable estimate under one-to-many ambiguity. Suppose the robot captures an image with 10 lines detected, and two rotation hypotheses $\mathbf{R}_1$ and $\mathbf{R}_2$ are evaluated by (4). Under hypothesis $\mathbf{R}_1$, lines l_1 to l_9 each has one inlier while line l_{10} has no inlier, resulting a value of 9. Under hypothesis $\mathbf{R}_2$, lines l_1 to l_9 have no inlier, while line l_{10} has 10 inliers, yielding a value of 10. Although $\mathbf{R}_2$ achieves a higher value in (4), one would intuitively consider $\mathbf{R}_1$ to be the better hypothesis since more 2D lines are settled under $\mathbf{R}_1$, indicating that a more diverse set of constraints—arising from the projections of these lines—are possibly satisfied. In case of $\mathbf{R}_2$, while l_{10} yields 10 inliers, at most one of them is valid, and the inliers from fake associations do not contribute valid constraints on the rotation.

In pursuit of a systematic solution to the above issue, we adopt Sat-CM (3) and formulate the following problem:

$$\max_{\mathbf{R} \in SO(3)} \sum_{k=1}^K \sigma_k \left(\sum_{i=1}^{M_k} \mathbf{1}\{|(\mathbf{R}^C \vec{\mathbf{n}}_k) \bullet \mathcal{W}\vec{\mathbf{v}}_{m_k^{(i)}}| \leq \epsilon_r\} \right). \quad (5)$$

The intuition from the previous toy example—that settling more 2D lines is better—may lead one to adopt the following ‘truncated’ saturation function, which essentially counts the number of settled 2D lines: let $\sigma_k(N) = \sum_{n=1}^N w_k(n)$,

$$\sigma_k(N) = \mathbf{1}\{N \geq 1\}, \quad w_k(n) = \mathbf{1}\{n = 1\}. \quad (6)$$

However, it performs poorly in highly ambiguous scenarios where bad hypotheses may tie (or even beat) the good ones in the number of settled 2D lines. This occurs because the intuition underlying (6) fails to account for other important factors such as the number of associations M_k for each 2D line l_k . For example, settling 2 lines each with 1 inlier out of 1000 putative associations is less likely to contribute to an accurate orientation than settling a single line with 1 inlier out of 2 putative associations. This underscores the necessity of principled saturation function design methodology.

C. Likelihood-Based Saturation Function Design

From a maximum likelihood standpoint, we propose a saturation function as follows: let $\sigma_k(N) = \sum_{n=1}^N w_k(n)$:

$$\sigma_k(N) = \log(1 + C \frac{N}{M_k}), \quad w_k(n) = \log(\frac{M_k + nC}{M_k + (n-1)C}), \quad (7)$$

where C is a scaling constant to be derived in likelihood later. For each 2D line l_k , we introduce a hidden variable ι_k to indicate index of the real association: $\iota_k = i$ for association $(l_k, L_{m_k^{(i)}})$ to be real, while $\iota_k = 0$ for all associations to be fake. Denote residual resulting from a hypothesis $\mathbf{R}$ and association $(l_k, L_{m_k^{(i)}})$ as $r_{k,i}(\mathbf{R}) := |(\mathbf{R}^C \vec{\mathbf{n}}_k) \bullet \mathcal{W}\vec{\mathbf{v}}_{m_k^{(i)}}|$. Our derivations are based on two assumptions:

Assumption 1: For each 2D line l_k , the association set $\{(l_k, L_{m_k^{(i)}})\}$ contains the real association with probability $q > 0$, and each association is equally likely to be real, i.e.,

$$\begin{cases} Pr(\iota_k = 0) = 1 - q, \\ Pr(\iota_k = i) = q/M_k, \quad 1 \leq i \leq M_k. \end{cases}$$

Assumption 2: With the ground truth rotation $\mathbf{R}^o$,

$$\begin{cases} r_{k,i}(\mathbf{R}^o) \sim U(0, \epsilon_r), & \text{conditioned on } \iota_k = i \\ r_{k,i}(\mathbf{R}^o) \sim U(0, u_r), & \text{conditioned on } \iota_k \neq i. \end{cases}$$

where ϵ_r is the same tolerance used in (5), u_r is the residual upper bound, and $u_r = 1$ in the PnL rotation problem.

Note that Assumption 1 applies when the associations have equal confidence level, and Assumption 2 extends the probabilistic justification of CM [25]. We calculate the likelihood for line l_k under a hypothesis $\mathbf{R}$ by marginalizing the hidden variable ι_k based on the above two assumptions. Denote the index set of the N_k inliers under $\mathbf{R}$ as $\mathcal{N}_k$, with $N_k = \#\mathcal{N}_k$, and the vector stacking all residuals $r_{k,i}(\mathbf{R})$'s as $\mathbf{r}_k$, where $i = 1, \dots, M_k$. Then, the likelihood for line l_k writes

$$\begin{aligned} l_{\mathbf{R}}(\mathbf{r}_k) &= p(\mathbf{r}_k, \iota_k = 0) + \sum_{\iota_k \in \mathcal{N}_k} p(\mathbf{r}_k, \iota_k) + \sum_{\iota_k \notin \mathcal{N}_k} p(\mathbf{r}_k, \iota_k), \\ &= (1 - q)u_r^{-M_k} + N_k q / M_k u_r^{-(M_k - 1)} \epsilon_r^{-1} + 0 \\ &= u_r^{-M_k} (1 - q) (1 + C N_k / M_k), \end{aligned}$$

where $C := u_r \epsilon_r^{-1} q (1 - q)^{-1}$. We arrive at (7) by taking logarithm and subtracting a common constant (given $u_r = 1$) in the likelihood of different l_k : $\log(1 - q) - M_k \log(u_r)$.

We highlight two key observations on (7). **First**, the weight $w_k(n)$ assigned to the n th inlier association decreases w.r.t the total association number M_k , essentially penalizing the level of ambiguity. **Second**, (7) assigns a diminishing weight $w_k(n)$ to inliers, and the total assignment saturates to $\log(1 + C)$ despite of M_k . This mechanism ensures that additional inliers contribute less to the overall consensus score, and prioritizes settling more lines over inflating the number of insignificant inliers. Although the diminishing rate of $w_k(n)$ is controlled by a hyperparameter q , estimators based on (7) are not sensitive to the choice of q in an appropriate range, as supported by our relocalization experiments. To enclose our discussion on function design, we present the global landscapes of the CM rotation problem (4) and the Sat-CM one (5) in Fig. 3, where the 2D/3D associations come from one of the query image in our experiments. As indicated by Fig. 3, adopting Sat-CM with a likelihood-justified function design structurally reinforce the true value as the global optimum while suppressing ambiguous solutions.

Finally, given the global optimum $\hat{\mathbf{R}}_\sigma^*$ of (5) and the corresponding inliers, we further formulate the translation estimation problem using Sat-CM:

$$\max_{\mathbf{t} \in \mathbb{R}^3} \sum_{k=1}^K \sigma_k \left(\sum_{i=1}^{M_k} \mathbf{1} \left\{ |\mathcal{W} \hat{\mathbf{n}}_k^* \bullet (\mathcal{W} \mathbf{p}_{m_k^{(i)}} - \mathbf{t})| \leq \epsilon_t \right\} \right), \quad (8)$$

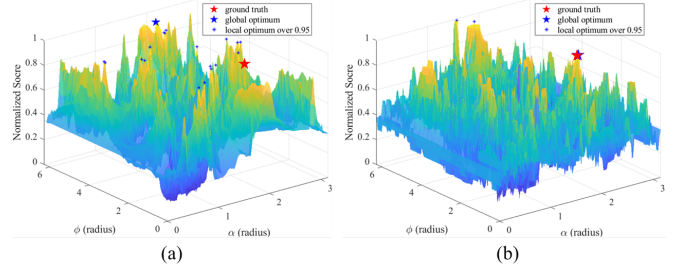


Fig. 3: The normalized global landscapes under (a) the CM problem (4) and (b) the Sat-CM problem (5). Each point is evaluated with a fixed rotation axis under polar coordinates (ϕ, α) and the ad-hoc optimal rotation amplitude. We choose (7) as the saturation function.

where $\mathcal{W} \hat{\mathbf{n}}_k^* := \hat{\mathbf{R}}_\sigma^* \mathbf{C} \hat{\mathbf{n}}_k$ is the rotated normal vector¹.

V. SOLVING SAT-CM EFFICIENTLY WITH DIMENSION-REDUCED BRANCH-AND-BOUND

As an extension of the NP-hard CM problem [26], Sat-CM is also computationally challenging to solve. In this section, we propose an accelerated branch-and-bound (BnB) algorithm for Sat-CM problems, which reduces the branching dimensions by one through novel bounding techniques as inspired by a recent accelerating framework [27]. Next, we deploy this general algorithm to solver the PnL problems (5) and (8). While a recent work [28] proposes a similar global rotation solver for PnL, our FGO-PnL (Fast and Globally Optimal) solver provides three key advances. **Firstly**, FGO-PnL accommodates the more general Sat-CM formulation. **Secondly**, rigorous interval analysis stands behind FGO-PnL to guarantee bound validity, while [28] uses heuristic approximation in their implementation. **Lastly**, FGO-PnL also includes an accelerated global translation solver. We release in our Git-Hub repository² Matlab and parallelized C++ implementations for FGO-PnL, which provide modular saturation function interfaces for easy customization and backward compatibility with CM via choosing $\sigma(N) = N$.

A. Accelerated Branch-and-Bound for Sat-CM

For clarity, we use $\mathbf{a}^{(k, m_k^{(i)})} := (\mathbf{x}_k, \mathbf{y}_{m_k^{(i)}})$ to denote an association. We start with a 1D Sat-CM problem:

$$\max_{\theta} \sum_{k=1}^K \sigma_k \left(\sum_{i=1}^{M_k} \mathbf{1} \{ |f(\theta | \mathbf{a}^{(k, m_k^{(i)})})| \leq \epsilon \} \right). \quad (9)$$

Assume the residual function is continuous in the parameter θ , we obtain by interval analysis:

$$|f(\theta | \mathbf{a}^{(k, m_k^{(i)})})| \leq \epsilon \Leftrightarrow \theta \in [\underline{\theta}(k, m_k^{(i)}), \bar{\theta}(k, m_k^{(i)})],$$

in which we assume a single interval is obtained for notational clarity. In practice, an association $\mathbf{a}^{(k, m_k^{(i)})}$ may

¹Note that since $\mathcal{W} \hat{\mathbf{n}}_k$ is not perfectly orthogonal to the direction vector $\mathcal{W} \hat{\mathbf{p}}_{k_j}$, choosing different point $\mathcal{W} \hat{\mathbf{p}}_{k_j}$ on a map line influences the residual. In order to make problem (8) invariant with the choice of parameterizing point, we project $\mathcal{W} \hat{\mathbf{n}}_k^*$ onto the null space of $\mathcal{W} \hat{\mathbf{p}}_{k_j}$ in implementation.

²<https://github.com/LIAS-CUHKSZ/SCORE>

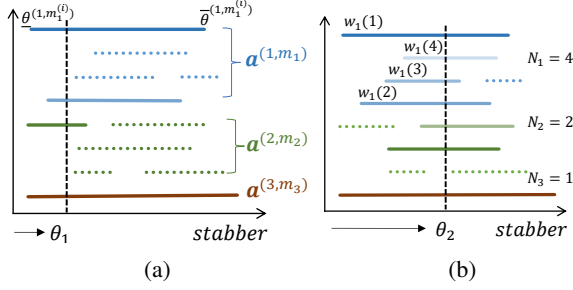


Fig. 4: Saturated interval stabbing. The increased transparency of intervals represents adaptively decreased weights for each additional inlier. Algorithm 1 sweeps the range of stabber θ from left to right and record the highest value achieved along the way. (a): $v(\theta_1) = \sigma_1(2) + \sigma_2(1) + \sigma_3(1)$, (b): $v(\theta_2) = \sigma_1(4) + \sigma_2(2) + \sigma_3(1)$.

induce multiple disjoint intervals, while it does not affect the proposed algorithm. Based on these intervals, we arrive at an equivalent problem for (9), which we refer to as saturated interval stabbing (Sat-IS) and illustrate in Fig. 4:

$$\max_{\theta} \sum_{k=1}^K \sigma_k \left(\sum_{i=1}^{M_k} \mathbf{1}\{\theta \in [\underline{\theta}^{(k,m_k^{(i)})}, \bar{\theta}^{(k,m_k^{(i)})}]\} \right). \quad (10)$$

Denote $M := \sum M_k$, we can solve (10) with $O(M)$ space and $O(M \log M)$ time using Algorithm 1.

Algorithm 1 Saturated Interval Stabbing

Inputs: Intervals $\mathcal{I}_k = \{[\underline{\theta}^{(k,m_k^{(i)})}, \bar{\theta}^{(k,m_k^{(i)})}]\}_{m_k^{(i)}=1}^{M_k}$ for $k = 1, \dots, K$ and functions $\sigma_k(N) = \sum_{n=1}^N w_k(n)$.

Outputs: optimal stabbers $\theta_{\{ \} }^*$, optimal value v^* .

- 1: $I = \text{sort endpoints in } \{\mathcal{I}_k\}_{k=1}^K$.
 - 2: $v^* = 0, v = 0, N_k = 0$ ($k = 1, \dots, K$).
 - 3: **for** $i = 1$ to $\text{len}(I)$ **do**
 - 4: **if** $I(i)$ is a left endpoint from $\mathcal{I}_k$ **then**
 - 5: $N_k = N_k + 1, v = v + w_k(N_k)$.
 - 6: **if** $c > v^*$ **then**
 - 7: $v^* = v, \theta_{\{ \} }^* = [I(i), I(i+1)]$.
 - 8: **end if**
 - 9: **else**
 - 10: $v = v - w_k(N_k), N_k = N_k - 1$.
 - 11: **end if**
 - 12: **end for**
 - 13: **return** $\theta_{\{ \} }^*, v^*$
-

As for a high-dimensional problem (3), we distinguish one parameter θ with others: $\Theta = (\Theta_{:-1}, \theta)$, and branches only over the space of $\Theta_{:-1}$, as motivated by [27]. Denote the optimal value of (3) for $\Theta_{:-1}$ constricted in a sub-cube $\mathcal{C}$ and θ free as $r^*(\mathcal{C})$. A **lower bound** for $r^*(\mathcal{C})$ writes:

$$\max_{\theta} \sum_{k=1}^K \sigma_k \left(\sum_{i=1}^{M_k} \mathbf{1}\{|f(\theta|\Theta_{:-1}^{(c)}, \mathbf{a}^{(k,m_k^{(i)})})| \leq \epsilon\} \right), \quad (11)$$

where $\Theta_{:-1}^{(c)}$ is the center point of sub-cube $\mathcal{C}$. Notice that the lower bound (11) corresponds to a 1-D Sat-CM problem, and it can be efficiently solved by Algorithm 1. As for the upper bound, we manage to find two bounding functions for each residual term:

$$f_L^{(k,i)}(\theta) \leq f(\theta|\Theta_{:-1}, \mathbf{a}^{(k,m_k^{(i)})}) \leq f_U^{(k,i)}(\theta),$$

where the inequality holds for any $\Theta_{:-1} \in \mathcal{C}$. Given f_L and f_U , we find an **upper bound** for $r^*(\mathcal{C})$ as follows:

$$\max_{\theta} \sum_{k=1}^K \sigma_k \left(\sum_{i=1}^{M_k} \mathbf{1}\{f_L^{(k,i)}(\theta) \leq \epsilon \text{ and } f_U^{(k,i)}(\theta) \geq -\epsilon\} \right). \quad (12)$$

Similar to the lower bound, we obtain the upper bound (12) by solving a Sat-IS problem with Algorithm 1. Equipped with the above lower and upper bounds, we search the global optimum of a N-D Sat-CM problem by branching and bounding only a subspace of $N - 1$ dimensions. For details in the BnB procedure, one can refer to our implementation and [27]. Next, we focus on finding the upper bounding functions f_L and f_U for the PnL problem. We put analysis corresponding to the translation part in Appendix IV, and focus on the challenging rotation problem. For conciseness, in the rest of this section, we refer to a general association as $\mathbf{a} := (l, L)$, and denote the corresponding normal, direction vector and a 3D point on L respectively as $\bar{\mathbf{n}}_a, \bar{\mathbf{v}}_a$ and $\mathbf{p}_a$.

B. Find Bounding Functions for Rotation Problem in PnL

We parameterize rotation with a rotation axis $\bar{\mathbf{u}} \in \mathbb{S}^2$ and an amplitude $\theta \in [0, \pi]$. We choose θ as the distinguished parameter, and further parameterize $\bar{\mathbf{u}}$ by polar coordinates:

$$\bar{\mathbf{u}} = (\sin \alpha \cos \phi, \sin \alpha \sin \phi, \cos \alpha) \quad \alpha \in [0, \pi] \quad \phi \in [0, 2\pi].$$

We denote a sub-cube for axis $\bar{\mathbf{u}}$ in the polar coordinate as

$$\mathcal{C}_{\bar{\mathbf{u}}} := \{(\alpha, \phi) | \alpha \in [\underline{\alpha}, \bar{\alpha}], \phi \in [\underline{\phi}, \bar{\phi}]\},$$

and denote its boundary as $\partial \mathcal{C}_{\bar{\mathbf{u}}}$.

Rewrite the residual function $f(\theta, \bar{\mathbf{u}}|\mathbf{a})$ for (5) as

$$\bar{\mathbf{n}}_a^\top \bar{\mathbf{v}}_a + \sin \theta \bar{\mathbf{n}}_a^\top (\bar{\mathbf{u}} \times \bar{\mathbf{v}}_a) + (1 - \cos \theta) \bar{\mathbf{n}}_a^\top [\bar{\mathbf{u}}]_{\times}^2 \bar{\mathbf{v}}_a, \quad (13)$$

where $[\bar{\mathbf{u}}]_{\times}$ denotes the skew matrix of $\bar{\mathbf{u}}$. Notice that as a function of $\theta \in [0, \pi]$, the residual (13) is monotone w.r.t:

$$h_1(\bar{\mathbf{u}}|\mathbf{a}) := \bar{\mathbf{u}}^\top (\bar{\mathbf{v}}_a \times \bar{\mathbf{n}}_a), \quad h_2(\bar{\mathbf{u}}|\mathbf{a}) := \bar{\mathbf{n}}_a^\top [\bar{\mathbf{u}}]_{\times}^2 \bar{\mathbf{v}}_a.$$

Based on monotonicity, we obtain the bounding functions for $f(\theta, \bar{\mathbf{u}}|\mathbf{a})$, $\bar{\mathbf{u}} \in \mathcal{C}$ by finding the lower and upper values of spherical functions h_1 and h_2 for $\bar{\mathbf{u}} \in \mathcal{C}_{\bar{\mathbf{u}}}$, denoted as h_1^L, h_1^U, h_2^L and h_2^U . The bounding functions write:

$$f_L(\theta) = \bar{\mathbf{n}}_a^\top \bar{\mathbf{v}}_a + h_1^L \sin \theta + h_2^L (1 - \cos \theta), \quad (14a)$$

$$f_U(\theta) = \bar{\mathbf{n}}_a^\top \bar{\mathbf{v}}_a + h_1^U \sin \theta + h_2^U (1 - \cos \theta). \quad (14b)$$

We summarize our results in the following two theorems. For conciseness, we first introduce several notations:

$$\bar{\mathbf{m}}_a = \frac{\bar{\mathbf{v}}_a + \bar{\mathbf{n}}_a}{\|\bar{\mathbf{v}}_a + \bar{\mathbf{n}}_a\|}, \quad \bar{\mathbf{m}}_a^\perp = \frac{\bar{\mathbf{v}}_a - \bar{\mathbf{n}}_a}{\|\bar{\mathbf{v}}_a - \bar{\mathbf{n}}_a\|}, \quad \bar{\mathbf{c}}_a := \frac{\bar{\mathbf{v}}_a \times \bar{\mathbf{n}}_a}{\|\bar{\mathbf{v}}_a \times \bar{\mathbf{n}}_a\|}.$$

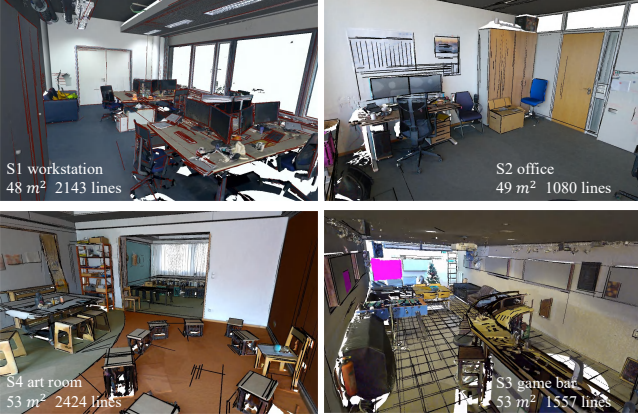


Fig. 5: Meshes of the selected four scenes from ScanNet++ and overlapped with the constructed line map.

Theorem 1 (Extreme Points for $h_2(\vec{u}|\mathbf{a})$):

- 1) If $\pm \vec{m}_a \in \mathcal{C}_{\vec{u}}$, $\arg \max_{\vec{u} \in \mathcal{C}_{\vec{u}}} h_2(\vec{u}|\mathbf{a}) = \pm \vec{m}_a$.
- 2) If $\pm \vec{m}_a^\perp \in \mathcal{C}_{\vec{u}}$, $\arg \min_{\vec{u} \in \mathcal{C}_{\vec{u}}} h_2(\vec{u}|\mathbf{a}) = \pm \vec{m}_a^\perp$.
- 3) Otherwise, the extreme points fall on $\partial \mathcal{C}_{\vec{u}}$.

Theorem 2 (Extreme Points for $h_1(\vec{u}|\mathbf{a})$):

- 1) If $\vec{c}_a \in \mathcal{C}_{\vec{u}}$, $\arg \max_{\vec{u} \in \mathcal{C}_{\vec{u}}} h_1(\vec{u}|\mathbf{a}) = \vec{c}_a$.
- 2) If $-\vec{c}_a \in \mathcal{C}_{\vec{u}}$, $\arg \min_{\vec{u} \in \mathcal{C}_{\vec{u}}} h_1(\vec{u}|\mathbf{a}) = -\vec{c}_a$.
- 3) Otherwise, the extreme points fall on $\partial \mathcal{C}_{\vec{u}}$.

We present the proof for Theorem 1 in Appendix I and omit the proof for Theorem 2 which is similar. For case 3) in Theorem 2, we actually can explicitly write out the extreme points without the need to transverse $\partial \mathcal{C}_{\vec{u}}$, and more explanation is provided in Appendix II.

VI. EXPERIMENTS ON SCANNET++ DATASET

We implement relocalization experiments on the ScanNet++ dataset [15], mainly for the convenience to construct a semantic line map. ScanNet++ provides for each scene an image sequence captured by an iPhone 13 Pro with default camera setting, and the user can render depth and semantic mask for each image from a human-annotated mesh. For each scene, we split the sequence of images into a reference set and a query set at a 7:1 ratio. We propose a pipeline to construct semantic line maps based on the reference images, and relocalize the query images in the constructed maps. Due to page limit, we present relevant content in Appendix III. As shown in Fig. 5, the quality of the constructed line maps is compromised when compared to the ones constructed by mature point-based pipelines like COLMAP [29], while, on the other hand, it is an ideal testbed for effectiveness of the Sat-CM mechanism and robustness of SCORE.

A. Settings for Relocalization Experiments

We test SCORE with both ground-truth and predicted semantic labels for the query images. Our customized segmentation pipeline integrates RAM++ [30] for semantic tagging and Grounded-SAM [31] for object detection: RAM++ tags images within the line map’s dictionary, and Grounded-SAM segments objects based on the recognized words. And we use

the released weights without fine-tuning for both models. In the line map, non-stationary object labels are excluded from the dictionary, such as *bottle* and *jacket*. While including these labels might improve relocalization accuracy, their transient nature in real-world environments renders such gains unrealistic. Additionally, we merge semantically ambiguous labels such as *shelf* and *bookshelf* to increase segmentation accuracy. These design choices underscore the critical role of dictionary engineering—a topic meriting deeper investigation in balancing accuracy with practical applicability.

We evaluate our work from two perspectives. From the methodology perspective, we evaluate the proposed Sat-CM mechanism with different saturation functions and compare with the classic CM method. From the practicability perspective, we compare SCORE with baselines in terms of runtime, storage, and accuracy. We respectively choose hloc [5] and PixLoc [6] as representatives for the structure-based methods relying on high-dimensional visual descriptors, and the learning-based method based on deep image features. For fair comparison, all methods use the same splitting of reference and query images and the same image retrieval (IR) results from NetVLAD [17], relocalizing query images within the sub-map observed by 12 retrieved images. We use consistent error thresholds for the FGO-PnL solver across settings: $\epsilon_r = 0.015$ and $\epsilon_t = 0.03$. We run all experiments on a PC equipped with an AMD Core 7950x CPU@4.5GHz, RAM@64GB and a GeForce RTX 5070Ti graphic card.

B. Sat-CM v.s. CM

We present here results of rotation estimation, which encounters the most severe ambiguity as the first step of a cascaded estimation procedure. In FGO-PnL, we utilize the IR results to effectively prune the space of rotation axis $\vec{u} \in \mathbb{S}^2$ according to that of the first retrieved image. Specifically, we divide $\mathbb{S}^2$ into cubes with equal side length and restrict our search within the cube where the retrieved rotation axis resides in. The side length is chosen according to IR accuracy, and we present the results with side length π (binary division) and $\pi/2$ (octal division), corresponding to a vague and moderate belief in the retrieval accuracy respectively. We use ‘PR’ and ‘GT’ in the legends to distinguish using predicted or ground truth semantic labels for the query images. As for the saturation functions, we use SCM0 in the legends for the naive truncated function (6), and SCM1 for the likelihood-based function (7). We choose $q = 0.9$ for SCM1-GT and $q = 0.5$ for SCM1-PR. We record the maximal error when FGO-PnL returns multiple global optimum, while in the complete pipeline, subsequent translation estimation helps select from the rotation candidates. For comparison in translation estimation and sensitivity evaluation in parameter q , we refer the readers to Appendix V.

As evidenced by Table II, SCM1 demonstrates consistent superiority across scenes and settings, and the performance lead is particularly pronounced under harsh conditions involving large search spaces and predicted semantic labels. While SCM0, which uses the naive truncation function (6), yields lower accuracy than CM in most configurations,

	S1	S2	S3	S4
# map lines	2143	1080	1557	2424
# map points	164860	108726	155673	185040
wrong label ratio	17.2%	19.6%	19.4%	20.5%
missed label ratio	27.0%	30.4%	30.7%	30.0%

TABLE I: Scene detailed information.

primarily due to susceptibility to multiple global optima. These results substantiate that Sat-CM, when incorporating a justified saturation function, enables accurate estimation under one-to-many ambiguity when CM fails. Scene S3 (game bar) exhibits the poorest results, which we attribute to compromised quality of the line map due to a cluttered layout, less accurate image pose and depth. We further observe that a more confined search space of rotation axis with side length $\pi/2$ enhances accuracy universally, since ambiguous candidates are pruned ahead of estimation. At last, we highlight the estimation difficulty when using predicted semantic labels, which can not be fully conveyed by the extremely high outlier ratio (consistently over 99%). As presented in Table I, beyond erroneous predictions, missed detections also degrade performance by reducing usable 2D lines.

C. SCORE v.s. Classic Relocalization Pipelines

We report the memory consumption for storing map geometry (3D points or lines), cameras (reference image poses and 2D-3D associations) and features (visual descriptors or deep feature maps). We also report the median runtime which consists of feature extraction, feature association and estimation. For hloc [5], we adopt SuperPoint [11] for keypoint extraction and description (stored in float 16), and use Lightglue [16] for keypoint association. For PixLoc [6], we store the reference images and extract feature maps of the 12 retrieved images online. For SCORE, we adopt the truncated saturation function instead of the likelihood-based function in translation estimation. This is because we further utilize two physical constraints to prune the inliers obtained from solving (8): the truly-associated 3D line resides in front of the camera and its projection intersects with the image. We choose the truncated saturation function in order to keep more translation candidates before pruning, and indeed observe a more robust performance (presented in Appendix V). As the above methods all rely on IR, we include the accuracy achieved by NetVLAD [17] for reference.

As presented in Table III, SCORE achieves practical relocalization accuracy using the true semantic labels, with median errors below 8 cm and 1.5° , though performance degrades when using predicted labels, particularly in translation. While PixLoc and hloc achieve finer-grained accuracy due to their dense point-based representations (Table I), SCORE provides radical memory efficiency (0.01%–0.1% storage overhead) while maintaining comparable estimation runtime thanks to an accelerated global solver and a parallelized C++ implementation. Compared with IR results achieved by NetVLAD, SCORE consistently improve over

the median rotation error even when using predicted labels³. Given true semantic labels, SCORE achieves better translation accuracy than IR except in S4 which has the most complete reference image database. To sum up, SCORE proves ultra-compact (kB-scale) semantic line maps can sustain viable relocalization, while with semantic segmentation remaining the accuracy and runtime bottleneck.

VII. CONCLUSION AND FUTURE WORKS

We push the boundary of map compactness in visual relocalization through two key innovations: semantically labeled 3D line maps as an ultra-efficient scene representation, and Saturated Consensus Maximization as a foundational solution for one-to-many ambiguous associations. Experimental validation on ScanNet++ confirms Sat-CM’s robustness and demonstrates our pipeline’s practical viability despite extreme map compression. To transition toward real-world deployment, we identify three critical research directions. Firstly, develop fast and approximate solvers for Sat-CM to further reduce runtime. Secondly, explore low-dimensional descriptors combining semantic-level efficiency with improved accuracy and repeatability. Thirdly, incorporate both point and line features to offer a more resilient and fine-grained solution.

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³Notice that the median rotation error reported in Table III is slightly better than those in Table II, since we use translation estimation to assist selection among multiple rotation candidates.

TABLE II: Compare CM with Sat-CM in rotation estimation.

Method	S1(workstation)	S2(office) 25%/50%/75% error quantiles (°)	S3(game bar) error quantiles (°)	S4(art room)	S1	S2	S3	S4	S1	S2	S3	S4
					Recall at 5° (%)				Median Outlier Ratio (%)			
CM ^π -GT	1.3/3.0/146.9	0.7/1.2/8.3	1.2/12.0/161.5	0.8/1.3/111.0	54	75	45	65				
SCM0 ^π -GT	1.6/3.5/179.3	1.6/2.7/179.7	2.5/8.0/179.9	2.2/4.9/179.7	56	58	41	51	95.7	94.7	94.9	95.7
SCM1 ^π -GT	0.9/1.4/3.1	0.6/0.9/1.9	0.8/1.7/4.4	0.6/0.9/1.7	81	90	76	91				
CM ^{π/2} -GT	1.1/2.4/39.8	0.7/1.1/2.6	1.1/2.8/33.0	0.7/1.1/2.7	63	85	53	79				
SCM0 ^{π/2} -GT	1.6/2.9/5.4	1.5/2.2/3.3	2.2/4.7/63.8	1.9/3.2/7.1	72	78	51	68	95.7	94.7	94.9	95.7
SCM1 ^{π/2} -GT	0.8/1.3/2.5	0.6/0.9/1.5	0.8/1.6/3.7	0.6/0.9/1.5	90	93	80	95				
CM ^π -PR	2.4/92.5/168.3	1.0/3.9/177.0	7.9/138.8/179.1	1.1/22.5/174.3	31	52	24	45				
SCM0 ^π -PR	4.6/171.5/179.8	3.0/178.7/179.9	4.2/179.5/180.0	6.6/179.4/180.0	27	33	25	22	99.4	99.5	99.3	99.4
SCM1 ^π -PR	1.6/2.6/90.5	0.7/1.6/10.8	1.1/3.7/149.7	0.8/1.8/72.0	64	73	56	69				
CM ^{π/2} -PR	2.0/28.9/53.7	0.9/2.0/18.2	2.6/26.7/61.2	1.0/2.8/41.2	39	68	37	55				
SCM0 ^{π/2} -PR	2.8/8.7/89.2	2.2/3.6/17.1	2.9/80.3/103.0	3.0/13.2/102.7	40	61	32	36	99.4	99.5	99.3	99.4
SCM1 ^{π/2} -PR	1.4/2.3/5.3	0.7/1.4/2.8	1.0/2.4/4.6	0.7/1.3/3.6	74	82	76	80				

TABLE III: Baseline comparison in terms of runtime, memory consumption and accuracy.

Method	median runtime extract + match&est	S1	S2	S3	S4	S1	S2	S3	S4
		storage: geometry&cameras+feature				Median Pose Error (cm/°)			
NetVLAD [17]	25+5 ms	0+7 MB	0+5 MB	0+6 MB	0+11 MB	7.8/4.5	11.1/10.0	8.8/6.6	3.4/4.0
hloc [5]	30+260 ms	40+788 MB	38+611 MB	51+809 MB	61+951 MB	0.5/0.2	0.7/0.2	0.8/0.2	0.5/0.1
PixLoc [6]	1910+30 ms	40+132 MB	38+90 MB	51+120 MB	61+192 MB	0.5/0.2	0.6/0.2	0.8/0.2	0.5/0.1
SCORE ^π -PR	1525+245 ms					13.3/2.4	7.7/1.5	100.7/2.5	21.2/1.7
SCORE ^{π/2} -PR	1525+100 ms	158+17 KB	92+8 KB	107+12 KB	206+19 KB	10.6/2.2	6.8/1.2	32.3/2.3	14.4/1.2
SCORE ^π -GT	/+130 ms					5.9/1.4	3.8/0.8	7.1/1.5	5.0/0.9
SCORE ^{π/2} -GT	/+45 ms					5.9/1.3	3.8/0.8	6.9/1.4	4.9/0.8

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APPENDIX I PROOF FOR THEOREM 1

Recall that Theorem 1 characterizes extreme points of the spherical function $h_2(\vec{u}|\mathbf{a})$, with the rotation axis $\vec{u}$ belonging to a sub-cube $\mathcal{C}_{\vec{u}}$ in polar coordinates.

Proof: First, rewrite $h_2(\vec{u}|\mathbf{a})$ as:

$$\begin{aligned} h_2(\vec{u}|\mathbf{a}) &= \vec{n}_a^\top [\vec{u}]_\times \vec{v}_a = \vec{n}_a^\top (\vec{u}\vec{u}^\top - \mathbf{I}_3) \vec{v}_a \\ &= \vec{u}^\top (\vec{n}_a \vec{v}_a^\top) \vec{u} - \vec{n}_a^\top \vec{v}_a \\ &= \vec{u}^\top \left(\frac{\vec{n}_a \vec{v}_a^\top + \vec{v}_a \vec{n}_a^\top}{2} \right) \vec{u} - \vec{n}_a^\top \vec{v}_a. \end{aligned}$$

Next, we derive the critical points of $h_2(\vec{u}|\mathbf{a})$ for $\vec{u} \in \mathbb{S}^2$. Denote $\mathbf{M}_a := \frac{\vec{n}_a \vec{v}_a^\top + \vec{v}_a \vec{n}_a^\top}{2}$, and take derivative of $h_2(\vec{u}|\mathbf{a})$ w.r.t a perturbation $\delta \in \mathbb{R}^3$ on the lie algebra as follows:

$$\begin{aligned} & \frac{\partial h_2(\exp([\delta]_\times) \vec{u}|\mathbf{a})}{\partial \delta} \Big|_{\delta=0} \\ &= \frac{\partial (\exp([\delta]_\times) \vec{u})^\top \mathbf{M}_a \exp([\delta]_\times) \vec{u}}{\partial \delta} \Big|_{\delta=0}, \\ &= \frac{\partial \vec{u}^\top (-[\delta]_\times \mathbf{M}_a + \mathbf{M}_a [\delta]_\times) \vec{u} + o(\delta)}{\partial \delta} \Big|_{\delta=0}, \\ &= 2[\vec{u}]_\times \mathbf{M}_a \vec{u} = 2\vec{u} \times (\mathbf{M}_a \vec{u}), \end{aligned}$$

where $\exp(\cdot)$ is the matrix exponential operation, and we use the equality $[\mathbf{a}]_\times \mathbf{b} = -[\mathbf{b}]_\times \mathbf{a}$ in the last equation. To let the derivative equal zero, one has

$$\vec{u} \parallel \mathbf{M}_a \vec{u} \Leftrightarrow \vec{u} \text{ is an eigenvector of } \mathbf{M}_a.$$

Substitute and verify that $\vec{m}_a$, $\vec{m}_a^\perp$, and $\vec{c}_a$ are three orthogonal eigenvectors of matrix $\mathbf{M}_a$, corresponding to a positive, negative and zero eigenvalue respectively. And $h_2(\vec{u}|\mathbf{a})$ achieves the global maximum at $\pm \vec{m}_a$, the global minimum at $\pm \vec{m}_a^\perp$, and saddle points at $\pm \vec{c}_a$. Given that $h_2(\vec{u}|\mathbf{a})$ with $\vec{u} \in \mathbb{S}^2$ has finite (six) critical points, we conclude that its extreme values for $\vec{u} \in \mathcal{C}_{\vec{u}}$ are achieved either at $\pm \vec{m}_a$, $\pm \vec{m}_a^\perp$, or at $\partial \mathcal{C}_{\vec{u}}$. ■

We visualize $h_2(\vec{u}|\mathbf{a})$ on the sphere in Fig. 6.

APPENDIX II SUPPLEMENTARY RESULTS FOR THEOREM 2

Denote polar coordinates for $\vec{c}_a := \frac{\vec{v}_a \times \vec{n}_a}{\|\vec{v}_a \times \vec{n}_a\|}$ as (α_c, ϕ_c) :

$$h_1(\vec{u}|\mathbf{a}) = \sin \alpha_c \sin \alpha \cos(\phi_c - \phi) + \cos \alpha_c \cos \alpha + \text{const.}$$

The partial derivative of h_1 with respect to α and ϕ writes

$$\frac{\partial h_1}{\partial \alpha} = \sin \alpha_c \cos \alpha \cos(\phi_c - \phi) - \sin \alpha \cos \alpha_c, \quad (15a)$$

$$\frac{\partial h_1}{\partial \phi} = \sin \alpha_c \sin \alpha \sin(\phi_c - \phi). \quad (15b)$$

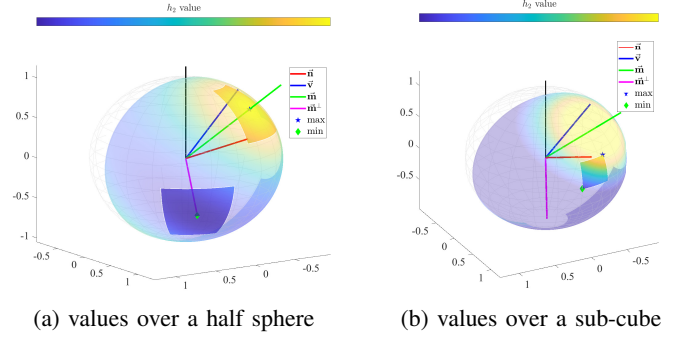


Fig. 6: Landscapes of function $h_2(\vec{u}|\mathbf{a})$.

For conciseness, we focus on the case where both the sub-cube $\mathcal{C}_{\vec{u}}$ and $\vec{c}_a$ are in the east-hemisphere, i.e., $\phi \in [0, \pi]$ and $\phi_c \in [0, \pi]$. The following arguments can be easily adapted to the other cases. Denote polar coordinates of $\vec{u}$ which minimizes $h_1(\vec{u}|\mathbf{a})$ as $\alpha_{\min}$ and $\phi_{\min}$, and of $\vec{u}$ which maximizes $h_1(\vec{u}|\mathbf{a})$ as $\alpha_{\max}$ and $\phi_{\max}$. We further denote

$$\alpha_{\text{near}} := \arg \min_{\alpha \in [\alpha_l, \alpha_r]} |\alpha - \alpha_c|, \quad \phi_{\text{near}} := \arg \min_{\phi \in [\phi_l, \phi_r]} |\phi - \phi_c|.$$

$$\alpha_{\text{far}} := \arg \max_{\alpha \in [\alpha_l, \alpha_r]} |\alpha - \alpha_c|, \quad \phi_{\text{far}} := \arg \max_{\phi \in [\phi_l, \phi_r]} |\phi - \phi_c|.$$

We first give a lemma for $\phi_{\max}$ and $\phi_{\min}$:

Lemma 1: If $(\alpha, \phi) \in \partial \mathcal{C}_{\vec{u}}$ is an extreme point for $h_1(\vec{u}|\mathbf{a})$ on the boundaries of cube, one must have $\phi_{\max} = \phi_{\text{near}}$ and $\phi_{\min} = \phi_{\text{far}}$.

Proof: As the ϕ dimension disappears for $\alpha = 0$ or π , we can focus on $\alpha \in (0, \pi)$. The result naturally arises from two observations on the partial derivative (15b). Firstly, for a fixed $\alpha \in (0, \pi)$, we have $\frac{\partial h_1}{\partial \phi} > 0$ if $\phi < \phi_c$, and $\frac{\partial h_1}{\partial \phi} < 0$ if $\phi > \phi_c$. Secondly, the partial derivative takes the same absolute value for ϕ_1 and ϕ_2 equally distant from ϕ_c . ■ After fixing the value of ϕ according to Lemma 1, we can focus on α by studying (15a).

Lemma 2: Given a fixed value of ϕ and for $\alpha_c \neq \pi/2$, the partial derivative (15a) has a unique zero point $\alpha^* \in [0, \pi]$, and α^* is a **global maximizer** of $h_1(\vec{u}|\mathbf{a})$ if $|\phi_c - \phi| < \pi/2$, and a **global minimizer** of $h_1(\vec{u}|\mathbf{a})$ if $|\phi_c - \phi| > \pi/2$.

Proof: Note that (15a) can be organized into a form of $A \sin(\alpha + \beta)$, with β a fixed angle. For $\alpha_c \neq \pi/2$, we have $\beta \neq 0$ or π , and thus zero point α^* is unique within $[0, \pi]$.

Given $\alpha_c \neq \pi/2$, $\alpha = \pi/2$ is not a zero point of (15a), and thus we can safely rewrite it as:

$$\frac{\partial h_1}{\partial \alpha} = \cos \alpha_c \cos \alpha (\tan \alpha_c \cos(\phi_c - \phi) - \tan \alpha). \quad (16)$$

Based on (16), we list in Table IV four cases for the zero point α^* which completes the proof. ■

Notice that we omit special cases where $\alpha_c = \pi/2$ or $|\phi - \phi_c| = \pi/2$ in Lemma 2 for conciseness, while interested readers can refer to our implementation for details. Based on Lemmas 1 and 2, we propose an efficient procedure to find extreme points of $h_1(\vec{u}|\mathbf{a})$ on $\partial \mathcal{C}_{\vec{u}}$ as follows. We introduce

TABLE IV: Four different cases of α_c and $\Delta\phi$

α_c	$ \phi_c - \phi $	α^*	$\cos \alpha_c \cos \alpha^*$	Extrema
$< \pi/2$	$< \pi/2$	$\in (0, \alpha_c)$	> 0	max
$> \pi/2$	$< \pi/2$	$\in (\alpha_c, \pi)$	> 0	max
$< \pi/2$	$> \pi/2$	$\in (\pi - \alpha_c, \pi)$	< 0	min
$> \pi/2$	$> \pi/2$	$\in (0, \pi - \alpha_c)$	< 0	min

the notation $\alpha^*(\delta\phi)$ for the unique zero point of α given $\delta\phi := |\phi_c - \phi|$, and denote

$$\alpha_{\text{near}}[\alpha^*(\delta\phi)] := \arg \min_{\alpha \in [\alpha_l, \alpha_r]} |\alpha - \alpha^*(\delta\phi)|,$$

$$\alpha_{\text{far}}[\alpha^*(\delta\phi)] := \arg \max_{\alpha \in [\alpha_l, \alpha_r]} |\alpha - \alpha^*(\delta\phi)|.$$

Find the maximizer with $\phi = \phi_{\text{near}}$, $\delta\phi_{\text{near}} := |\phi_c - \phi_{\text{near}}|$:

- 1) If $\delta\phi_{\text{near}} = 0$, we have $\alpha_{\text{max}} = \alpha_{\text{near}}$;
- 2) if $\delta\phi_{\text{near}} = \pi/2$,

$$\alpha_{\text{max}} = \begin{cases} \alpha_l & \text{if } \alpha_c \leq \pi/2, \\ \alpha_r & \text{if } \alpha_c > \pi/2; \end{cases}$$

- 3) if $\delta\phi_{\text{near}} > \pi/2$, we have $\alpha_{\text{max}} = \alpha_{\text{far}}[\alpha^*(\delta\phi_{\text{near}})]$
- 4) if $\delta\phi_{\text{near}} < \pi/2$, $\alpha_c < \pi/2$, and $\alpha_l \geq \alpha_c$, we have

$$\alpha_{\text{max}} = \alpha_l;$$

- 5) if $\delta\phi_{\text{near}} < \pi/2$, $\alpha_c > \pi/2$ and $\alpha_r \leq \pi - \alpha_c$:

$$\alpha_{\text{max}} = \alpha_r;$$

- 6) otherwise, $\alpha_{\text{max}} = \alpha_{\text{near}}[\alpha^*(\delta\phi_{\text{near}})]$.

Find the minimizer with $\phi = \phi_{\text{far}}$, $\delta\phi_{\text{far}} := |\phi_c - \phi_{\text{far}}|$:

- 1) If $\delta\phi_{\text{far}} < \pi/2$, we have $\alpha_{\text{min}} = \alpha_{\text{far}}[\alpha^*(\delta\phi_{\text{far}})]$
- 2) if $\delta\phi_{\text{far}} = \pi/2$,

$$\alpha_{\text{min}} = \begin{cases} \alpha_r & \text{if } \alpha_c \leq \pi/2, \\ \alpha_l & \text{if } \alpha_c > \pi/2; \end{cases}$$

- 3) if $\delta\phi_{\text{far}} > \pi/2$, $\alpha_c < \pi/2$ and $\alpha_r \leq \pi - \alpha_c$:

$$\alpha_{\text{min}} = \alpha_r;$$

- 4) if $\delta\phi_{\text{far}} > \pi/2$, $\alpha_c > \pi/2$ and $\alpha_l \geq \pi - \alpha_c$:

$$\alpha_{\text{min}} = \alpha_l;$$

- 5) otherwise, $\alpha_{\text{min}} = \alpha_{\text{near}}[\alpha^*(\delta\phi_{\text{near}})]$

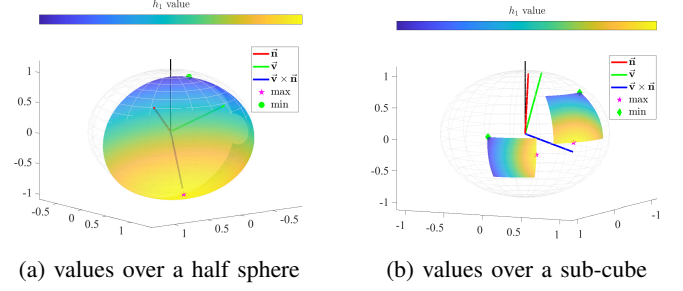
We finally illustrate Theorem 2 and the above results by visualizing $h_1(\vec{u}|\mathbf{a})$ on the sphere, as presented in Fig. 7.

APPENDIX III

SEMANTIC LINE MAP CONSTRUCTION

We propose to construct a semantic 3D line map based on posed RGB-D images and their semantic segmentation. The procedure consists of three steps:

- 1) Extract 2D lines in each image, and assign a semantic label according to segmentation.
- 2) Inversely project a 2D line l based on depth and regress a 3D line L , assign L with the same semantic as l .
- 3) After processing all images, cluster and prune the regressed 3D lines.


 Fig. 7: Landscapes of function $h_1(\vec{u}|\mathbf{a})$.

For the first step, we adopt ELSSED [32] to extract 2D lines, and use the rendered semantic mask provided by ScanNet++. For the second step, we dedicate to tackling the background interference issue. As shown in Fig. 8, the inversely projected 3D points can fall on the background while the extracted edge belongs to the foreground object, due to pose or depth error. We propose to evaluate multiple line hypotheses generated from perturbing the 2D line, and select one with a small average depth and mild perturbation.

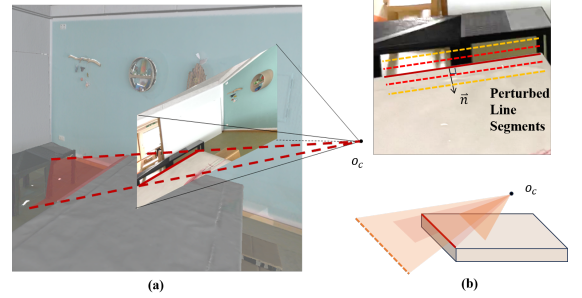


Fig. 8: (a) Regressing a 3D line based on points found by inverse projection is prone to the interference of background points. (b) We select the foreground 3D line from multiple hypotheses obtained from perturbed 2D line segments.

Since a 3D line can be observed by multiple images from different viewpoints, there exist a lot of redundant 3D lines after processing all images. In the last step, we cluster these lines based on geometric constraints. We treat each 3D line $L_m : (\mathbf{p}_m, \vec{\mathbf{v}}_m, s_m)$ as a vertex $\mathcal{V}_m$ on a graph $\mathcal{G}$. For two lines L_m, L_n satisfying the following **parallel** and **proximity** conditions, we connect the two vertices with an edge:

$$\angle(\vec{\mathbf{v}}_m, \vec{\mathbf{v}}_n) < \delta_r \text{ and } \|(\mathbf{I}_3 - \vec{\mathbf{v}}_m \vec{\mathbf{v}}_m^\top)(\mathbf{p}_m - \mathbf{p}_n)\| < \delta_t.$$

Denote the set of vertex $\mathcal{V}_m$ and its neighbors as $\mathcal{N}(\mathcal{V}_m)$. We summarize the clustering algorithm in Algorithm 2. The overall procedure is fast and efficient with only several parameters to set, which rarely require tuning across different scenes. In fact, we use the same set of parameters across scenes in experiments. As a demonstration of effectiveness, we present local line maps obtained w/ and w/o the proposed multiple hypothesis and clustering algorithms in Fig. 9.

Algorithm 2 3D Line Clustering

Inputs: graph $\mathcal{G}$, degree threshold δ_d
 $\mathcal{V}_{\max}, d_{\max} = \text{Maximal Degree}(\mathcal{G})$
1: **while** $d_{\max} \geq \delta_d$ **do**
2: Remove $\mathcal{N}(\mathcal{V}_{\max})$ from $\mathcal{G}$.
3: **for** each unique label s_m in $\mathcal{N}(\mathcal{V}_{\max})$ **do**
4: Register a 3D line $L : (\vec{v}_{\max}, \mathbf{p}_{\max}, s_m)$.
5: **end for**
6: **end while**

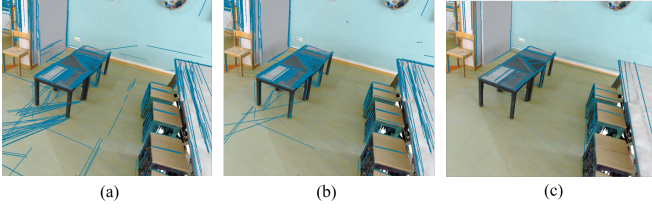


Fig. 9: Regressed 3d lines (a) w/o applying the multi-hypotheses and clustering algorithms, (b) w/o clustering, and (c) with both algorithms applied. Note that Thanks to the multi-hypothesis algorithm, there are much less lines in (b) which fall on the background compared to (a). The clustering algorithm clusters the proximate lines and prune the less-observed lines in (b), giving a much neater map in (c).

APPENDIX IV FIND BOUNDING FUNCTIONS FOR TRANSLATION PROBLEM IN PNL

For the three degrees of freedom t_x , t_y and t_z in translation, we distinguish the one with largest range, as determined by the environment size. Without generality, we assume t_x is the distinguished parameter. Denote $\mathbf{R}_{\sigma}^* \vec{\mathbf{n}}_a$ as $\vec{\mathbf{n}}_a^* := (n_{a,x}^*, n_{a,y}^*, n_{a,z}^*)$, and denote $\text{const}_a := \vec{\mathbf{n}}_a^* \bullet \mathbf{p}_a$, we rewrite the residual function $f(t_x, t_y, t_z | \mathbf{a})$ in (8) as follows:

$$t_x n_{a,x}^* + t_y n_{a,y}^* + t_z n_{a,z}^* - \text{const}_a. \quad (17)$$

For t_x in its range $\mathcal{I}_x$ and (t_y, t_z) in a sub-cube $\mathcal{C}_{yz}$, the bounding functions $f_L(t_x)$ and $f_U(t_x)$ for (17) can be easily found by solving a linear programming problem about (t_y, t_z) . Take $f_L(t_x)$ as an example. We obtain it by solving

$$\arg \min_{t_y, t_z} t_y n_{a,y}^* + t_z n_{a,z}^* \quad (t_y, t_z) \in \mathcal{C}_{yz},$$

and substituting the solution into (17). Since a global optimum of linear programming occurs at the vertices, we simply compare the value achieved by the four vertices of $\mathcal{C}_{yz}$.

APPENDIX V SUPPLEMENTARY EXPERIMENT RESULTS

A. Sat-CM v.s. CM in translation estimation

We compare translation estimation accuracy between the Sat-CM and CM method. Since the translation problem (8) is formulated based on a rotation estimate, the orientation accuracy directly governs the precision of translation. In this experiment, we use the most accurate upstream rotation

estimates, i.e., those based on the likelihood saturation function (7), to formulate the translation problem. In the legend we use ‘ π –’ and ‘ $\pi/2$ –’ to distinguish the upstream rotation estimates. We set the translation residual upper bound (used in Assumption 2) at $u_t = 1$, and choose $q = 0.9$ for SCM-GT, $q = 0.5$ for SCM-PR. After solving (8), we improve the output translation candidates with two steps. Firstly, we prune the inlier associations by imposing two physical constraints: the truly-associated 3D line resides in front of the camera and its projection intersects with the image. Next, we fine-tune the translation estimates by minimizing least squares error or the pruned inliers. As present in Table V, Sat-CM performs consistently better than CM across all scenes and settings. Notably, the truncated saturation function performs as good as the likelihood saturation function when using ground truth labels, and presents better robustness under predicted labels by keeping more potentially good candidates for further refinement.

B. parameter sensitivity evaluation

We evaluate performance sensitivity in parameter q by plotting the error curves. For rotation estimation, we choose $q \in \{0.6, 0.7, 0.8, 0.9, 0.99\}$ when using the ground truth labels and choose $q \in \{0.2, 0.35, 0.5, 0.65, 0.8\}$ when using the predicted labels. The results are present in Fig. 10 and 11. For translation estimation, we choose the same set of q values for both ground truth and predicted semantic labels $q \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. The results are present in Fig. 12 and Fig. 13.

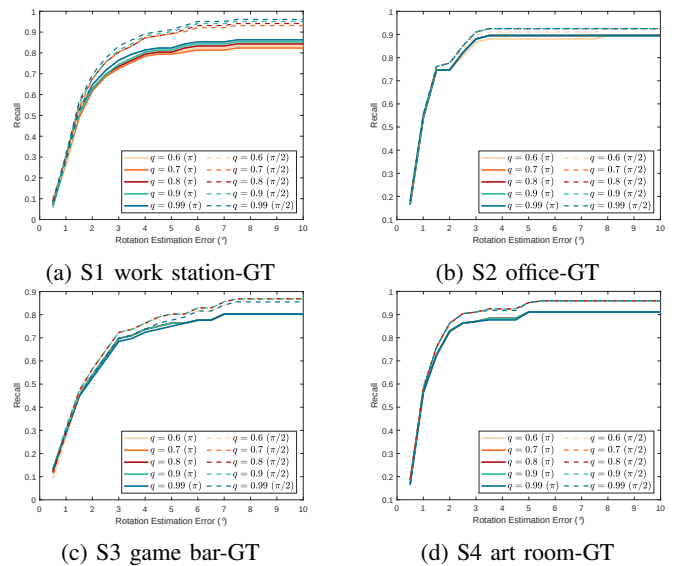
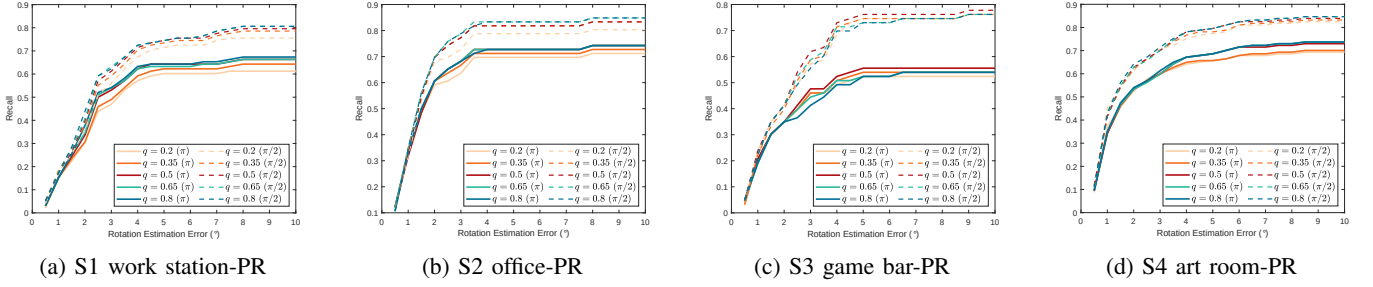
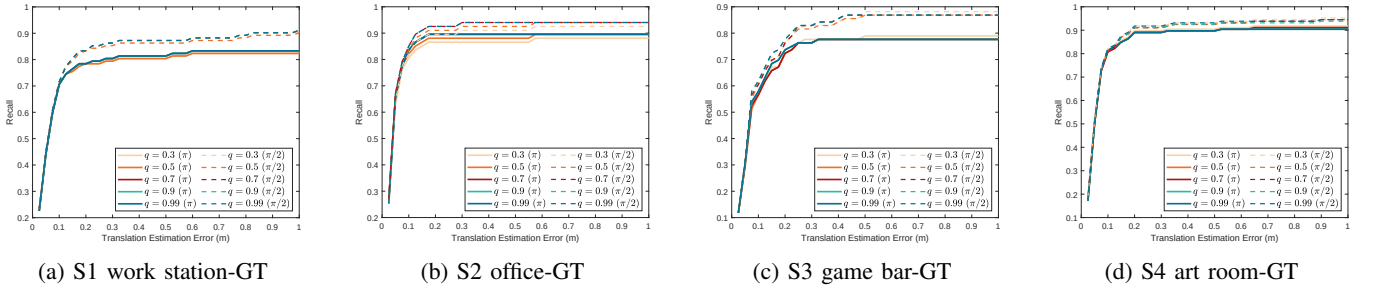
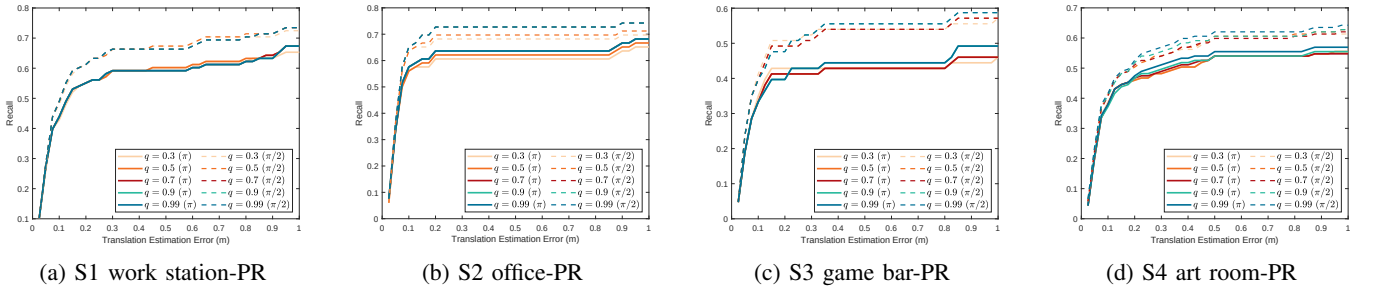


Fig. 10: Rotation error recall using ground truth semantic labels and different values of q .

TABLE V: Compare CM with Sat-CM in translation.

Method	S1(workstation)	S2(office)	S3(game bar)	S4(art room)	S1	S2	S3	S4
	25%/50%/75% error quantiles (cm)				Recall at 5cm/10cm/15cm (%)			
π -CM-GT	3.0/5.7/33.9	2.8/4.0/7.9	4.6/9.9/78.4	3.3/5.5/34.9	47/65/72	60/78/81	25/50/61	42/68/71
π -SCM0-GT	2.9/5.9/11.0	2.4/3.8/7.1	4.0/7.1/22.2	2.9/5.0/8.5	45/73/80	63/84/88	30/58/68	50/77/84
π -SCM1-GT	3.0/5.8/14.1	2.5/4.0/6.7	4.0/6.8/22.6	3.0/5.0/8.2	44/71/76	66/84/88	30/58/68	50/82/85
$\pi/2$ -CM-GT	3.0/5.5/15.6	2.8/4.0/7.9	4.2/8.6/28.1	3.3/5.5/19.7	48/66/75	60/78/82	26/51/63	42/68/71
$\pi/2$ -SCM0-GT	2.8/5.9/10.8	2.4/3.8/6.7	4.0/6.9/19.4	2.9/4.9/8.2	46/74/85	64/85/91	32/61/72	51/78/86
$\pi/2$ -SCM1-GT	2.7/5.6/11.3	2.5/3.8/6.6	4.0/6.5/18.9	3.0/4.9/8.0	45/72/80	67/85/91	32/62/72	51/82/88
π -CM-PR	5.2/77.4/198.4	5.2/63.0/214.9	8.2/129.1/241.2	6.9/121.5/226.9	24/37/42	23/41/44	17/27/35	18/29/34
π -SCM0-PR	4.8/13.3/150.7	3.7/7.7/197.5	7.2/100.7/232.0	5.9/21.2/195.5	28/44/53	35/59/62	17/29/37	20/39/45
π -SCM1-PR	4.5/13.8/150.7	3.7/7.3/194.5	6.4/135.8/226.6	5.9/33.8/208.3	28/44/53	33/58/61	19/33/41	20/38/45
$\pi/2$ -CM-PR	5.2/16.7/175.2	4.7/11.8/194.5	7.7/61.1/234.9	6.4/107.0/208.1	24/42/48	26/47/52	17/29/37	19/31/36
$\pi/2$ -SCM0-PR	4.4/10.6/90.1	3.7/6.8/49.2	6.6/32.3/199.0	5.5/14.4/119.8	30/49/60	36/65/70	19/33/43	23/43/51
$\pi/2$ -SCM1-PR	4.5/10.2/113.2	3.6/6.3/105.9	5.2/26.7/224.5	5.7/18.0/192.1	29/49/59	36/65/70	24/40/49	23/41/48


 Fig. 11: Rotation error recall using predicted semantic labels and different values of q .

 Fig. 12: Translation error recall using ground truth semantic labels and different values of q .

 Fig. 13: Translation error recall using predicted labels and different values of q .

Occlusion-Aware Multi-Object Tracking for Point-of-Care Semen Analysis Using Smartphone Imaging

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Abstract—Semen analysis is the gold standard method for assessing male reproductive capacity. Although clinical semen analysis routinely uses Computer-Assisted Semen Analysis (CASA) systems to measure sperm concentration and motility, clinical visits for semen analysis are not always feasible due to the unavailability of the high-cost CASA systems and the mental stress for clinical visits. Here, we present a point-of-care semen analysis method using smartphone imaging. It follows the same visual tracking scheme as laboratory CASA systems, aiming to provide semen analysis results that are accurate and consistent with clinical standards. To address the challenge of robustly tracking multiple sperm posed by smartphone imaging and the point-of-care nature, we propose an Occlusion-Aware Joint Probabilistic Data Association Filter (OA-JPDAF). It combines sperm head contour obtained from a boundary-sensitive segmentation network and a kinematics-based probabilistic model to determine the occlusion status of both targets and measurements, excluding unreasonable association events to enable robust data association. Experimental results showed that the OA-JPDAF outperformed the state-of-the-art multi-sperm tracking algorithms, and achieved a success rate of 95.14% for tracking occluded sperm, a mean error of 2.03 million/ml for sperm concentration measurement, and a mean error of 1.58% for sperm motility measurement. Clinical tests involving 50 participants revealed that the point-of-care method showed good agreement in sperm concentration and motility measurement (Spearman rank correlation coefficients of 0.94 and 0.89, respectively) with that of the laboratory CASA, even when used by inexperienced users.

Index Terms—Joint Probabilistic Data Association, Point-of-care, Semen Analysis, Smartphone imaging

I. INTRODUCTION

THE World Health Organization (WHO) estimates over 48 million couples worldwide suffer from infertility [1]. Male factor infertility accounts for 40%-50% of all infertility

cases in clinical practice [2], mostly attributing to the failure to produce a sufficient quantity of motile sperm. For clinical diagnosis of male infertility, semen analysis is routinely performed to evaluate sperm concentration and motility. Sperm concentration measures the number of sperm per unit volume of the ejaculate, and a low sperm concentration (oligospermia) may indicate undescended testicles [3], genetic disorders [4], dysfunction in spermatogenesis [5], or even azoospermia [6]. Sperm motility measures the percentage of sperm with progressive movement, and a low sperm motility (asthenozoospermia) is an indicator for varicocele [7], sperm mitochondrial dysfunction [8], or testicular damage [9]. Quantitative measurement of sperm concentration and motility and accurate semen analysis results are the basis for diagnosis and treatment of male infertility.

Computer-Assisted Semen Analysis (CASA) has been gradually recognized as a gold standard for laboratory semen analysis [10]. A CASA system visually recognizes and locates sperm in microscopic images of a semen sample, and tracks multiple sperm over a period of time to obtain sperm counts as well as kinematics and thus to calculate concentration and motility. Since CASA systems use a high-grade microscope with an integrated high-performance computer for data processing, visualization and storage, CASA instruments are expensive (tens of thousands of dollars) and require professional technicians to operate.

Availability, affordability, and quality of the diagnosis of male infertility remain a challenge globally [1]. The demand for low-cost, professional-free, accurate semen analysis, or in other words, point-of-care semen analysis is high worldwide. In the past decade, several point-of-care semen analysis methods have been proposed, for instance, paper-based colorimetric assay [11], lateral flow immuno-chromatography [12], centrifugal separation analysis [13], and smartphone-based visual tracking [14],[15]. Among these methods, smartphone-based visual tracking is considered a promising approach for point-of-care semen analysis. It shares the same technical nature as a laboratory CASA, i.e., automated multi-sperm tracking based on sperm detection and location. Differently, it uses a smartphone camera and magnification accessories to image semen samples instead of a high-grade microscope and certified medical camera. The advantages over laboratory CASA systems include significantly lower cost, ease of use, and high

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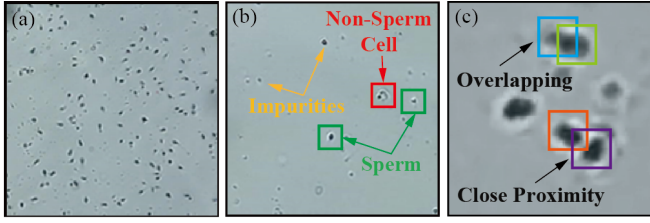


Fig. 1. Technical challenges of point-of-care semen analysis with smartphone imaging: (a) sperm are small in size, making the sperm tail ($<1 \mu\text{m}$) even invisible under smartphone imaging; (b) non-sperm impurities are prevalent and need to be distinguished by the algorithm; (c) semen analysis uses undiluted raw semen where sperm are in close proximity and overlapping/occlusion of sperm is frequent (e.g., >100 sperm moving simultaneously within a field of view of $1024 \text{ pixels} \times 1024 \text{ pixels}$), thus requiring robust multi-sperm tracking algorithms to track individual sperm.

accessibility that mentally helps mitigate embarrassment and stress caused by clinical visits.

However, constrained by cost and portability requirements, smartphone-based microscopic imaging typically has lower magnification and inferior imaging quality compared to laboratory microscopy imaging. Thus, the following technical challenges must be overcome for smartphone-based semen analysis to achieve as accurate sperm concentration and motility measurement as a laboratory CASA.

1) The size of a sperm is small (**Fig. 1a**). Typical tail diameter is smaller than $1 \mu\text{m}$, making it hardly visible under smartphone imaging ($200\times$). The small size of sperm requires the algorithm to rely mainly on features of the sperm heads to identify and distinguish each sperm.

2) Non-sperm impurities are prevalent in raw semen samples (**Fig. 1b**), whose quantity may exceed the number of sperm by more than two-fold. These impurities often exhibit similar sizes, shapes, and colors to sperm, rendering them challenging to differentiate from sperm.

3) Sperm frequently overlap and occlude each other (**Fig. 1c**). Under a typical concentration of 50-80 million sperm/ml [18], hundreds of sperm move simultaneously within a field of view. As shown in **Section V.D**, failure to track individual sperm leads to significant errors in semen motility evaluation, calling for a robust multi-sperm tracking algorithm.

This paper reports a novel point-of-care semen analysis method using smartphone imaging to address the technical challenges above and improve robustness in tracking occluded targets to achieve accurate sperm concentration and motility measurements. Contributions include:

1) A novel smartphone-based imaging modality for semen analysis was developed. Through this modality, continuous bright-field imaging at an equivalent microscope $200\times$ magnification of a controlled volume of raw semen samples can be obtained at 4K 30 fps.

2) A boundary sensitive segmentation network was proposed for specifically identifying, segmenting the sperm head, and separating different sperm in close proximity under smartphone imaging.

3) An occlusion-aware multi-object tracking algorithm was developed to reduce occlusion-caused tracking errors. By incorporating information from both the track (sperm kinematics) and measurements (segmented contours), the occurrence,

persistence, and termination of an occlusion event between sperm targets can be predicted and determined. With occlusion awareness, the constraints in enumerating feasible joint events are dynamically relaxed during occlusion to allow the same measurement to match multiple targets. Experiments confirmed the occlusion-aware strategy significantly increased accuracy for tracking occluded sperm.

4) Theoretical analysis was performed to reveal how failures in multi-sperm tracking affect sperm concentration and motility evaluation. Tracking failures were classified into six types, and the effect of each failure type on sperm concentration and motility measurement was quantified. To the best of our knowledge, this is the first work to quantify the effect of tracking failures, especially failures in distinguishing occluded sperm, on semen analysis, which is applicable for both point-of-care and clinical semen analysis.

II. RELATED WORKS

A. Methods for Sperm Identification

Accurate identification of sperm cells is the basis for sperm counting and subsequent tracking, which can be achieved by either object detection methods or semantic segmentation methods. Compared to detection methods, target segmentation algorithms enable higher localization precision and provide additional target features (e.g., target contour) that can facilitate robust sperm tracking in complex environments. In this subsection, existing or potential segmentation methods for sperm identification are reviewed. The CNN-based semantic segmentation algorithms, including U-Net [16], DeepLabv3 [17], and Mask-RCNN [18] are the state-of-the-art approaches for sperm segmentation. Compared with conventional thresholding methods such as binary pixel intensity thresholding [19], enhanced-Otsu thresholding [20], and adaptive local threshold [21], CNN-based methods are capable of extracting high-level features to distinguish the background and impurities, resulting in significant improvement in segmentation accuracy. Chen et. al. [22] evaluated popular methods for sperm segmentation on the public Sperm Videos and Images Analysis dataset, among which U-Net demonstrated the top dice score. Although improvements to U-Net and other CNN-based methods for sperm segmentation have been proposed [23]-[25], these methods have not been specifically optimized for smartphone imaging conditions. Moreover, U-Net shares a prominent mis-segmentation issue with other CNN-based methods that multiple sperm in close proximity tend to be segmented as a cluster [22] although they are distinguishable by human eyes. This problem is worsened under smartphone imaging, due to the poorer imaging quality of non-diluted raw semen samples. In this paper, a boundary sensitive segmentation network is proposed to address this problem, as described in **Section IV**.

B. Vision Methods for Sperm Motility Measurement

As stipulated by WHO [26], sperm motility measurement is to calculate the percentage of motile sperm (velocity $> 5 \mu\text{m/s}$) in a semen sample. The technical requirement for sperm motility measurement is to simultaneously track multiple sperm to obtain their trajectories, which is essentially a multiple-object

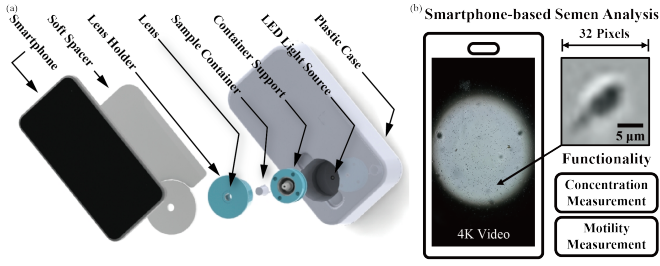


Fig. 2. (a)Diagram of parts breakdown of the developed smartphone semen imaging modality; (b) Raw images captured by the modality.

tracking (MOT) problem. Tracking-by-detection is the most extensively utilized paradigm for such purpose. Its nature is a data association problem, i.e., to associate sperm positions (measurements obtained by the front-end detection or segmentation method) to an existing target in previous frames.

To date, most commercial CASA and point-of-care semen analysis systems use Nearest Neighbor for data association in sperm tracking [27]. It associates a target to a measurement according to minimum inter-frame Euclidean distances. Advances have been achieved by Global Nearest Neighbor (GNN) tracking, which adopts distance-based global optimal association. However, limited by the greedy data association principle, both NN and GNN-based tracking are occlusion sensitive, i.e., the only measurement produced by two or more targets in occlusion cannot be assigned to multiple targets, leading to increased association errors and risk of target loss. Although point-of-care semen analysis with smartphone imaging has been attempted, the NN and GNN-based tracking algorithms applied in such attempts resulted in large errors in motility measurement, up to 100% [14], [28].

Multi-Hypothesis Tracking (MHT) [29], Markov Chain Monte Carlo Data Association (MCMCDA) [30] and Joint Probabilistic Data Association (JPDAF) [31] employ advanced data association principles to accommodate higher tolerance for temporary loss of measurements due to occlusion. MHT uses all past measurements to perform data association, whose heavy computational complexity makes it impractical to track sperm in high concentration samples, despite its high tracking accuracy. MCMCDA uses Markov Chain Monte Carlo to merge track partitions originating from the same target, caused by occlusion. However, this method does not track well for high sperm concentrations (e.g., >60 million/ml) [20] due to the difficulty of correctly grouping too many similar tracks.

The JPDAF tracker represents the state-of-the-art technique for multiple sperm tracking [32]. Its high accuracy is attributed to its enumeration mechanism, i.e., feasible target-measurement association events are enumerated, then the probability of each event is calculated for weighted update of target states. The enumeration strictly follows the one-to-one correspondence assumption, i.e., a target generates at most one measurement and a measurement originates from at most one target. However, this assumption is not always valid, e.g., during sperm overlapping a single measurement (contour) originates from multiple targets. This assumption prevents the correct association event to be enumerated, causing significant association bias and even target loss during occlusion.

Another type of data association strategy is to employ deep

learning networks to extract features for feature similarity-based data association [33]–[35]. However, low magnification, lack of color features under smartphone microscopic imaging, and high similarity in sperm appearance (especially during occlusion) may hinder the strength of feature similarity-based data association. Highly similar features also make it challenging for joint-detection-and-tracking methods [36],[37]. Although existing methods [38]–[41] incorporating Transformer may provide enhanced feature extraction ability, under smartphone imaging mainly the sperm heads are visible, and it remains unknown whether these methods can keep distinguishing a specific sperm using the limited features of sperm heads during occlusion. To the best of our knowledge, no studies have validated the performance of such methods for the challenging task of tracking multiple sperm with highly similar appearance.

In addition to multi-sperm tracking, another type of method uses deep learning to take sperm videos as input to directly predict/classify sperm motility, e.g., CNN-based prediction [42] and motilitAI [43]. Such methods bypass multi-sperm tracking, and thus fail to provide sperm trajectories. The black-box classification approach lacks interpretability and makes it challenging to identify potential errors in motility predictions, thus increasing the risk of misdiagnosis.

In this work, we advance the state-of-the-art multi-sperm tracking algorithm, JPDAF. Through occlusion-awareness, the original constraint is dynamically relaxed to ensure robust tracking under occlusion. Note that occlusion-aware JPDAF is not a black box. By analyzing the obtained trajectories, how tracking failure affects motility measurement is revealed.

III. SMARTPHONE MICROSCOPIC IMAGING MODALITY FOR SEMEN ANALYSIS

A novel imaging modality was developed to enable smartphone-based microscopic imaging of a fixed volume of semen samples. As shown in **Fig. 2**, the proposed imaging modality comprises a smartphone, a soft spacer, an objective lens and its holder, a semen sample container and its support, an LED light source, and a plastic case. In contrast to other smartphone-based semen analysis systems that rely on microfluidic devices to maintain a consistent sample volume for imaging, our system utilizes two small, identical, and transparent containers that mimic the glass slide-cover slip structure. The semen sample containers were engineered to accommodate a fixed volume ($0.02\mu\text{L}$) of semen and allow light to pass through for imaging. To prepare the sample, an excess of raw semen is first added to one of the containers, and the other container is then inserted, causing excess semen to be squeezed out, thus leaving a consistent volume of semen confined to the space between the two containers. The objective lens is mounted in a holder with a small magnet that holds it in place with the container support and keeps the optical path centered. During imaging, the rear camera of the smartphone is aligned with the optical axis of the objective lens by the operator. To accommodate various smartphone models, a soft spacer is custom-designed to provide adequate clearance for the protruding smartphone lenses during imaging. The built-in video recording software of the smartphone enables the

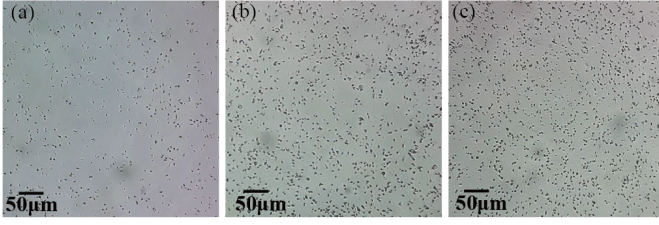


Fig. 3. The FoVs of samples with different sperm concentrations using the smartphone imaging modality. The size for each FoV is 1024 pixels $\times$ 1024 pixels. Each FoV contains (a) 326 sperm (concentration: 84 million/ml); (b) 597 sperm (concentration: 154 million/ml); (c) 912 sperm (concentration: 235 million/ml).

recording of 4K (3648 by 2056 pixels) videos at 30 frames per second. As calibrated using a commercial calibration slide (MR095, AmScope, USA), the smartphone microscopic imaging modality achieved a magnification that is equivalent to 200 $\times$ of a standard laboratory microscope. The subsequent concentration and motility analysis in this study was based on these settings for a video duration of three seconds.

IV. OCCLUSION-AWARE JPDAF FOR MULTI-SPERM TRACKING

A. Problem Formulation

Accurate visual measurement of sperm concentration and motility is essentially a multi-object (multiple sperm) tracking problem. Multi-sperm tracking using smartphone imaging modality is challenging due to 1) the use of raw, undiluted semen samples in which sperm frequently cross over and occlude each other, as shown in Fig. 3; 2) the inherent limitations in imaging quality of smartphone technology, which further exacerbates the challenge of accurately tracking each sperm.

As discussed in II. Related Work, among existing data association algorithms, the joint probability data association filter (JPDAF) achieves the highest accuracy for tracking multiple sperm under occlusion. It achieves high accuracy mainly because it enumerates all feasible joint events (association of targets and measurements). However, JPDAF was originally designed for radar-based tracking of multiple flying objects such as aircrafts. In such scenarios, measurement (i.e., physical position) of each target is obtained from radars and such measurement is rarely affected by target occlusion. The JPDAFs basic assumption, i.e., the one-to-one association between measurement and target, is a global optimization hypothesis to enumerate a feasible joint event corresponding to the real situation.

To address these limitations, a novel occlusion-aware JPDAF (OA-JPDAF) is proposed, aiming to realize robust multi-sperm tracking under occlusion for smartphone-based semen analysis. As shown in the workflow in Fig. 4a, a dynamic constraint relaxation mechanism is proposed to relax the constraint in the original assumption of one-to-one association between measurement and target. An occlusion evaluator, which incorporates information from both measurements (sperm contour) and targets (kinematic model), is utilized as a front module to enable efficient occlusion awareness. This allows for the initial filtering of feasible joint events that do not match reality. Feasible joint event exclusion are then applied to further refine the enumerated feasible joint events and improve the tracker's robustness against clutter.

B. Relaxation of Feasible Joint Events Enumeration Constraint

JPDAF enumerates feasible joint events based on a validation matrix Ω , which is defined as:

$$\Omega = [\omega_{ij}], \quad i = 1, \dots, m, \quad j = 1, \dots, n \quad (1)$$

where m and n are the number of measurements and targets in the validation matrix, ω_{ij} is given by:

$$\omega_{ij} = \begin{cases} 1, & \text{if } M_i \in G_j \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

where G_j is the gating region of T_j (target j), which means ω_{ij} equals to 1 only when M_i (measurement i) falls into the gating region of T_j .

Feasible joint events of a validation matrix are enumerated based on the basic assumption of original JPDAF that measurement and target are one-to-one corresponding, which can be converted to the feasible enumeration constraints as

$$\sum_{i=1}^m \omega_{ij} \leq 1 \quad \text{and} \quad \sum_{j=1}^n \omega_{ij} \leq 1 \quad (3)$$

where $\sum_{i=1}^m \omega_{ij} \leq 1$ constrains that one target can generate at most one new measurement, and $\sum_{j=1}^n \omega_{ij} \leq 1$, constrains that one measurement can originate from at most one target.

When occlusion occurs, however, the contours of two or more sperm merge into one larger contour, thus providing only one measurement. Using the original constraint (3), no enumerated feasible joint events correspond to the real case. As a result, target states would be incorrectly updated with biased association. As one can expect, if sperm concentration become higher and occlusion between targets becomes more frequent, biased associations would become more common, leading to increased tracking errors.

Assuming a single measurement might originate from multiple targets, the constraints in (3) is relaxed to

$$\sum_{i=1}^m \omega_{ij} \leq 1 \quad (4)$$

However, directly relaxing the constraints by using (4) causes the total number of enumerated feasible joint events to significantly increase, causing more biased associations. With increased biased associations, the target position prediction becomes less accurate with a larger gating region, leading to increased size of validation matrix (increased computational efforts in enumeration which is already exponential time complexity [44]) and wrong state updates. Further restrictions should be imposed on (4) for practicable enumeration of feasible joint events.

This side effect stems from the fact that it is unknown whether a target is occluded. Even for targets that are not occluded, it may share the same measurement with other targets in certain feasible joint events. Assume the occlusion status of the targets can be obtained (e.g., by using methods in Section IV.C), the relaxation of the constraint can be strictly limited to occluded targets, and the number of feasible joint events can be considerably reduced, also reducing the risk of biased associations. Hence, the first restriction is proposed.

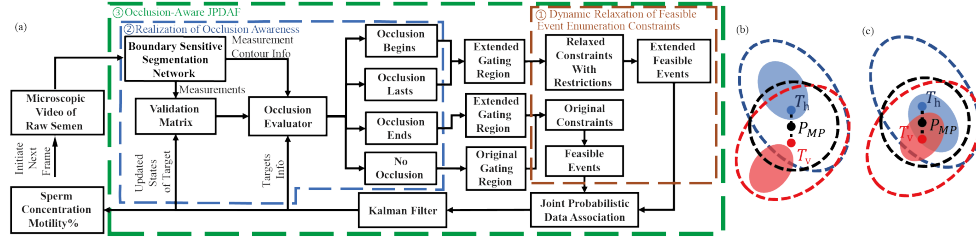


Fig. 4. (a) Workflow of our proposed occlusion-aware JPDAF for robust sperm tracking in the scenario of smartphone-based point-of-care semen analysis. (b) demonstrates an example that no occlusion/occlusion ends is confirmed from measurements by the occlusion evaluator, whereas occlusion begins/ lasts is confirmed in (c). The blue and red dots are the predicted position of $\hat{T}_h$ and $\hat{T}_v$, surrounded by an individual ellipsoidal gating region. Red and blue shadowed areas are the contour measurement of target T_h and T_v from the segmentation network. The circular region Λ is denoting as the area inside the black dashed line, whose center is at P_{MP} (midpoint of $\hat{T}_h$ and $\hat{T}_v$).

Restriction 1: The relaxed constraint (4) is only applied to occluded targets.

The side effect of significantly increased feasible joint events also originates from the lack of information to determine if a measurement is generated by a single target or multiple occluded targets. Even for a measurement that is not caused by target occlusion, it could be shared by two or multiple targets in certain feasible joint events. If the occlusion status of a measurement can be obtained (e.g., by comparing contour areas across frames which detailed in Section IV.C), the feasible joint events enumerated by the (4) with **Restriction 1** can be further refined. Hence the second restriction is proposed.

Restriction 2: The relaxed constraint (4) is only applied to measurements that originate from occluded targets.

With the **Restriction 2**, for a measurement from a non-occluded target, feasible joint events that associate this measurement with two or more targets would be excluded. Also, targets within the same occlusion cluster should avoid being associated with different measurements in feasible joint events, which is granted by the third restriction.

Restriction 3: Targets in the same cluster should be simultaneously associated with either a same measurement or null.

The determination of whether targets are within the same enclosed cluster can be obtained pairwise using the method in Section IV.C. When different pairs of targets have one or more shared target(s), these targets are obviously within the same cluster. This restriction further reduces the number of enumerated feasible joint events.

For the tracking of occluded targets, the position of an immotile sperm is apparently easier to determine than that of a motile sperm, even if it may present subtle movement due to collisions. For this reason, immotile targets that have been confirmed as in occlusion should be excluded for feasible joint event enumeration until occlusion ends. Here the fourth restriction is proposed.

Restriction 4: Immotile targets in occlusion are excluded for feasible joint event enumeration.

During occlusion, the immotile target is updated using the state before the occlusion. This design further reduces the number of targets in validation matrix and thus contributes to fewer feasible joint events to be enumerated. Moreover, it helps avoid incorrect position updates for immotile targets that are in occlusion, and avoid tracking errors such as target ID switching with moving targets. Following the end of occlusion, such design also facilitates correct re-matching of occluded

targets. The above four restrictions can be summarized as the new constraint (C') in (5)

$$C' = \begin{cases} \sum_{i=1}^m \omega_{ij} \leq 1 \text{ and } \omega_{ih} = \omega_{iv}, \text{ if} \\ M_i \in S_{OM}, T_j, T_h, T_v \in S_{OT} \cap S_{MT} \\ \sum_{i=1}^m \omega_{ij} \leq 1 \text{ and } \sum_{j=1}^n \omega_{ij} \leq 1, \text{ otherwise} \end{cases} \quad (5)$$

where T_h and T_v are pair of occlude targets in the same cluster at the current frame. S_{OT} is the set of targets in occlusion, S_{MT} is the set of targets that are motile, and S_{OM} is the set of measurements that originate from the occlusion of two or more targets.

With the above restrictions, the number of feasible joint events can be significantly reduced and the risk of biased association is reduced. Give that the **Restriction 1,2** and **3** rely on the awareness of occlusion of target and measurement, it is necessary to realize occlusion-awareness, which will be detailed in the following section.

C. Realization of Occlusion-awareness

Here we propose an occlusion evaluator driven by multi-frame information fusion to realize predicting, monitoring, and determining the occurrence, persistence, and termination of an occlusion event of multiple sperm targets. To detect when an occlusion event occurs, from the perspective of measurement, the contour of a single sperm differs from that of a cluster of sperm. Another approach is to predict the occlusion of a moving target based on its kinematic model. Hence, the proposed occlusion evaluator assesses an occlusion event from two aspects, i.e., occlusion prediction from existing targets and occlusion validation from measurements.

(i) Occlusion-awareness from Targets: Using the kinematic model and the existing target states, the region where the target appears in the next frame is predictable and the probability of the targets location follows a two-dimensional normal distribution. Therefore, the probability of occlusion is obtained by probabilistically integrating the elliptical overlap part of the predicted position probabilities between targets. For example, for occlusion prediction for target h (T_h) and target v (T_v), the possibility of occlusion can be derived as

$$P_{Occul, T_h T_v} = P(T_{h, pred}) P(T_{v, pred}) \iint f_{T_h} f_{T_v} dx dy \quad (6)$$

where $P(T_{h, pred})$ and $P(T_{v, pred})$ are the probability of successful gating a correct measurement for targets T_h and T_v , which are preset empirical parameters for the formation of the validation matrix. f_{T_h} and f_{T_v} are the probability density function of the target threshold ellipse, which follows

$$f_{T_h}(x, y) = \frac{e^{-\frac{1}{2}[(x, y) - (\mu_x, \mu_y)]^T \Sigma^{-1} [(x, y) - (\mu_x, \mu_y)]}}{2\pi \sqrt{\det(\Sigma)}} \quad (7)$$

where Σ is covariance-dependent eigenvalue matrix. When $P_{Occul, T_h T_v}$ exceeds the set threshold, T_h and T_v will be proposed as occluded target pair in the next frame.

ii) Occlusion-awareness from Measurements: Measurements are first obtained using image segmentation for subsequent occlusion evaluation. This image segmentation step serves two purposes. The first is to detect/locate sperm from a FoV mixed with non-sperm cells and other impurities. The second is to extract contour of a spermatozoa or a sperm cluster to provide measurement for occlusion evaluation.

For the first purpose of detecting/locating sperm, an enhanced U-Net network using ResNet-34 as the backbone is proposed. Segmentation is preferred over object detection because in smartphone imaging, sperm are densely distributed small objects, and accurate target bounding box for object detection method relies on a good selection of non-maximal suppression parameters, while image segmentation methods do not require this delicate parameter adjustment. To improve the capacity of segmentation network for identifying sperm, the segmentation network is desired to extract the more generalizable features to exclude sperm-like impurities but to include sperm groups containing two or more overlapping sperm (see **Fig. 1c**).

Hence, our network adopts ResNet-34 as the backbone. The introduction of ResNet-34 is expected to solve the problem of vanishing gradients and mitigate the degradation (accuracy saturation) problem of deeper neural network models, leading to better learning of generalizable sperm features. For the second purpose, accurately segmenting and distinguishing single sperm and sperm clusters is critical, because it determines the accuracy of subsequent occlusion evaluation. Hence, the enhanced U-Net should be further improved in terms of boundary sensitivity to separate single sperm that is close to other sperm but do not form a cluster due to occlusion. As shown by our results, the original U-Net sometimes fails. To address this issue, a loss function based on boundary-weighted maps is introduced. While the original U-Net does not elevate weights for boundaries of isolated objects (objects that are far away from other objects), the weight map generation method consistently enables the highest elevated weights on the sperm contour boundary line, followed by pixels close to the contour line, and no elevation for other pixels. This enables the segmentation network to learn sperm boundaries with higher weights even if there are no other sperm nearby, facilitating the network to learn more boundary information. The boundary-sensitive loss function is

$$\mathcal{L} = \frac{1}{N} \sum_{i=1} -w_i [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (8)$$

where y_i is the labeled value of pixel (1 for sperm, 0 for others). p_i is the predicted probability of a pixel belonging to sperm, w_i is the weight value of the pixel i in the pre-generated weight map, and N is the total number of pixels in an image.

Based on precise segmentation results, an occlusion-awareness method utilizing information from both targets and

measurements powered by multi-frame information fusion is proposed. Initially for an occlusion event, in the $(t-1)^{th}$ frame, $P_{Occul, T_h T_v}$ is first calculated followed by (6) and (7). Once T_h and T_v are proposed as a potential occluded target pair in the t^{th} frame, an examination of contour area changes in a circular searching region Λ is subsequently performed to confirm. The searching region Λ (illustrated in **Fig. 4b&c**) centers at the midpoint (P_{MP}) of the predicted positions of T_h and T_v in t^{th} frame, with a radius that is proportional to the velocity of T_h or T_v , which is defined as

$$\Lambda = \left\{ p \mid (p - P_{MP})^2 < R^2, R \propto \max(v_h, v_v) \right\} \quad (9)$$

where p is a point on and within the region Λ , and v_h and v_v are the absolute velocity of T_h and T_v , respectively.

There are two potential circumstances for T_h and T_v in the t^{th} frame: 1) occlusion begins and 2) no occlusion. The probability of occlusion p_t in the t^{th} frame is calculated by comparing the size of the largest contour in the circular region Λ with the size of the contour of the target sperm in the preceding frame, which is given by

$$p_t = c \sum_{i=t-2}^{t-1} \left[\log\left(\frac{A_t}{A_{h,i}}\right) \log\left(\frac{A_t}{A_{v,i}}\right) \right] \quad (10)$$

where A_t is the measurement of the largest contour area of sperm or sperm cluster within the circular searching region Λ , and $A_{h,i}$ and $A_{v,i}$ are the measurement of sperm contour area in i^{th} frame of target h and v , and c is a scaling coefficient.

The change of occlusion conditions alters the value of p_t . Thus, by comparing p_{t-1} and p_t , different occlusion conditions (Occlusion begins: when $p_t \geq T_H$; No occlusion: when $p_t \geq T_H$; Occlusion ends: when $p_{t-1} \geq T_L$ and $p_t < T_L$ after occlusion begins; Occlusion lasts: when $p_{t-1} \geq T_L$ and $p_t \geq T_L$ after occlusion begins. p_{t-1} is the possibility of occlusion occur in $(t-1)^{th}$ frame, T_L is the low threshold and T_H is the high threshold).

After confirmation by the above method, the pair of targets proposed to be in occlusion by (6) can be finally labeled as occluded pair of targets (for **Restriction 1** and **3**). The measurement with largest contour in the circular region Λ can be identified as a measurement generated by occluded targets (for **Restriction 2**). By the proposed method, accurate awareness of occlusion state of both targets and measurements is obtained.

D. Calculation of Sperm Concentration and Motility

The sperm concentration is calculated by

$$\text{Concentration} = \frac{N_{sperm}}{L \times W \times \mu^2 \times H} \quad (11)$$

where N_{sperm} is the final number of targets in the tracker, L (unit: pixel) is the length of the FoV, W (unit: pixel) is the width of the FoV, μ is a calibrated coefficient converting the length of each pixel from image coordinate to real world length, H is the depth of semen container. In the current setup, $L = W = 1024$ pixels, $\mu = 0.43 \text{ pixel}/\mu\text{m}$, $H = 20 \mu\text{m}$. This method is used to calculate concentration instead of averaging the number of measurements provided by the segmentation network in each frame because it avoids the reduction of sperm

count due to occlusion, and also avoids the effect of clutter on the count that only presents intermittently for a few frames.

The motility is calculated as the percentage of motile sperm (velocity $> 5 \mu\text{m/s}$ as defined by WHO) in a semen sample:

$$\text{Motility} = \frac{N_{\text{sperm}|v>5\mu\text{m/s}}}{N_{\text{sperm}}} \quad (12)$$

where $N_{\text{sperm}|v>5\mu\text{m/s}}$ is the number of sperm with velocity greater than $5\mu\text{m/s}$.

V. EXPERIMENTAL RESULTS AND DISCUSSION

Human sperm samples used in experiments were obtained from Shanghai Ninth Peoples Hospital (Shanghai, China). Informed consent was obtained from all subjects (50 patients enrolled). The study was approved by the Institutional Review Board of the Shanghai Ninth Peoples Hospital (IRB: SH9H-2021-T14-1). In experiments, all semen videos were recorded using an Android phone (Huawei P20, China) and the developed imaging modality. Following the WHO guideline, the duration of each video was set to 3 seconds.

A. Evaluation of the Segmentation Network for Sperm Recognition

TABLE I
PERFORMANCE OF THE SEGMENTATION NETWORK FOR
DISTINGUISHING SPERM AND NON-SPERM CELLS

Sample	1	2	3	4	5	6	7	8	9	10	Overall
Sperm Count	114	137	83	165	158	295	260	150	105	141	1608
Non-sperm Count	201	69	119	126	123	129	120	192	154	247	1480
True Positive	106	128	77	159	153	279	251	132	97	131	1513
False Positive	5	3	8	14	16	14	16	8	12	13	109
False Negative	8	9	6	6	5	16	9	18	8	10	95
Precision	0.95	0.98	0.91	0.92	0.91	0.95	0.94	0.94	0.89	0.91	0.93
Recall	0.93	0.93	0.93	0.96	0.97	0.95	0.97	0.88	0.92	0.93	0.94

To evaluate the performance of the proposed segmentation network for distinguishing sperm and non-sperm impurities, ten raw semen samples with high impurities were used for analysis, in which non-sperm impurities accounted for at least 33% and up to 64% of the total identifiable objects. The number of sperm and non-sperm impurities of each tested sample were manually counted by experienced lab technicians.

In these challenging scenarios, the proposed segmentation network achieved a mean precision of 93% and a mean recall rate of 94% among the 10 samples (Table I). As for error cases, totally 109 false recognitions (false positive) and 95 missing recognitions (false negative) were found in the evaluation, accounting for 3.1% and 3.5% of total identifiable objects in all samples, with the worst case 5.7% and 5.3%, respectively. In addition, no trend of increase in precision and recall rate was found as non-sperm rate or sperm count increases. The results demonstrated that our proposed algorithm remained robust even in the case of high impurities.

The capability of our boundary-sensitive segmentation network was also qualitatively evaluated and compared to the original U-Net. For comparison purposes, the boundary sensitive network and the original U-Net were both trained on the same dataset containing 50 images (12,229 sperm head contours labeled by experienced technicians). As shown in Fig. 5, our boundary-sensitive segmentation network trained with boundary weighted map was able to obtain separated contours of sperm with close distances. While for the original U-Net, these sperm in close proximity were segmented as

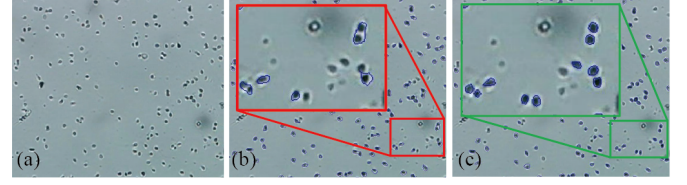


Fig. 5. An example segmentation result of the boundary sensitive segmentation network. (a) Original image of a semen sample obtained by the smartphone imaging modality; (b) Segmentation result of the original U-Net, sperm in close proximity were segmented as a whole contour; (c) Segmentation result of our proposed boundary sensitive segmentation network, sperm in close proximity were segmented as individual and separated contours.

a cluster with a large contour. Note that the goal in semen analysis is to separate, as much as possible, all sperm that are at a close distance without occlusion. The improvement in boundary sensitivity of the proposed segmentation network contributes to a lower computational complexity and a more accurate occlusion evaluation for the subsequent Occlusion-Aware JPDAF tracker.

B. Tracking Occluded (Overcrossing) Sperm

Most existing multi-sperm trackers can robustly track sperm in non-occluded scenarios (i.e., individual sperm swimming freely). Hence, we focused on evaluating the tracking performance of the proposed Occlusion-Aware JPDAF tracker (OA-JPDAF) for occluded sperm (i.e., multiple sperm crossing over and overlapping each other). The performance of OA-JPDAF was compared to that of the original JPDAF (a state-of-the-art multi-sperm tracking benchmark) and Nearest Neighbor (NN, the most commonly used tracking method for commercial CASA systems). Our boundary sensitive segmentation network was used to provide accurate sperm head contours and head centroid coordinates for all the three trackers.

To quantitatively evaluate the success rate of the three tracking algorithms, a total 329 cases of overcrossing of sperm from 8 video clips were included, with results summarized in Table II. A success case is defined as the sperm continuing to be tracked by the tracker with the same ID as before and after occlusion. The proposed OA-JPDAF showed the highest success rate and was capable of tracking 95.14% of sperm through occlusions. The original JPDAF was ranked next, with a success rate of 82.98%. While the NN algorithm performed the worst among the three, with a tracking success rate of 75.08%. It is worth noting that the achieved tracking success rate of 95.14% is for occlusion cases only. In more common samples that include both occlusion and non-occlusion cases, the overall success rate would be even higher. To further analyze tracking failure cases of each algorithm, the failure cases were classified into six categories according to the type of errors and occlusions, as detailed in the next section.

C. Analysis of Tracking Failure Cases

Depending on how the target sperm ID is affected, failure cases were divided into the six cases, as shown in Fig. 6.

(1) Extra sperm ID generation: an extra sperm ID is generated after the sperm has been occluded, and the existing sperm ID exists simultaneously with the new sperm ID. Obviously, this type of error results in a higher than actual sperm concentration. If this type of error occurs between

TABLE II
PERFORMANCE OF TRACKING OCCLUDED TARGET

Method	Overcrossing Case Count	Error Case Count						Success Case Count	Success Rate
		a	b	c	d	e	f		
Occlusion-aware JPDAF	329	6	3	4	1	0	2	313	95.14%
Original JPDAF	329	24	8	7	1	12	2	275	83.59%
Nearest Neighbor	329	16	7	36	14	5	4	247	75.08%

motile and immotile sperm (tracking error **a**), it results in a statistically higher motility rate. If it occurs between motile sperm (tracking error **b**), the motility values may either increase or decrease depending on the original speed of the two sperm. The main reason for the occurrence of type **a** error is the biased association caused by sperm occlusion. One of the targets in occlusion may associate with clutter or measurements originated from other sperm, altering the kinematic model. This causes the measurement that should have been matched to exceed the gating area of the target when the occlusion ends, and the measurement is established as a new target. Similarly, type **a** error happens when motile sperm collides with non-motile sperm, causing the non-motile sperm to move and altered the velocity vector of the motile sperm. Although it is the most frequent error in OA-JPDAF, its frequency was only 37.5% of the NN algorithm and 25% of the original JPDAF algorithm. This significant improvement was enabled by the improved data association mechanism for occlusion targets supported by occlusion awareness. When occlusion is predicted for a target and confirmed based on the size change of the contour, OA-JPDAF stops updating the state of the immotile target with new measurements until the end of occlusion (**Restriction 4** of the relaxed constraint) to ensure that the states of the motile and immotile targets do not converge due to moving together followed by collisions. Moreover, OA-JPDAF essentially alleviates biased association during occlusion by using the relaxed constraint with extra restrictions (**5**), ensuring significantly reduced tracking error when handling occluded targets.

(2) Sperm ID switch: the occluding sperm IDs are switched or exchanged after occlusion. The total number of sperm remains unchanged, and this would not cause a statistical error in sperm concentration. However, when this error occurs between motile and immotile sperm (tracking error **c**), the motility values are incorrectly increased. When the error occurs between motile sperm (tracking error **d**), although the change in motility value cannot be determined qualitatively, it changes the true trajectory of the sperm.

Type **c** and **d** errors are mainly prevalent in NN due to the fact that the data association of the NN algorithm is based on the nearest distance principle only. Therefore, sperm ID exchange often occurs when the targets are occluded. The JPDAF algorithm introduces the Mahalanobis distance and kinematic state updated by Kalman Filter to facilitate correct target-measurements matching. As a result, it can significantly reduce target ID exchange between motile and immotile sperm, thus leading to fewer type **c** and **d** errors than NN (**Table II**). However, when the motile sperm push the immotile sperm to move, their kinematic states may be still

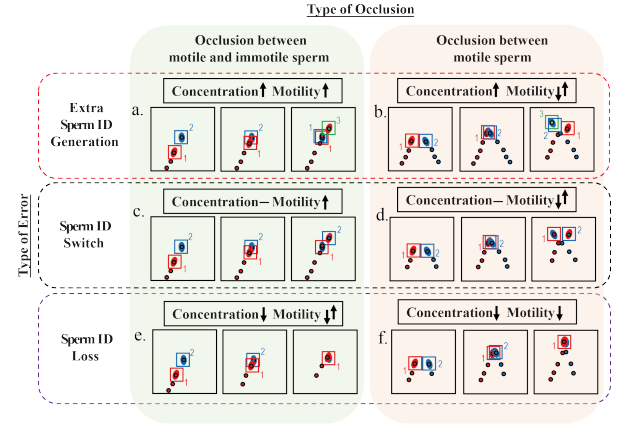


Fig. 6. Schematics of six common tracking errors when tracker handling occluded targets as in Table II. The effect of different tracking errors on concentration and motility measurement are given by $\uparrow$, $\downarrow$, $\updownarrow$, and $-$, which represents increase, decrease, indeterminable, and no effect.

prone to converge, and then sperm ID exchange still occurs in the original JPDAF. OA-JPDAF inherits the advantages of the original JPDAF while it stops updating the state vectors of stationary targets in occlusion state based on occlusion awareness (**Restriction 4** of the relaxed constraint), and alleviates biased data association by implementing an occlusion-aware based relaxed constraint (**5**), ensuring that the ID switching is further reduced. Accordingly, OA-JPDAF further reduced the occurrence of type **c** error by over 40% compared to the original JPDAF.

(3) Sperm ID loss: after occlusion, one of the sperm IDs is deleted and the two sperm share a single ID. Evidently, this type of error leads to a decrease of the sperm concentration. For this type of error occurring between motile and immotile sperm (tracking error **e**), depending on whether the disappearing ID originally belonged to a motile or immobile sperm, the motility value can be either incorrectly decreased or increased. For such errors occurring between motile sperm (tracking error **f**), the concentration will be incorrectly reduced because the percentage of motile sperm is decreased.

Type **e** and **f** errors are mainly due to the inability of the trackers data association mechanism to correctly assign a single measurement caused by occlusion to two or more sperm. For the NN algorithm, this single measurement would be assigned to any target that was closer to the measurement in the previous frame.

If the occlusion duration is long, a target has not associate with any measurements for a period of time will be deleted, leading to type **e** and **f** errors. For the original JPDAF algorithm, ideally, when only a single measurement falls into the gating regions of two occluded targets at the same time, both targets can use the same measurement for state update based on probability weighted data association. However, when targets are dense or clutters exist in the surrounding area, the gating region of the occluded target may include clutter or measurement of other targets, causing the incorrect updates of target position. If no other measurement falls into its gating region after incorrect state updates for certain period of time, it would be deleted, resulting in errors of type **e** and **f**. OA-JPDAF employs an occlusion-aware based relaxed constraint (**5**), which senses the occlusion status of

both targets and measurements, and only retains feasible joint events that are most likely corresponding to the true target-measurement matching relationship. The application of this novel data association mechanism of OA-JPDAF resulted in a 50% reduction in the number of type **f** errors relative to the original JPDAF and NN. Combining with the mechanism of stopping state updates of stationary occluded targets with measurements, this resulted in no occurrence of type **e** errors in all of the counted overcrossing cases.

D. Effects of Tracking Failure on Sperm Concentration and Motility Evaluation

To further quantify the effect of these tracking errors on sperm concentration and motility measurement, another eight videos were captured from eight participants and the results of smartphone measurement were benchmarked to manual counting and measurement results. To provide accurate manual benchmarking, experienced laboratory technicians counted the number of sperm in each video with best care and then converted to concentration and motility according to (11) and (12). Similarly, the trajectory of each sperm presented in the last frame was retrospectively reviewed by the technician to calculate sperm velocity and motility results.

For concentration (Fig. 7a), the OA-JPDAF showed the least mean error of 2.03 million/ml, followed by NN (2.90 million/ml) and original JPDAF (3.70 million/ml). All the three tracking algorithms gave a higher concentration than manual measurement. This is in agreement with the analysis of tracking failure cases in Section V.C because error type **a** and **b** increase the concentration, while type **e** and **f** decrease the concentration. All three algorithms have more type **a** and **b** errors than type **e** and **f** errors. Despite this, no significant differences were observed among the three groups. This could be because the same segmentation network was used to identify sperm and to provide head contours for all the three trackers.

As for motility (Fig. 7b), the OA-JPDAF showed a significantly lower mean motility error than original JPDAF (1.58% vs. 7.73%, $p < 0.001$) and NN (1.58% vs. 21.40%, $p < 0.001$). These data also align with the tracking error case analysis in Section V.C. Both Original JPDAF and NN had significantly more type **a** and **c** errors than OA-JPDAF. Given that type **a** and **c** errors lead to an increase in motility, the average motility obtained by Original JPDAF and NN was significantly higher than that of OA-JPDAF. These statistical results illustrate that OA-JPDAF outperformed the other two algorithms in motility measurement accuracy. Furthermore, concentration and motility results of the algorithms were highly correlated with their tracking failure cases, suggesting that tracking errors in occluded targets had a significant impact on the measurement of concentration and motility.

E. Comparison of Smartphone-Based Semen Analysis by Inexperienced End-Users with Laboratory CASA on 50 Clinical Samples

In order to fully test the performance under point-of-care scenarios, all samples for smartphone-based semen analysis were prepared by users without relevant experience, and the

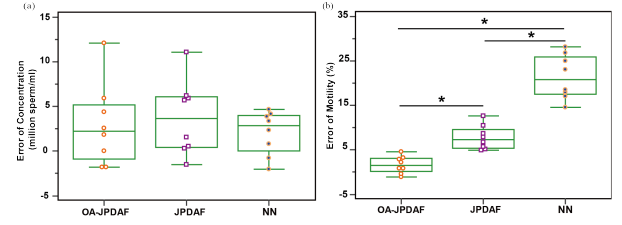


Fig. 7. Sperm concentration and motility errors of our proposed smartphone-based semen analysis system benchmarked to manual measurement, where * denotes significant difference ($p < 0.001$) is found between two groups.

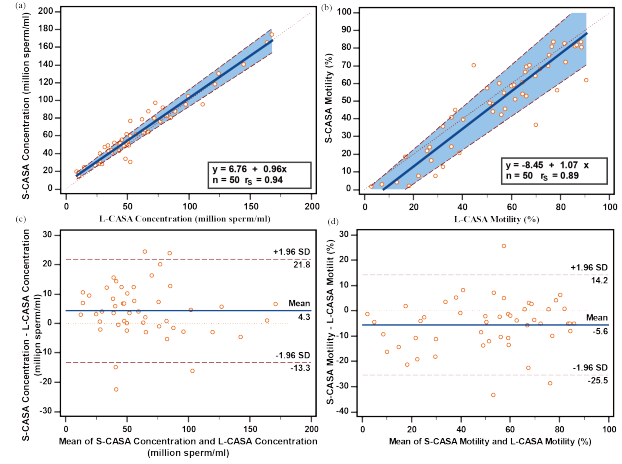


Fig. 8. Evaluation of clinical samples ($n = 50$). First row: Passing-Bablok regression for the concentration (a) and motility results (b) between the proposed smartphone-based CASA and laboratory CASA; Second row: Bland-Altman analysis comparing the sperm concentration (c) and motility (d) obtained by smartphone-based CASA and laboratory CASA

obtained results were compared to laboratory CASA. From each sample, one aliquot was tested using our smartphone-based system by an inexperienced user, while another aliquot was tested using laboratory CASA (BEION S5, BEION, China) by an experienced laboratory technician, for side by side comparison. Fifty patients were enrolled in this study.

Passing-Bablok regression ($n = 50$) was conducted to evaluate the agreement of sperm concentration (Fig. 8a) and motility (Fig. 8b) calculation between the proposed smartphone-based semen analysis system and laboratory CASA. For the regression of concentration, a slope of 0.96 (95% Confidence Interval CI: 0.90 to 1.02) and an intercept of 6.76 (95% CI: 2.64 to 9.64) were achieved. Model linearity was validated by Cusum test with $P = 0.99$. A Spearman rank correlation coefficient of 0.94 (95% CI: 0.9 to 0.97) with $P < 0.0001$ was obtained. For the regression of motility, a slope of 1.07 (95% CI: 0.95 to 1.20) and an intercept of -8.45 (95% CI: -15.95 to -1.44) were achieved. Model linearity was validated by Cusum test with $P = 0.89$. A Spearman rank correlation coefficient of 0.89 (95% CI: 0.81 to 0.94) with $P < 0.0001$ was obtained. The above results confirmed that our point-of-care semen analysis system for sperm concentration and motility calculation based on smartphone raw semen images agreed well with conventional laboratory CASA.

The Bland-Altman analysis ($n = 50$), which compared the sperm concentration obtained by smartphone-based CASA and laboratory CASA (Fig. 8c), showed a mean bias of 4.25 million sperm/ml with an SD of 8.95 million sperm/ml. The limits

of agreement ranged from 13.34 to 21.85 million sperm/ml. The Bland-Altman analysis on sperm motility between two methods (**Fig. 8d**) showed a mean bias of -5.62% with an SD of 10.13%. The limits of agreement ranged from 25.45 to 14.20. As shown in **Fig. 8c&d**, as sperm concentration or motility increased, no fixed difference or trend of error was found, suggesting the proposed method can accommodate samples with a wide range of sperm concentrations to provide accurate sperm concentration and motility measurement.

VI. CONCLUSION

This paper presented a novel semen analysis method with smartphone imaging to address the challenges of robust point-of-care sperm concentration and motility measurement. A smartphone imaging modality was developed to enable clear imaging of a fixed volume of raw semen samples. A boundary sensitive segmentation network was proposed to segment sperm contours and to distinguish sperm with non-sperm cells or impurities in raw semen. To tackle the challenging of frequent overlapping and occlusion between sperm, an Occlusion-Aware JPDAF was developed for robust tracking of multiple sperm. Comparative studies confirmed that the proposed Occlusion-Aware JPDAF outperformed state-of-the-art tracking algorithm, making significantly less tracking errors when handling occluded targets. The effect of tracking failure on sperm concentration and motility measurement was quantified, which is applicable to both point-of-care and clinical semen analysis for male infertility diagnosis.

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Revisiting Log Parsing: The Present, The Future, and The Uncertainties

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Abstract—In the recent decade, the amount of software runtime logs has increased rapidly and spawned a line of automated log analysis research using machine learning or data mining algorithms. In the typical workflow of log analysis, log parsing, which aims to transform unstructured or semi-structured logs into structured logs, is crucial to various downstream algorithms. While the state-of-the-art (SOTA) parsers achieve extremely high accuracy, recent research shows that these parsers are far from being useful under stricter evaluation metrics. Thus, researchers and practitioners are unclear about the current state of log parsing research and what might be important to explore in the future. To this end, we conduct an empirical study to revisit log parsing by running extensive experiments of SOTA parsers on sixteen widely-used log datasets under five evaluation metrics with different preprocessing settings. Our results show that the performance of log parsers varies significantly under different evaluation metrics. In addition, preprocessing plays an important role in the evaluation. In particular, preprocessing with common regular expressions can cause a 0.38 performance difference in group accuracy (GA), highlighting the importance of reporting preprocessing details in parsing research. We also generalize the word-level regular expressions in preprocessing and try to use them to parse the whole logs, which leads to surprisingly decent accuracy. These results imply that formulating log parsing as a word-level classification task is a feasible future direction. Moreover, we find out that the most widely-used dataset (i.e., LogHub) is suboptimal because it contains labeling errors. To address this issue, we make an extensive manual effort to fix the errors in the log dataset, providing a revised Ground Truth for future log parsing research. On the revised log dataset, our simple parser (word-level regex-based) achieves 0.97 Precision-template accuracy (PTA) on the Spark dataset and an average Recall-template accuracy (RTA) of 0.93 on 16 datasets, which outperforms all existing parsers. We gathered log data from actual companies and evaluated all parsers. Our regex-based parser achieved a PTA value of 0.9565, surpassing other parsers by 0.7642.

Index Terms—Log parsing, Log analysis, Empirical study, Dataset

I. INTRODUCTION

SOFTWARE logs are a chronological collection of specific system operations and their operation outcomes. Logs are generated by logging statements written by developers in the source code, and thus, logs often contain structured information (e.g., timestamp) and unstructured natural language

text. As the system generates much more logs, managing and analyzing them becomes increasingly challenging. To address this problem, recent research has focused on designing automated log management and analysis approaches.

Log parsing is an essential aspect of system reliability engineering. It helps practitioners identify and troubleshoot system failures by analyzing log files generated by software systems. These log files contain a wealth of information that can provide insights into system behavior, performance, and errors. Previous research has highlighted that minor parsing errors can have a significant impact on the efficacy of downstream processes [1]. Furthermore, IBM utilizes log analysis for monitoring network outages¹. Therefore, accurate and efficient log parsing is critical for ensuring system reliability. In this context, the study of log parsing methods, their accuracy, and efficiency is an important area of research within the broader field of reliability engineering. The use of log parsing tools and techniques can help reduce the time and effort required to diagnose system issues, thereby improving system reliability. Furthermore, as software systems become increasingly complex and larger volumes of data are generated, log parsing is becoming even more critical for ensuring system reliability. In summary, log parsing is an integral part of reliability engineering, and the study of log parsing accuracy and efficiency is crucial for improving system reliability. The paper's contribution to the field lies in evaluating the performance of various log parsing methods and providing insights into how these methods can be improved to enhance system reliability.

In the workflow of log management and analysis, log parsing, which transforms unstructured or semi-structured logs into structured logs, is the first and foremost step. Typically, a log message contains constants, which are tokens written by developers in logging statements (e.g., a description of system operation), and variables, which exhibit different values during runtime (e.g., timestamps and program states). A constant, which is the same for each occurrence of the event, displays the event template for the log message, while a variable includes dynamic information that can change depending on the circumstances of certain events. The output of a log parser is a log event template and a list of structured log messages. Fig. 1 shows an example of log parsing. Date, Time, User, Component, PID, Content, EventId, and Event Template are all included in the parsed log message from the Mac dataset. The variables are converted into user-defined symbols (e.g., ARPT:

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¹<https://developer.ibm.com/blogs/how-mining-log-templates-can-help-ai-ops-in-cloud-scale-data-centers/>

TABLE I
SUMMARY OF LOG DATASETS.

Dataset	Time Span	Data Size	Messages	Templates (total)	Templates (2k)
HDFS	38.7 hours	1.47 GB	11,175,629	30	14
Hadoop	N.A	48.61 MB	394,308	298	114
Spark	N.A	2.75 GB	33,236,604	456	36
ZooKeeper	26.7 days	9.95 MB	74,380	95	50
OpenStack	N.A.	60.01 MB	207,820	51	43
BGL	214.7 days	708.76 MB	4,747,963	619	120
HPC	N.A.	32.00 MB	433,489	104	46
Thunderbird	244 days	29.60 GB	211,212,192	4040	149
Windows	226.7 days	26.09 GB	114,608,388	4833	50
Linux	263.9 days	2.25 MB	25,567	488	118
Mac	7.0 days	16.09 MB	117,283	2,214	341
Android	N.A.	3.38 GB	30,348,042	76,923	166
HealthApp	10.5 days	22.44 MB	253,395	220	75
Apache	263.9 days	4.90 MB	56,481	44	6
OpenSSH	28.4 days	70.02 MB	655,146	62	27
Proxifier	N.A.	2.42 MB	21,329	9	8

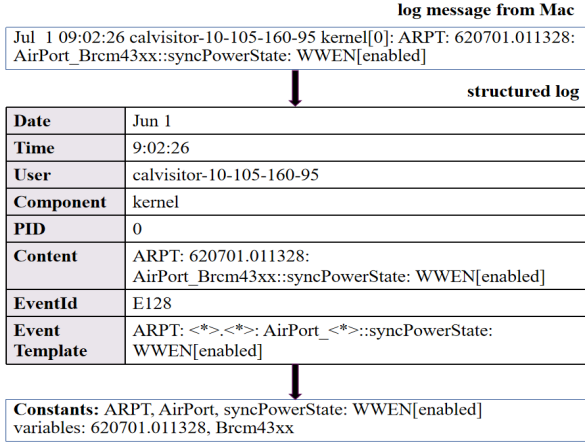


Fig. 1. An example of log parsing.

620701.011328 is converted into ARPT: <*>). The benefit of log parsing is that the structured logs and templates can be easily adopted by subsequent tasks such as log processing, analysis, and querying, which largely facilitates the whole workflow.

At present, there are two main trends in research for log parsing: (1) design a novel log parser that achieves SOTA accuracy, such as Drain [2], MoLFI [3], and LogClass [4]; (2) propose a new evaluation metric and claim that the current one is not good enough (such as the paper proposed by Khan et al. [5], Logram [6] and Uniparser [7]). Zhu et al. performed a thorough examination of automated log parsers and then published the results as reusable benchmarks [8]. More specifically, the authors tested thirteen log parsers against sixteen log datasets from distributed systems, supercomputers, operating systems, mobile systems, server applications, and standalone

software. Khan et al. [5] assessed log parsers in terms of metrics based on this research. The authors believed that these metrics have a few flaws, Khan et al. evaluated the properties of existing metrics and developed their own (PTA and RTA) for a more thorough evaluation of log parsers. The outcomes of all parsers considerably declined in these rigorous metrics testing. Thus, the authors indicated that log parsers are still far from perfect and there are a lot of opportunities for further development. Moreover, the authors discussed possible correlations between various metrics. In addition, the authors mention that the dataset (i.e., event templates and structured logs) might contain labeling errors (e.g., a wrong event template).

These existing papers have largely facilitated the development of log analysis research. However, some important issues have not been well explored or addressed. First, to the best of our knowledge, none of the existing papers systematically explored the impact of preprocessing on parsing accuracy, which we argue is important. Second, most of the papers [9] in this field formalize log parsing as a clustering problem that clusters log messages with the same event template into the same group (log message-wise). Meanwhile, a few recent papers [10] start to regard log parsing as a classification problem that classifies each token in a log message as either a *constant* or a *variable* (token-wise). While log message-wise approaches outperform manual approaches that construct regular expressions (regex) for every possible log message, it is unclear whether regex can be extensively used in token-wise approaches. Third, none of the existing papers systematically fix the labeling errors in LogHub [11], the most widely-adopted dataset in this field, which could lead to the unreliability of the results reported by many log parsing papers [7].

To fill these significant gaps, in this work, we conduct an empirical study to revisit log parsing by running extensive

experiments of thirteen well-known parsers on sixteen widely-used datasets under five evaluation metrics. The results reveal that preprocessing has a significant impact on parsing accuracy (e.g., 0.38 performance difference in group accuracy), highlighting the importance of reporting preprocessing details in this field. In addition, we construct a token-wise toy parser that consists of only 23 simple token-level regexes, which achieve surprisingly decent performance on most of the datasets. We believe these results demonstrate the potential of the idea that regards log parsing as a classification problem and leverage token-level regexes. The parsing results have been greatly improved under all metrics, especially PTA and RTA. Moreover, we invested a huge amount of manual effort (21 man-months) to manually revisit the labels in LogHub datasets and spotted 587 templates errors that affect over 14,200 log messages. The revised datasets have been merged with LogHub in its newest release.

The evaluation metrics and log parsing techniques proposed in the paper can help improve the reliability of systems that generate large amounts of log data in several ways. First, these techniques can help identify and diagnose errors, warnings, and other issues in log data more accurately and quickly, allowing system administrators and developers to address them before they become more serious problems. Second, by improving the accuracy and efficiency of log parsing, these techniques can help reduce the amount of time and effort required to analyze log data, enabling faster and more effective troubleshooting and maintenance of systems. Finally, by enabling better log data analysis and management, these techniques can help improve overall system performance and reliability by identifying patterns and trends in log data that may indicate emerging issues or areas for optimization. Overall, the use of effective log parsing techniques and evaluation metrics can help organizations better understand and improve the reliability of their systems, leading to increased uptime, reduced downtime, and improved user satisfaction. The main contributions of this paper are as follows:

- It is the first empirical study that systematically explores the impact of preprocessing in log parsing.
- It studies a simple log parser that consists of 23 token-level regexes, which outperforms the current popular parsers in all metrics, demonstrating the potential of token-level parsers with regexes.
- It provides a brand-new version of sixteen widely-used log datasets with extensive manual effort, which provide a better benchmark for log parsing research.²

The paper has the following structure: Section 1 (Introduction) explains the importance of log parsing, the problems it solves, and the scope and main contributions of the paper. Section 2 (Related work) presents the log parsers to be evaluated in the article, including their methods and characteristics. Section 3 (Present) presents the experimental data, the effect of preprocessing on log parsing, evaluation metrics, and the results of our experiments on parsing errors. Section 4 (Future) describes how we modified the original ground truth and designed a regex-based parser, followed by the results of our

experiments to verify their effectiveness. Section 5 (Uncertainties) discusses uncertainties in the modification process and experiments. Section 6 (Threats) identifies open issues for log parsing technology. Finally, Section 7 (Conclusions) concludes the paper and suggests future work directions.

II. RELATED WORK

In this section, we talk about different ways to parse logs, such as fixed time windows, sliding time windows, session windows, and other methods. Log parsing is important for log analysis and it can find related event templates from unstructured logs.

A. Log parsing

Log parsing is the process of extracting related event templates from unstructured logs, each of which is made up of many parameters and serves as the foundation for feature extraction. Fixed time windows, sliding time windows, session windows, and other log segmentation methods are now employed due to the exploding number of logs. The detection of these logs is conducted by segmentation into many portions or by sampling [12]. In this paper, we chose several parsers that are currently popular and commonly used. They are SLCT [13], AEL [14], IPLoM [15], LKE [16], LFA [17], LogSig [18], SHISO [19], LogCluster [20], LenMa [21], LogMine [22], Spell [23], Drain [2], MoLFI [3].

The parsers used in the study have different algorithms and methods for log parsing. LogMine is a parser that extracts structured information from logs by using regular expressions and heuristics. It is capable of handling a wide range of log formats and can automatically learn new patterns from unlabeled data. Drain is a parser that groups similar log messages together using a log clustering algorithm and then applies a set of predefined templates to extract structured information from each cluster.

Spell is a parser that learns the structure of log messages using a probabilistic model and applies this model to parse new logs. It is designed to handle noisy and heterogeneous logs with varying levels of structure. AEL is a parser that combines multiple parsing techniques, including regular expressions, decision trees, and support vector machines, using an ensemble learning approach. It is designed to handle complex logs with multiple levels of structure. SLCT uses a frequency-based approach to identify constant and variable parts of log messages. It has low computational complexity but may generate inaccurate templates for logs with low frequency or variable length. IPLoM uses a partitioning-based approach to split log messages into groups based on length, token position, and token frequency. It can handle high-dimensional logs but may produce redundant or incomplete templates for complex logs.

MoLFI uses an evolutionary-based approach to finding the optimal set of templates for log messages based on multiple objectives. It can handle complex and diverse logs but may require high computational complexity and parameter tuning. LKE uses a clustering-based approach to group log messages based on string similarity and extract templates using longest

²<https://github.com/logpai/logparser/tree/master/logsrevised>

common substrings. It can handle noisy logs but may require manual intervention to merge clusters or refine templates. LFA uses a factorization-based approach to decompose log messages into a dictionary of words and a matrix of coefficients. It can handle large-scale logs but may be affected by noise or outliers in the log data.

LogSig uses a signature-based approach to generate unique identifiers for log messages based on their punctuation and length features. It can handle online log parsing but may fail to capture semantic similarities or differences among log messages. SHISO uses a self-supervised learning approach to train a neural network to generate templates for log messages. It can handle complex and diverse logs but may require a large amount of training data and computational resources. LogCluster uses a density-based approach to cluster log messages based on word frequency and extract templates using the longest common substrings. It can handle variable-length and high-dimensional logs but may be sensitive to parameter settings and noise in the log data. LenMa uses a length-based approach to split log messages into groups based on length and extract templates using the longest common substrings. It can handle variable-length logs but may produce inaccurate templates for logs with similar lengths but different structures.

Our proposed parser uses a combination of data-based features and rules to extract structured information from logs. It is flexible and adaptable to different log formats and can be trained on labeled data. The study evaluated these parsers on multiple datasets using various metrics to identify their strengths and weaknesses and to gain insights into their relative performance.

Furthermore, there have been other proposed log parsers. Yu et al. [24] propose an approach and system framework to use documentation knowledge for log parsing. It can improve the parsing accuracy and discover the linkages between event templates. Rand et al. [25] explore the possibility of automating the parsing task by employing machine translation. It creates a tool that generates synthetic Apache log records and trains recurrent neural network-based models. It shows that the models can learn Apache log format and parse individual log records. Fastlogsim [26] realizes fast log parsing by calculating text similarity and merging similar templates. The paper [27] is mainly about the execution log of two or more systems, outputting a model of their differences. LogHound [28] mainly uses a frequent itemset mining algorithm to discover frequent patterns from event logs. To summarize, current log parsing technology is relatively mature, primarily through the definition of rules, code analysis, and machine learning methods to achieve rapid parsing of various log structures, removing irrelevant content, extracting key fields of various events, and storing information for later analysis.

B. Log Analysis

Log analysis is also commonly used for anomaly identification. The use of artificial intelligence systems to automatically examine system logs to uncover and diagnose errors is at the heart of log anomaly detection. A lot of studies have been done on existing log-based anomaly detection algorithms [29],

[30], [31]. Manually inspecting logs to discover abnormalities becomes infeasible as the volume and complexity of existing systems grow. As a result, the necessity for automatic anomaly detection for log data is becoming increasingly critical, and several solutions in related disciplines have been presented [29], [32], [33]. The use of artificial intelligence systems to automatically examine system logs to uncover and diagnose errors is at the heart of log anomaly detection. Machine learning-related technologies have exploded in popularity, resulting in widespread adoption. LogClass is a data-driven system for detecting and categorizing anomalies in device logs [4]. LogClass builds bag-of-words vectors, then trains anomaly detection and classification models using the PU learning approach and SVM, and lastly conducts online anomaly detection. Semi-supervised algorithms work well with labeled data that is entirely normal. Yang et al. researched this [34]. Anomaly detection techniques based on deep learning are becoming increasingly popular, and their use in a range of problem sets outperforms previous methods [35]. LADRA [36] is an abnormality detection tool based on GRNN neural network model.

III. THE PRESENT

In this section, we discuss the experimental data, how preprocessing affects log parsing, how we measure the results, and what we found about parsing errors.

A. Datasets

The data format is versatile and adaptable. For parser evaluation, selecting adequate datasets is critical. We use the datasets and benchmarks gathered and proposed by Zhu et al. [2] in this study. All the log datasets are presented in Table I. Time duration, data size, message count, and template count are all included in the content. The log datasets come from a variety of sources, including distributed systems, operating systems, and mobile devices. HDFS [37] is a distributed file system under Hadoop. It is the lowest-level distributed storage service and is one of Hadoop's essential components. It's a storage system that connects several computers. Hadoop [38] is an Apache open-source Java framework for distributed data processing across clusters of computers utilizing a simple programming model. Spark [39] is an open-source platform for interactive inquiry, machine learning, and real-time workloads. It does not have own storage system and will instead rely on HDFS and other storage systems. Spark processes data in RAM utilizing a concept known as a Resilient Distributed Dataset, while Hadoop reads and writes files to HDFS. BGL [40] is a powerful supercomputer led by IBM. MacOS [41] is a graphical user interface-based operating system launched by Apple and is the main operating system for Macintosh series computers. Microsoft Windows [42] is a family of proprietary commercial software operating systems with a graphical user interface developed by Microsoft. HealthApp³ is a mobile application that collects and manages personal health information. Proxifier [11] is a sophisticated socks5

³<https://github.com/logpai/loghub/tree/master/HealthApp>

client that allows network programs that don't support proxy servers to pass HTTPS or SOCKS proxies or proxy chains. Data from different systems can better assess the generality of the parsers, and the outcomes of the experiments are also more persuasive.

B. Preprocessing

The following are the primary goals of log analysis data preprocessing: (1) Filter out the non-compliant data and remove the meaningless data. (2) Regularization and format conversion. (3) Filter and segregate the various data columns according to the later use requirements.

In this research, we use a log data preparation approach. To make the parsers' processing easier, we first filter the data, then create rules to replace parameters in log data with regular expressions. To further understand the impact of preparation on log parsing, we go through three steps.

- We evaluate the selected log parsers without using any preprocessing algorithms.
- We filter the data without using regular expression techniques.
- We use regular expressions after filtering all log data.

Data filtering. As shown in Fig. 1, there are generally multiple columns in log data, such as Date, Time, ID, and Component. Templates, on the other hand, are just concerned with the content. The content element is crucial for log parsers, while the other parts are less important. We only preserve the content column in this section and delete the other columns.

Regular expression. Regular expressions are string-matching patterns that can be used to check if a string contains a specific substring, replace a matching substring, extract a substring that meets a specific condition from a string, and so on. As a result, we use the symbol $\langle * \rangle$ to replace the specified strings. We developed the following broad string selection principles based on our observations of experimental data:

- All strings with numbers in them (date, IP, numbers, hexadecimal numbers etc.).
- URLs.
- Strings to the right of the equal sign.
- Paths.
- String after special nouns (e.g., user, admin)

The examples of rules can be found in Table II. The broad guidelines, however, cannot cover all data.

TABLE II
EXAMPLES OF RULES.

Log data	Regular expression
Register: 0x0002f900	Register: $\langle * \rangle$
URL: https://tools.google.com/service/update2?	URL: $\langle * \rangle$
productID= com.google.Keystone	productID= $\langle * \rangle$
file /home/pakin1/sweep3d-2.2b/results/random1-8x32x32x2.map	file $\langle * \rangle$
user admin	user $\langle * \rangle$

In this paper, we choose 13 existing log parsers. Table III shows the effective log parsing tools. Some of these methods, such as Drain and Spell, use preprocessing, while others do not. If a log parsing technique includes a pretreatment phase, we mark it with a $\checkmark$, and if it does not, we mark it with an $\times$. These parsers cover a wide range of methodologies and can represent the current industry standard, making them useful research tools. Other research work [2] [5] have also utilised these log parsers. A log keeps track of a series of events in a system or application. A record usually represents the content of an event. The time, source, tags, and information are often contained in log records, but they are not the only ones. In most cases, log parsing entails evaluating log records, deleting unnecessary information, and then understanding the log content using related methods.

TABLE III
SUMMARY OF AUTOMATED LOG PARSERS.

Parser	Year	Preprocessing
SLCT	2003	$\times$
AEL	2008	$\checkmark$
IPLoM	2012	$\times$
LKE	2012	$\checkmark$
LFA	2010	$\times$
LogSig	2011	$\times$
SHISO	2013	$\times$
LogCluster	2015	$\times$
LenMa	2016	$\times$
LogMine	2016	$\checkmark$
Spell	2016	$\checkmark$
Drain	2017	$\checkmark$
MoLFI	2018	$\checkmark$

C. Evaluation Metrics

Different definitions for correctly processed log messages and templates are used in the existing accuracy measures. The following are the metrics that were picked to address the issue in the section.

Grouping Accuracy (GA). The GA metric [5] is used to assess the accuracy of log template identification, we follow the name proposed by Khan et al. It's calculated as the proportion of correctly parsed log messages to total log messages. Each log message has an event template after it has been parsed, which corresponds to a set of messages with the same template [2]. The GA metric does not consider whether the identified templates are the same as the ground truths but only considers if they are grouped correctly.

Parsing accuracy (PA). The PA metric [8] is similar to GA in that it measures the proportion of correctly processed log messages to total log messages. It is more stringent than GA because it takes into account static text and variables. In Table IV, we offer an example of a comparison of different metrics. The sample set of logs messages are $M = m_1, m_2, m_3, \dots, m_n$. The obtained templates after parsing are represented

as $T = t_1, t_2, t_3, \dots, t_n$, and the corresponding templates of the ground truth are $G = g_1, g_2, g_3, \dots, g_n$. We use $\checkmark$ if the parsing result is determined to be correct; otherwise, we use $\times$. The metric GA of m_4 and m_8 are correct, while PA are wrong. We can deduce that if $PA=1$, then $GA=1$, but not the other way around. GA gets a 4/7 because five out of seven messages (m_1, m_2, m_3, m_7) can be correctly parsed under the GA criterion; PA gets a 3/7 because four out of seven messages (m_1, m_2, m_3) can be correctly parsed under the PA criterion.

Template Accuracy (TA) Metrics. To test the successfully parsed templates considerably more carefully, the TA metrics [5] are proposed. These are the most stringent measures in comparison to the others. There are more variables to consider. Khan et al. introduce two TA metrics: **Precision-TA (PTA)** and **Recall-TA (RTA)**. On the basis of the metrics, we add one more metric **F1 score-TA (F1TA)** which is dependent on the metrics PTA and RTA and can be calculated by them. The TA metrics not only consider the correctness of template classification but also the correctness of template content. In Table IV, the correctly identified templates are t_1 and t_2 , then the PTA can be calculated as 2/5. The RTA also can be calculated as 2/5 because two templates (g_1, g_2) out of five are correct. There are variances between the metrics, as can be shown.

$$F_1TA = \frac{2PTA * RTA}{PTA + RTA} = \frac{2TP}{2TP + FP + FN} \quad (1)$$

In this paper, we selected five evaluation metrics, namely PA, GA, PTA, RTA, and FTA, which are commonly used in previous log parsing research. Each metric has unique characteristics, as well as its own strengths and weaknesses. GA is a straightforward metric that measures how well a log parser groups or clusters log messages. It is useful for understanding log structures and detecting potential issues or anomalies. It is particularly helpful for evaluating parsers that cluster or group log messages. However, it only considers the grouping of log messages and not their content, which may result in inaccurate results when identifying specific log events or issues. It also assumes that the ground truth labeling or clustering is correct and may not be sensitive to differences in the distribution or prevalence of different log categories or clusters, which can impact its performance in real-world situations.

PA is a commonly used metric in log parsing that measures the proportion of accurately parsed messages in a log file. Its strengths include being simple to understand and widely used in research, making it useful for comparing parser performance. However, it has limitations, such as not considering the extracted message's content, being sensitive to the quality of the ground truth labels, and not accounting for parsing performance on a per-line basis. As such, it should be used alongside other metrics and qualitative analysis to fully assess a parser's performance.

TA is a metric that evaluates a log parser's ability to identify and extract log message templates based on their similarity to the ground truth templates. It is particularly useful for parsers using template-based approaches and provides insights into

the parser's ability to generalize across different log files. However, it requires a ground truth template set, which may not always be available, and does not evaluate the parsing accuracy of log messages that do not match any ground truth template. Additionally, it may not accurately reflect the quality of the log parser's output in terms of identifying specific log events or issues, so it should be used in conjunction with other metrics and qualitative analysis for a complete evaluation of the parser's performance.

Overall, GA, PA, and TA are commonly used metrics for evaluating the performance of log parsers because they provide different perspectives on the performance of a parser in different aspects of log parsing. By evaluating parsers using these three metrics, we can gain a comprehensive understanding of the performance of the parser in different aspects of log parsing, including message template extraction, parsing accuracy, and grouping accuracy.

D. Experiments

We use 13 existing log parsing approaches on 16 benchmark datasets in this part to conduct various experiments on the difficulties outlined above. The benchmarking results are presented in terms of three research questions.

- **RQ1:** How do the existing approaches perform on different metrics?
- **RQ2:** How likely would preprocessing improve the performance?
- **RQ3:** How do the existing approaches perform under different preprocessing conditions?

To answer RQ1, we compare and contrast the 13 log parsers on 16 datasets using various metrics (GA, PA, PTA, RTA, and F1TA). The test environment and equipment types employed in each parser's evaluation process are the same. The findings of each parser were averaged across all datasets and are displayed in the first column of Table V. Since different assessment measures have distinct criteria, the outcomes of the same parser under various metrics are quite different. Table V shows experimental findings ranging from 0.07 to 0.87. The Drain parser receives the highest score and the LogSig parser receives the lowest score in the GA metric; however, this does not imply that the same is true for other metrics. The SLCT and Drain parsers receive the greatest PA scores, while the MoLFI parser receives the lowest. Actually, the experimental findings of the same parser on the same dataset vary dramatically. We used Drain as an example and checked each measure on the HDFS dataset; the result of GA is 1, but all other metrics are 0. Furthermore, the results of multiple datasets for the same metric indicate significant differences. Drain, for example, is used to analyze all datasets using the GA measure. Although Apache's result is 1, Proxifier's GA score is only 0.51. Despite the fact that the ratings vary greatly, the Drain parser consistently ranks first in each category.

Since different assessment measures have distinct criteria, the outcomes of the same parser under various metrics are quite different. This means that some parsers may perform well on one metric but poorly on another metric. As a result,

TABLE IV
AN EXAMPLE OF COMPARISON OF DIFFERENT METRICS.

Message	Template	Ground truth	GA	PA
(m_1) PacketResponder 1	(t_1) PacketResponder <*>	(g_1) PacketResponder <*>	✓	✓
(m_2) PacketResponder 0	(t_1) PacketResponder <*>	(g_1) PacketResponder <*>	✓	✓
(m_3) blk-12330058	(t_2) blk- <*>	(g_2) blk- <*>	✓	✓
(m_4) blk-8736461	(t_3) <*>	(g_2) blk- <*>	✗	✗
(m_5) updated: 10:50010	(t_4) updated: <*>	(g_3) updated: <*>: <*>	✗	✗
(m_6) updated: 10:50010:46509	(t_4) updated: <*>	(g_4) updated: <*>: <*>: <*>	✗	✗
(m_7) route = 0x0	(t_5) <*>= 0x0	(g_5) route = <*>	✓	✗

TABLE V
RESULTS OF LOG PARSERS ON DIFFERENT DATASETS.

	Original parsing results					Parsing without preprocessing					Parsing with data filtering					Parsing with regular expressions				
Parser	GA	PA	PTA	RTA	FITA	GA	PA	PTA	RTA	FITA	GA	PA	PTA	RTA	FITA	GA	PA	PTA	RTA	FITA
SLCT	0.64	0.29	0.22	0.16	0.19	0.64	0.29	0.22	0.16	0.19	0.64	0.30	0.22	0.16	0.19	0.64	0.31	0.23	0.18	0.20
AEL	0.75	0.25	0.25	0.27	0.26	0.72	0.21	0.21	0.23	0.22	0.75	0.26	0.25	0.27	0.26	0.75	0.25	0.25	0.27	0.26
IPLoM	0.77	0.19	0.16	0.15	0.15	0.77	0.19	0.16	0.15	0.15	0.75	0.19	0.16	0.16	0.16	0.73	0.10	0.11	0.10	0.11
LKE	0.56	0.12	0.12	0.14	0.13	0.34	0.06	0.09	0.07	0.07	0.34	0.06	0.09	0.06	0.06	0.53	0.08	0.16	0.11	0.13
LFA	0.65	0.20	0.14	0.14	0.14	0.65	0.20	0.14	0.14	0.14	0.65	0.21	0.14	0.15	0.14	0.65	0.16	0.16	0.17	0.16
SHISO	0.68	0.13	0.16	0.17	0.16	0.68	0.13	0.16	0.17	0.16	0.68	0.13	0.16	0.17	0.16	0.68	0.13	0.16	0.17	0.16
LogSig	0.48	0.12	0.08	0.07	0.07	0.48	0.12	0.08	0.07	0.07	0.48	0.12	0.08	0.07	0.07	0.53	0.15	0.10	0.08	0.09
LogCluster	0.67	0.14	0.11	0.22	0.15	0.67	0.14	0.11	0.22	0.15	0.67	0.15	0.11	0.22	0.15	0.67	0.18	0.15	0.22	0.18
LenMa	0.72	0.19	0.17	0.23	0.20	0.72	0.19	0.17	0.23	0.20	0.72	0.19	0.17	0.23	0.20	0.83	0.24	0.30	0.30	0.29
LogMine	0.69	0.22	0.16	0.20	0.18	0.31	0.15	0.05	0.23	0.07	0.31	0.15	0.05	0.23	0.07	0.69	0.22	0.16	0.20	0.18
Spell	0.75	0.19	0.15	0.17	0.16	0.70	0.14	0.14	0.14	0.13	0.75	0.19	0.15	0.17	0.16	0.75	0.19	0.15	0.17	0.16
Drain	0.87	0.29	0.27	0.29	0.28	0.87	0.25	0.25	0.26	0.25	0.87	0.25	0.25	0.26	0.25	0.87	0.29	0.27	0.29	0.28
MoLFI	0.61	0.11	0.09	0.09	0.09	0.63	0.08	0.09	0.11	0.09	0.60	0.09	0.08	0.10	0.09	0.59	0.11	0.09	0.09	0.09

we observed that there is significant variation in parser performance under different evaluation metrics. Therefore, while some parsers may perform well overall, their performance may be inconsistent across different metrics. This highlights the importance of carefully selecting appropriate evaluation metrics when comparing log parsers and interpreting their results.

Summary The results of some metrics are very inconsistent. They give different results when used on different types of data.

To answer RQ2 and RQ3, in this section, we look at how preprocessing affects parsers in this part. The tests are separated into four parts, as previously stated: (1) All parsers with no preprocessing, (2) All parsers with only data filter but no regular expressions, and (3) All parsers with both data filter and regular expressions. The experimental results are displayed in the second, third, and fourth columns of Table V respectively. We notice that there is a very small gap between the results in the second and third columns. Many of the outcomes are nearly identical. It can therefore be deduced that data filtering has little impact on parsers. On the other hand, the obvious difference reflected in the third and fourth columns

indicates that regular expressions have a certain impact on the parsers. Preprocessing has varying effects on different parsers. We highlight some obvious examples in Table V. Details of them are described in the following paragraph.

The LogMine parser itself comes with preprocessing. When the preprocessing part is deleted, the results drop significantly. The GA score dropped from 0.69 to 0.31; the PA score changed from 0.22 to 0.15; the FITA score dropped from 0.18 to 0.07; the scores of other metrics fall to some extent. The LKE parser behaves similarly. It is assumed that these two parsers both rely on regular expressions to some extent. Preprocessing can also greatly enhance the outcomes of some other parsers, such as the LenMa parser. The GA score went up from 0.72 to 0.83; the PA score went up from 0.19 to 0.24; the FITA score went up from 0.20 to 0.29. However, not all parsers benefit from preprocessing. The impact is negative for the IPLoM parser. When preprocessing is used, the GA score drops from 0.77 to 0.75. The construction of regular expressions does not encompass all data; the exceptions are just disregarded. If the data formats and contents are consistent, preprocessing may have good consequences, and general regular expressions can cover all or most of the data; on the other hand, if the data formats and contents differ, and numerous special circumstances are present, preprocessing

TABLE VI
RESULTS OF PARSERS ON DIFFERENT TEMPLATE FORMS. "P" REFERS TO "PARAMETER".

Parser	Start with P	End with P	continuous P	Overall accuracy
IPLoM	5.99% (230/3837)	14.18% (2345/16,543)	1.71% (11/643)	18.58% (5944/32,000)
AEL	6.10% (234/3837)	18.27% (3022/16,543)	3.11% (20/643)	25.72% (8229/32,000)
Spell	5.99% (230/3837)	14.18% (2158/16,543)	0% (0/643)	18.30% (5855/32,000)
Drain	6.18% (237/3837)	20.81% (3443/16,543)	24.41% (157/643)	29.18% (9388/32,000)
LenMa	0.10% (4/3837)	3.27% (541/16,543)	3.27% (21/643)	17.68% (5658/32,000)
LFA	5.99% (230/3837)	16.33% (2701/16,543)	0.47% (3/643)	19.51% (6243/32,000)
LKE	0% (0/3837)	0% (0/16,543)	0% (0/643)	8.63% (2762/32,000)
LogMine	6.07% (233/3837)	13.37% (2211/16,543)	20.99% (135/643)	20.34% (6509/32,000)
LogSig	0% (0/3837)	0% (0/16,543)	0% (0/643)	12.67% (4054/32,000)
MoLFI	0% (0/3837)	0% (0/16,543)	0% (0/643)	9.04% (2892/32,000)
SHISO	0.10% (4/3837)	2.13% (353/16,543)	1.71% (11/643)	12.72% (4069/32,000)
SLCT	4.80% (184/3837)	18.43% (3049/16,543)	3.58% (23/643)	30.31% (9700/32,000)
LenMA	0.10% (4/3837)	3.27% (541/16,543)	3.27% (21/643)	17.68% (5658/32,000)

may have negative impacts.

The reason for this could be that some log messages contain complex or irregular patterns that are difficult to capture using regular expressions or other preprocessing techniques. In such cases, the parser may not be able to accurately extract all the relevant information from the log message, even with preprocessing. Even if a preprocessing technique is designed for a specific log format, errors can occur during the preprocessing stage. For example, if the preprocessing technique relies on a regular expression to extract relevant information, the expression may not match all instances of the pattern in the log data, resulting in incomplete or inaccurate preprocessing. Another possible reason is that some parsers may already have built-in preprocessing techniques that are optimized for certain types of log messages, such as Drain. In such cases, additional preprocessing may not provide any significant improvement in performance. Apart from the dimension of accuracy, preprocessing can initially filter the data, decreasing the load on the later parser, reducing runtime, and therefore improving the parser's efficiency.

Summary (1) Preprocessing usually helps most parsers, it can boost the GA value by up to 0.38 and the PA value by up to 0.07. But there are some exceptions where it does not work. (2) Regular expressions are mainly used in preprocessing, but data filtering is not.

E. Incorrect Parser Results

The trials mentioned above show that the outcomes for several metrics are not optimal. The outcomes of the experiment have been examined. We evaluate the outcomes of the experiment in an effort to identify the problem and its cause. The current parsers do a good job of classifying log templates, but it needs to parse log templates better. We choose several parsers for analysis. In parallel, a number of

template forms are chosen to gauge the parser's effectiveness. The results under examination are (1) The proper parsing rate (percentage of correctly parsed templates) of templates that start with a parameter, (2) The proper parsing rate of templates that end with a parameter, (3) The proper parsing rate of templates that contain continuous parameters, (4) The size of the templates' parameters list, and (5) The average distance between parameters in templates. We examine the parsers using 16 datasets and show the summarized results in Table VI and Fig 2.

When parsing templates with successive parameters, most parsers are ineffective, except Drain and LogMine. Most parsing templates with successive parameters have a fixed order of parameters, which makes them ineffective in handling the high variability and complexity of log messages. In other words, the templates require a certain order of parameters to extract information, but log messages often have inconsistent or varied orders of parameters. This can result in parsing errors and make it difficult to extract structured information from logs. However, Drain and LogMine are effective in handling this issue because they use log clustering and regular expressions, respectively, to extract structured information from logs. They do not rely on a fixed order of parameters and can handle logs with varying parameter orders. Additionally, the Proxifier dataset basically starts with "c.baidu.com:80" and "proxy.cse.cuhk.edu.hk:5070", making it very difficult for IPLoM to recognize them, which results in very low performance.

Diverse datasets produce wildly different parsing results. AEL, Drain and LenMa score 100% on the Android dataset using the "Start with parameter" metric, while other datasets do not perform as well. This is most likely caused by the quality and format of the data. Fig 2 shows the analytical outcomes for different log message categories: for the number of tokens, we define six ranges, which are 1-2, 3-4, 5-6, 7-8, 9-10, and more than 11; for the number of parameters, the ranges are 0, 1, 2, 3, 4, and over 5; for the average

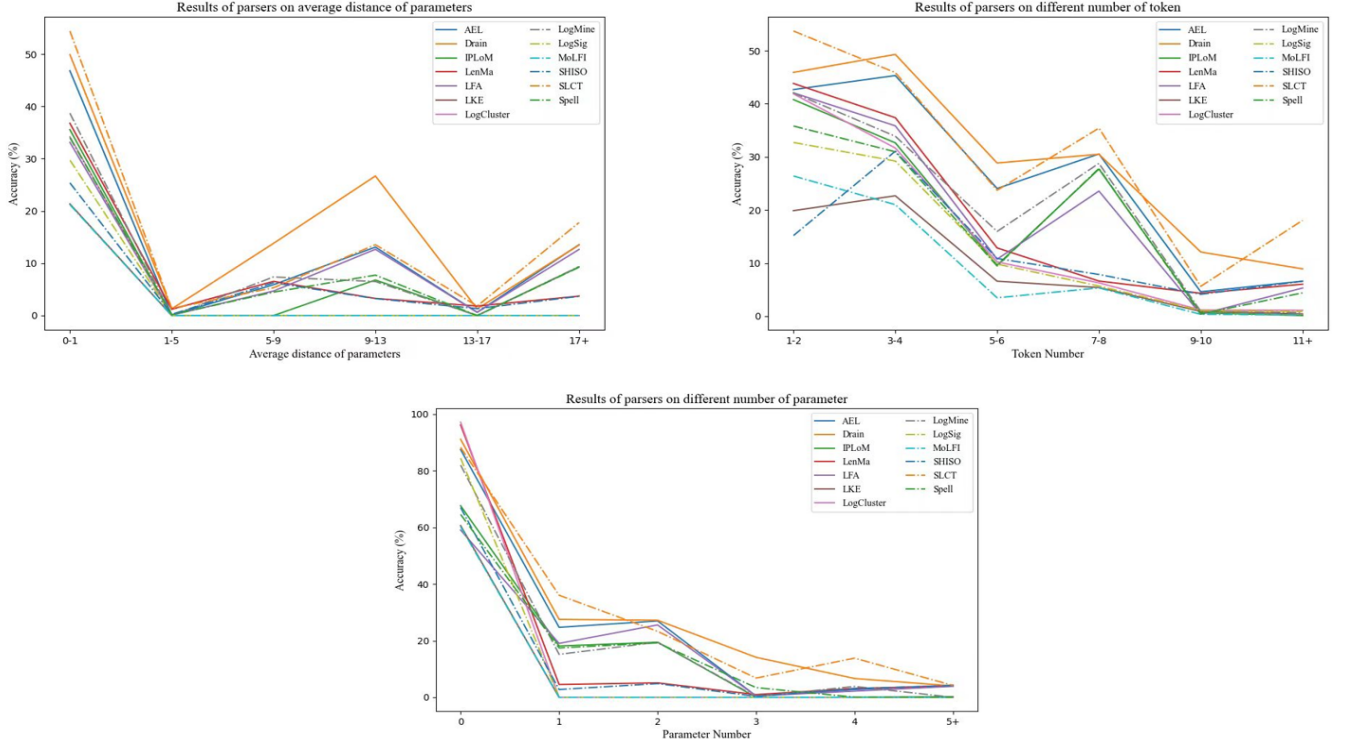


Fig. 2. Results of parsers on different template forms.

distance of parameters, the ranges are 0-1, 1-5, 5-9, 9-13, 13-14, and over 17. The fewer the number of tokens or parameters a parser deals with, the better accuracy it can achieve. When the parameters in the log message have an average distance of zero (which is actually the case where the number of parameters is 0 or 1), the parser performs better. When there are parameters in the log template, LKE and LogSig cannot recognize them. The paper only summarizes the results, while more thorough and in-depth findings are put in GitHub.⁴ Here, we list a few instances of parsing mistakes from Drain in Table VII. Our finding is that Drain has poor parsing of numeric classes, symbols, etc. Besides, it occasionally produces different parsing results for the same log due to the fact that the context content differs. However, we found that several bugs were brought on by wrongly labeled Ground Truths. As a result, we decided to entirely and thoroughly modify the Ground Truth files.

Summary (1) Most parsers do not work well with log messages that have consecutive variables. (2) The parsers perform better when there are fewer tokens or parameters.

IV. THE FUTURE

In this section, we talk about how we changed the ground truth and made a regex-based parser. We also show the results of our tests to check how well they work.

A. Datasets

Multiple errors in the current Ground Truth: (1) There are some strings that are misrepresented as arguments. (2) Poor consistency. Symbols are handled poorly; some are assigned as parameters while others are not. (3) Poor interpretability. Certain parameter settings are absurd. According to a set of rules, we modify the Ground Truth files. For the logs that can be confirmed by the source code, we alter the files in accordance with the source code. For other logs, we aim for consistency and common sense. If the guidelines are not applicable, we use our best judgment. To keep the consistency, the following regulations have been created. (1) Combining letters, integers, and symbols into a single parameter, if there is no space in between, symbols are not preserved. However, the symbol must be kept if there is a space between it and the character. Whatever the case, parentheses, square brackets, curly brackets, etc. need to be preserved. (2) The following are regarded as parameters: pure numbers, numbers only mixed with decimal points, and numbers only mixed with letters. (3) Strings with numerous decimal points between them are regarded as parameters (the number of decimal points is greater than 2). (4) It is treated as a parameter if it comes after the equal sign. (5) Under some circumstances, capital letters are parameters (e.g., from INITED to SETUP). (6) Paths and URLs are regarded as parameters. Whenever a string has more than two slashes, it is treated as a URL or path. (7) The use of certain special terms as parameters, such as null, true, and false. (8) Strings are treated as parameters if they follow certain specific terms, such as user and admin. In Table VIII, we provide various samples. The majority of the log content

⁴<https://github.com/logpai/logparser/tree/master/logsrevised>

TABLE VII
INSTANCES OF PARSING MISTAKES FROM DRAIN.

Content	Result from Drain	Original Ground Truth	What we deem to be the proper Ground Truth
Starting Socket Reader #1 for port 62260	Starting Socket Reader #1 for port <*>	Starting Socket Reader #<*>for port <*>	Starting Socket Reader <*>for port <*>
Scheduled snapshot period at 10 second(s).	Scheduled snapshot period at 10 second(s).	Scheduled snapshot period at <*>second(s).	Scheduled snapshot period at <*>second(s).
adding path spec: /mapreduce/*	adding path spec: <*>	adding path spec: /<*>/	adding path spec: <*>

TABLE VIII
EXAMPLE OF MODIFYING GROUND TRUTH TEMPLATES.

Content	Original template	Modified template
appattempt_1445144423722_0020_000001	appattempt_<*>	<*>
org.apache.hadoop.mapreduce.lib.output.File	org.apache.hadoop.mapreduce.lib.output.File	<*>
OutputCommitter set in config null	OutputCommitter set in config null	OutputCommitter set in config <*>
Transitioned from NEW to INITED	Transitioned from NEW to INITED	Transitioned from <*>to <*>
http://wiki.apache.org/hadoop/NoRouteTo	http://wiki.apache.org/hadoop/NoRouteTo	<*>
admin [preauth]	admin [preauth]	user [<*>]
Function=int	Function=int	Function=<*>
fpr29=0xffffffff ffffffff ffffffff ffffffff	fpr29=<*>	fpr29=<*><*><*><*>

is processed using these rules. We have not made any changes for some particular circumstances or issues with the original data. After adjustment, the type and quantity of templates will change, and the specifics will be shown in Table IX. We provide some details, such as how many template types have been edited, the number of original template types, and how many template types existed.

TABLE IX
MODIFICATION OF THE GROUND TRUTH TEMPLATE TYPES.

Dataset	Modified	Original	Now
HDFS	14	14	14
Hadoop	77	114	103
Spark	14	36	32
ZooKeeper	17	50	49
OpenStack	14	43	43
BGL	29	120	120
HPC	11	46	44
Thunderbird	58	149	149
Windows	12	50	49
Linux	48	118	115
Mac	161	341	324
Android	76	166	155
HealthApp	21	75	73
Apache	1	6	6
OpenSSH	19	27	27
Proxifier	15	8	15

B. Regular Expressions

Since the existing one performs poorly when parsing some elements of the log content, as khan et al. [5] also point out that some metrics do not favor existing approaches, in this section, we intend to present a straightforward and efficient parser. The regex-based parsers mentioned by Zhu et al.[2] rely on the manual construction of regexes for matching all possible logs, which could be hundreds of complex regexes (e.g., 120 regexes for the 2k BGL dataset). Differently, the regexes in this paper are designed to match tokens (instead of logs). Thus, we only need to manually construct several simple regexes for matching tokens in all datasets. [We use CSV files for both input and output.](#) We created a regular expression-based parser according to the modified Ground Truth rules and the log content. Here, we provide some pseudocode.

It has 23 regexes that match common log patterns, such as timestamps, process IDs, and error codes, and they fall approximately into five groups. We chose these groups based on our data and they can help us start developing regexes for log parsing. The first group has strings with numbers, like dates, IP addresses, and hex numbers. These strings can tell us when and where events happened. The second group has URLs, which are common in web server logs and can tell us what pages users visited on a website. The third group has strings after an equal sign in a log message. These strings often assign values to variables or parameters and can help us identify specific events or actions. The fourth group has paths, which often show file locations or directories in a log message. These strings can help us identify specific files or directories that were used or changed. The last group has strings after special nouns like “user” or “admin”. These

strings can help us identify specific users or administrators who did certain actions in a log message. These groups are not complete and may not cover all data in log parsing. But they can help us start making regexes and grouping information for log analysis. Some regular expressions are displayed in Fig. 3. The attachment contains all more information. Using both the original Ground Truth files and the modified Ground Truth files, we test the existing parsers and our parser. The benchmarking results are presented in terms of three research questions.

- **RQ4:** How do the existing approaches perform on the new Ground Truth? How is this performance different from that on the original Ground Truth?
- **RQ5:** How does our regex-based parser perform under different Ground Truths and metrics?

Input :
 $C = \{c_1, c_2, \dots, c_x\}$: log contents
 $I = \{i_1, i_2, \dots, i_y\}$: regular expressions for identifying a target string in the log content
 $S = \{s_1, s_2, \dots, s_y\}$: regular expressions to extract a substring from the identified string
 $R = \{r_1, r_2, \dots, r_y\}$: strings used to replace the substrings
 where $x = |C|$, $y = |I| = |S| = |R|$.

Function `fitRE` (C, I, S, R):

```

C' = list()
// For each log
for m from 1 to x do
    // For each RE rule
    for n from 1 to y do
        // Perform replacement until the search
        // fails
        while re.search(in, cm) do
            // Search for in in cm
            idfString ← re.search(in, cm)
            // Search for sn in idfString
            subString ← re.search(sn, idfString)
            // Replace subString by rn in
            // idfString
            repString ← idfString.replace(subString, rn)
            // Replace idfString by repString in
            // cm
            cm ← cm.replace(idfString, repString)
        end
    end
    end
    C'.append(cm)
end
return C'

```

Algorithm 1: Pseudocode of conducting the regular expressions

Under the original Ground Truth and the new Ground Truth, we list the ranking of parsers for each metric in Table X. The ranking of parsing results under each metric has changed using different Ground Truths. It is clear that the change to Ground Truth has an influence on the current parsers, whether it has a positive or negative impact. The RTA metric has resulted in a major change in the LogCluster rankings; RTA is ranked 4 with the original Ground Truth but dips to 11 with the new Ground Truth. Similar things arise when using different parsers and metrics. Table XI displays all of the outcomes. In Table XI, we present the evaluation outcomes of the regex-based parser for the initial Ground Truth and the current Ground Truth, respectively. We list the top outcomes from other parsers in the final row of Table XI. Except for

Prefix/Suffix Expressions
Prefix: non-number+dot, colon, space, parentheses(3 kinds), equal, comma, start, , <, quotation(single+double), semi-colon prefix = r'([0-9]\. :\(\) \[\]\ \}\ _ !^\'\"`\ )'`
Suffix: space, parentheses(3 kinds), comma, end, colon, dot+space/end, , >, equal, quotation(single+double), semi-colon suffix = r'(\(\) \[\]\ \}\ _!\$: \. \ .\$ \\ > = \'\"`\)'
Long Expressions
URL (http/hdfs + ://) param = r'[A-Za-z._]+://[A-Za-z0-9\.\^#@:_~]*(?![:.])'
Path start with (. /) param = r'([A-Za-z] \.){0,1}((\/)[0-9A-Za-z- _\.:/*\+\\$#@!\ \\?\=%&]+)(?![:.])'
Following the keyword
User param = r'user() [A-Za-z0-9]*(?<!request)(?! methods)' from ... to ... param = r'from [A-Za-z0-9_] + to [A-Za-z0-9_] +'
Short Expressions
Inside <> param = r'[A-Za-z0-9#@_\.]+'`
Days param = r'(Mon Tue Wed Thu Fri Sat Sun)'
Lastly, Deal With remaining improper <*>
Merge <*>;<*>, <*>:[end] param = r'(<!*>)(-<!*> [A-Za-z0-9]+)*(?<![*>:Got)'
Merge <*> KB/MB param = r'<!*> (B KB MB)'

Fig. 3. Formulation of regular expressions.

the value of GA, our regex-based parser outperforms all other parsers in terms of other metrics based on the original Ground Truth. Khan et al. [5] point out the ineffectiveness of current parsers and how poorly they perform on their proposed metrics (PTA and RTA). However, even with the original Ground Truth, our regex-based parser can still produce good results across the metrics. The regex-based parser’s performance has significantly improved since utilizing the new Ground Truth. The results of all metrics outperform those of all other parsers and are near to being full marks. Despite not utilizing machine learning, our parser can nevertheless produce excellent results.

To confirm the versatility of our parser, we gathered log data from the CodeArts PerfTest system of Huawei company. CodeArts PerfTest is a cloud-based service that enables performance testing for application interfaces and links. It supports protocols such as HTTP/HTTPS/TCP/UDP and offers a wide range of test model definition capabilities to simulate real-world business access scenarios. By identifying application performance problems beforehand, users can make necessary improvements and avoid potential issues. We utilized it in our experiments to investigate the generalizability of our approach. Without writing any additional regular expressions, we directly applied the regex-based parser to industrial data and achieved impressive results, outperforming other parsers. Our parser performed exceptionally well, with a PA value of 0.98, surpassing other parsers by 0.88. Additionally, the RTA reached 0.9778, outperforming other parsers by 0.8667.

TABLE X
THE RANKING RESULTS OF LOG PARSERS ON ORIGINAL AND NEW
GROUND TRUTH.

Parser	Rank	GA	PA	PTA	RTA	FITA
SLCT	Original	10	1	3	8	4
	New	11	3	3	8	5
AEL	Original	3	3	2	2	2
	New	2	2	2	2	2
IPLoM	Original	2	6	6	9	9
	New	3	7	6	9	8
LKE	Original	12	11	10	11	11
	New	12	13	12	12	12
LFA	Original	9	5	9	10	10
	New	8	4	5	7	7
SHISO	Original	7	10	9	6	6
	New	9	9	5	7	7
LogSig	Original	13	12	13	13	13
	New	13	11	13	13	13
LogCluster	Original	8	9	11	4	8
	New	10	10	10	11	11
LenMa	Original	5	4	4	5	3
	New	4	8	8	3	4
LogMine	Original	6	4	5	5	5
	New	6	5	9	4	9
Spell	Original	4	7	8	7	7
	New	5	6	4	6	3
Drain	Original	1	1	1	1	1
	New	1	1	1	1	1
MoLFI	Original	11	13	12	12	12
	New	8	12	11	10	10

Summary (1) Changing the Ground Truth files affects the current parsers, both good and bad. The rankings of the results can change by up to 7. (2) Parser technology is not bad. Our regex-based parser can still produce good results using the original Ground Truth across different metrics. (3) Our regex-based parser has improved a lot with the modified Ground Truth, and now all the metric evaluation values are almost perfect.

V. THE UNCERTAINTIES

In our work, there are various uncertainties that mostly stem from two aspects. (1) Some log templates are too complex for the current parsers to handle correctly, or too difficult to verify their correctness. The dataset itself has a number of issues that the parser finds challenging to identify and address. Multiple spaces or missing spaces frequently occur in logs, which has significant negative influence on log parsing. We give some examples in Fig. 4. Due to the absence of space, the parser will interpret "job_1445144423722_0020Job" as a whole parameter in the first example. The recognized template

- (1) **Content:** job_1445144423722_0020Job Transitioned from NEW to INITED
Parsing result: <*> Transitioned from <*> to <*>
Ground Truth: <*> Job Transitioned from <*> to <*>
 miss a space
- (2) **Content:** user inspir from 175.102.13.6 port 47130 ssh2
Content: user 0101 from 5.188.10.180 port 36279 ssh2
Template: user <*> from <*> port <*> <*>
 an extra space
- (3) **Content:** GET /v2/54fadb412c4e40cdbaed9335e4c35a9e/servers/detail HTTP/1.1
Template: GET <*> HTTP/1.1
- (4) **Content:** Failed password for root from 5.36.59.76 port 42393 ssh2
Template: Failed password for <*> from <*> port <*> <*>
Content: Disconnecting: Too many authentication failures for root [preauth]
Template: Disconnecting: Too many authentication failures for root [<*>]
- (5) **Content:** problem state (0=sup,l=usr).....0
Parsing result: problem state (0=<*>,l=<*>).....<*>
Ground Truth: state (<*>=sup,<*>=usr).....<*>

Fig. 4. Examples of uncertainties.

of the second example is different from the correct template because the content has more space. Also, the regex-based parser does not work well with dates, strings before and after equal signs (the fifth example). Additionally, there are particular regex rules that clash.

(2) It can be challenging to determine whether certain log templates are accurate while updating Ground Truth files. "HTTP/1.1" should be a parameter in the third example if our rules are followed, however, we believe that it should not be in this instance, since a great number of log messages hold this term unchanged while varying in other parts. It is quite difficult to determine logs like this. In addition, various logs may interpret the same string differently; in the fourth illustration, the root may or may not be the parameter. This is quite difficult for the parser to handle; consequently, more rules are required to construct a more comprehensive and accurate regex parser.

VI. THREATS TO VALIDITY

The validity of our study may be subject to some threats. The selection of the log datasets poses the biggest risk of external validity. We reviewed the literature on log analysis and identified 16 datasets that are widely used in the relevant papers. On 16 general datasets, all of the existing parsers and our proposed parser are compared. However, the data has not been updated in a while, and both the Ground Truth files and the datasets contain numerous inaccuracies. We completely altered the Ground Truth files to make this better. The analysis in RQ4 and RQ5 demonstrates how the accuracy of the Ground Truth affects the evaluation outcomes. Threats to internal validity are factored into our methodology and may affect the parsing results. Our parser is generated according to data-based features and rules, the regex-based parser may not be suitable for other datasets. Expanding the scope of the study to include more evaluation metrics could result in a more comprehensive understanding of log parsing performance. To mitigate this threat, we try to make general rules to cover more types of log content. Besides, we gathered 2000 log data from the system of Huawei company and utilized it in our experiments to investigate the generalizability of our

TABLE XI
RESULTS OF REGEX-BASED PARSING.

Parsing with Original Ground Truth						Parsing with new Ground Truth				
Dataset	GA	PA	PTA	RTA	FTA	GA	PA	PTA	RTA	FTA
HDFS	0.9975	0	0	0	0	0.9975	0.9985	0.88	1.0	0.93
Hadoop	0.88	0.158	0.2636	0.2543	0.2589	0.92	0.9195	0.8545	0.9038	0.8785
Spark	0.9965	0.7765	0.4643	0.52	0.4906	0.999	0.9995	0.9677	0.9375	0.9524
ZooKeeper	1.0	0.377	0.6512	0.6512	0.6512	0.9945	0.988	0.7321	0.8367	0.7810
OpenStack	1	0.377	0.6512	0.6512	0.6512	1	0.9685	0.9302	0.9302	0.9302
BGL	0.991	0.89	0.7521	0.75833	0.7552	0.991	0.9935	0.9504	0.9583	0.9544
HPC	0.9325	0.91	0.6604	0.7609	0.7071	0.978	0.99	0.7736	0.9318	0.8454
Thunderbird	0.6745	0.355	0.4220	0.4899	0.4534	0.6745	0.935	0.7341	0.8523	0.7888
Windows	0.6915	0.676	0.6364	0.7	0.6667	0.996	0.994	0.8727	0.9796	0.9231
Linux	0.3575	0.196	0.6087	0.5932	0.6009	1	0.938	0.9304	0.9304	0.9304
Mac	0.898	0.4415	0.4583	0.4516	0.4549	0.9525	0.8275	0.6935	0.7191	0.7061
Android	0.8465	0.526	0.5466	0.5301	0.5382	0.95	0.8515	0.9068	0.9419	0.9241
HealthApp	0.9025	0.6185	0.6712	0.6533	0.6622	1	0.8595	0.9041	0.9041	0.9041
Apache	1	0.7155	0.8333	0.8333	0.8333	1	1	1	1	1
OpenSSH	1	0.109	0.2593	0.2593	0.2593	1	1	1	1	1
Proxifier	0.0485	0	0	0	0	1	1	1	1	1
Average	0.8256	0.4623	0.4940	0.5024	0.4974	0.9658	0.9539	0.8828	0.9266	0.9032
Best of others	0.87	0.29	0.27	0.29	0.28	0.83	0.35	0.30	0.33	0.31

Parsing with Industrial Data (Regex-based)						Parsing with Industrial Data (Best of others)				
Dataset	GA	PA	PTA	RTA	FTA	GA	PA	PTA	RTA	FTA
Huawei	0.9975	0.98	0.9565	0.9778	0.9670	0.3055	0.1	0.1923	0.1111	0.1408

approach. We also have certain ambiguous issues that could be overlooked while editing the Ground Truth files, which could impact the outcomes. Modifying Ground Truth files is a very rigorous and meticulous work, even one space will affect the final result. To address this issue, we had three people to conduct these studies in order to reduce this threat and make sure the process of change was uniform. To look for inconsistencies, four people conducted manual inspections and compared the trial data. We also made the updated Ground Truth files accessible to everyone.

VII. DISCUSSION AND CONCLUSION

Our study showed that our parser performed well on both existing and industrial log data. However, there were limitations to our study that could be addressed in future research. For example, we only focused on word-level classification tasks and did not explore other approaches to log parsing. To address these limitations, potential avenues for research include exploring deep learning techniques, using domain-specific knowledge or ontologies, and developing log parsing methods for different log formats. Additionally, we suggest the need to develop standardized evaluation metrics and datasets to enable direct comparisons between parsers and promote progress in the field.

In this paper, we selected a diverse set of datasets of varying complexity, and source, each with an equal number of log messages to ensure fair and unbiased evaluation results. We evaluated each parser on all datasets and used multiple evaluation metrics to assess parser performance from different perspectives. Although other variables could have influenced our results, we believe our study design and methodology were robust enough to minimize their impact. However, we acknowledge that factors such as the complexity, variety, and quality of logs can still affect parser performance and should be considered in the evaluation process. To address this issue, future studies can take steps to better control for potential confounding variables. This could involve using more standardized datasets or statistical techniques such as randomization or matching to ensure differences in performance are not simply due to differences in datasets. Researchers can also be more transparent in reporting their methodology and results, including details about dataset selection and processing, and any preprocessing steps taken. This will help readers better understand the study's limitations and how the results can be applied in practice.

In conclusion, the empirical study conducted in this paper sheds light on the current state of log parsing research and provides important insights for future directions. The results showed significant variation in parser performance under dif-

ferent evaluation metrics and emphasized the importance of preprocessing in log parsing. The study also revealed that the commonly used LogHub dataset contains labeling errors and a revised Ground Truth was provided after extensive manual efforts. Finally, the results suggest that formulating log parsing as a word-level classification task may be a feasible future direction, as the simple parser based on word-level regular expressions outperformed existing parsers in terms of different metrics on various datasets. A number of conclusions can be summed up.

- Under the current metrics, we produce a wide range of findings. Different metrics behave very differently in evaluating log parsers, and may display very different results under a single parser.
- The analytical results for various datasets are quite disparate.
- Preprocessing can impact the log parsers. The amount and quality of the impact depend on factors like the dataset and regular expressions used.
- Parser technology is not poor. Our regex-based parser can still produce good results using the original Ground Truth across different metrics.
- Errors in the log content and Ground Truth greatly affect the performance of the regex-based parsing. But after revising the Ground Truth, the performance improved a lot. We have successfully posted the latest version on GitHub.
- Our regex-parser can process logs much more efficiently than other parsers. However, regular expressions can not handle all uncertainties, so we may need to consider other options.

Based on our results, we recommend that practitioners pay close attention to preprocessing steps when parsing logs, as these can have a significant impact on parser performance. In particular, we found that removing irrelevant or redundant information from logs prior to parsing can improve accuracy and reduce labeling errors. Additionally, we suggest that practitioners consider using word-level classification tasks as a feasible approach to log parsing, as this method outperformed existing parsers in our experiments. Overall, we believe that our study provides valuable insights into the current state of log parsing research and highlights areas for improvement in automated log analysis. By taking into account the recommendations outlined in our paper, practitioners can improve the accuracy and usefulness of their log parsing methods and ultimately enhance their ability to extract insights from software runtime logs.

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